HYPER-RIDGE PERSONALEVO: META-LEARNING THE ARCHIVE GEOMETRY FOR PROVABLE PERSONALIZED ROUTING

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ABSTRACT

Personalized agents maintain libraries of prompt variants, skill modules, or orchestration policies, which we call *candidates*, collected in an *archive*.

The simplest strategy is *uniform deployment*: pick the single candidate with the highest average reward across all users.

We propose PersonalEvo, a contextual-bandit method that learns to route each user to their best-matching candidate online, and optionally grows the archive by evolving new candidates after failures.

We give PersonalEvo a provable guarantee: its reward decomposes exactly into four interpretable terms, the *default reward* of the best single candidate served to everyone, a *routing gain* Δ_{route} from matching each user to their own best candidate, an *evolution gain* from newly discovered candidates, and an *exploration cost* paid while learning user preferences.

This decomposition yields a sharp condition for when personalization helps, routing provably outperforms uniform deployment whenever the routing and evolution gains together exceed the exploration cost, together with a plug-in regret bound on the exploration cost.

We evaluate PersonalEvo on nine benchmarks: three synthetic personalized-agent environments, two agent-framework substrates (GEPA prompt archives (Agrawal et al., 2025), Hermes skill configurations (Team, 2025b)), and four real-data preference datasets (BESPOKE (Team, 2025c), PAHF Shopping and Embodied (Labs, 2025), and PrefEval (Zhao et al., 2025)).

Across all nine, PersonalEvo achieves higher mean reward than both the static baseline and random candidate selection.

Our results show that: (1) the routing gain Δ_{route} is larger than the evolution gain on every benchmark; (2) on the four real-data benchmarks, Δ_{route} ranges from 0.036 to 0.251, indicating that the value of personalized routing varies substantially across application domains; (3) growing the archive without gating hurts performance, because the added observed-reward residual outweighs the evolution gain. The saved base experiments report $m_t - r_t$, not the mean-reward exploration cost $m_t - \mu(U, a_t)$ in our theorem.

Building on this base, we introduce *Hyper-Ridge PersonalEvo*, which replaces independent isotropic regularization with a shared archive geometry. For known geometry, we prove a pathwise exploration bound with an explicit confidence radius and an effective-dimension potential term. We also prove finite-sample covariance control for a calibration-split variant that estimates the geometry from fresh candidates and freezes it before routing. With fixed problem constants, its explicit calibration charge and width-inflation remainder are $o(\sqrt{T})$, while its full exploration bound is $O(\sqrt{T}\log T)$. Neither theorem covers the periodically re-estimated router. In diagnostics, the true geometry lowers cumulative exploration cost by 46–59%, whereas periodic re-estimation raises it by 7–19%.

1 INTRODUCTION

Modern LLM agents increasingly choose among collections of prompts, skills, tools, and workflows instead of relying on one fixed policy. We call such a collection an *archive* and each deployable entry a *candidate*. Personalization methods learn user profiles and preference-conditioned behavior (Salemi et al., 2023; Zhao et al., 2025; Team, 2025c); prompt and agent evolution methods search for stronger candidates (Fernando et al., 2023; Guo et al., 2024; Zhou et al., 2024; Agrawal et al., 2025); and routing methods choose among available models or policies at inference time (Ong et al., 2024; Hu et al., 2024; Li, 2025; Tsai & Tran, 2026). These lines of work address three related decisions: what a user prefers, how the archive should improve, and which candidate should handle the next request.

Consider a support agent whose archive contains formal and casual policies for refund and shipping requests. This example fixes the setting we study. At each round a user arrives, the agent deploys one candidate from the current archive, and only the deployed candidate's reward is observed; candidate selection is therefore a contextual bandit over the archive. The agent is also self-evolving: after poor feedback it mutates an existing candidate and adds the result to the archive, so the set of arms grows during deployment. Serving every user the best policy on average discards the archive's useful diversity. A personalized router can learn which policy works for each user, but it must first try candidates whose value is uncertain. Archive growth makes this cost recur: each new mutation is a new arm that a conventional bandit starts learning from scratch. Yet the candidates are related. What the router learns about formality from a refund policy should inform a formal shipping policy, even though the two policies should retain different reward models. Archive growth therefore creates a tension: it can add better candidates while also adding enough exploration to erase their benefit.

Existing methods resolve only parts of this tension. Profile and memory methods condition an agent on the user, but do not control exploration over a growing archive (Salemi et al., 2023; Team, 2025c; Zhang et al., 2026; Westhäußer et al., 2025). Evolution methods improve candidate quality, usually offline or at the population level, but do not learn which candidate to serve to each user online (Fernando et al., 2023; Guo et al., 2024; Zhou et al., 2024; Agrawal et al., 2025). Fixed-pool routers make that assignment, but do not account for archive growth (Ong et al., 2024; Li, 2025; Tsai & Tran, 2026). Standard disjoint linear bandits also estimate every candidate independently, so evidence about one candidate cannot reduce uncertainty about another (Li et al., 2010; Chu et al., 2011). Pooling all candidates into one predictor goes too far in the other direction and erases real candidate-specific differences. What is missing is the *geometry of variation across the archive*: which behavioral directions recur across candidates, which differences remain candidate-specific, and when evidence can be transferred between them.

These observations lead to the question studied in this paper:

> *Can a self-evolving agent transfer evidence across related archive candidates while retaining a measurable guarantee that personalized routing is worth its exploration cost?*

We address this question in two steps. First, *PersonalEvo* separates the value of personalization from the cost of learning it. Let A_1 be the initial archive and let R_T be the router's average reward over T interactions. We prove the exact decomposition

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \tag{1}$$

where $V_{\text{static}}(A_1)$ is the reward from serving everyone the best single initial candidate, $\Delta_{\text{route}}(A_1)$ is the gain from matching users to different initial candidates, $\bar{G}_T$ is the gain from candidates discovered after deployment, and $\bar{L}_T$ is the reward lost while the router learns. In the support example, these terms are the best universal reply policy, the value of matching tone and topic to each user, the value of newly evolved replies, and the cost of trying the wrong reply. The identity holds for any measurable router and gives the testable deployment criterion

$$\Delta_{\text{route}}(A_1) + \bar{G}_T \;>\; \bar{L}_T, \tag{2}$$

so personalized routing is worthwhile exactly when its routing and evolution gains cover its exploration cost.

Second, *Hyper-Ridge PersonalEvo* targets the only negative term in equation 1. Each candidate a has its own parameter vector θ_a, which maps d user and candidate features to expected reward.

A disjoint LinUCB router estimates each vector with the same spherical regularizer and therefore encodes no relation among candidates (Abbasi-Yadkori et al., 2011; Chu et al., 2011). We instead model the candidate vectors as draws with a shared covariance Ω. This covariance is the archive geometry. Its dominant directions capture recurring factors such as tone and topic, while every candidate keeps its own reward model. Hyper-Ridge uses this geometry to regularize candidate estimates, so a newly evolved candidate starts with uncertainty shaped by the archive rather than an isotropic blank slate.

The gain is largest when candidate parameters vary mainly along a few shared directions even though the feature representation is high-dimensional. Let $\bar{d}_{\mathrm{eff},T}(\Omega)$ be a deterministic upper bound on the effective dimension of every candidate design through round T, and let $c_T^{\Omega} = 1 + \log(1 + TL_{\Omega}^2/\lambda)$. For known Ω, our theorem gives the explicit exploration budget

$$B_T^{\Omega}(\delta) = 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\mathrm{eff},T}(\Omega)}, \tag{3}$$

where $\alpha_{\Omega,T}(\delta)$ is the confidence radius stated in Proposition 8. The base LinUCB budget has the same form with its own radius and the isotropic potential $d\log(1 + TL_x^2/(\lambda d))$. Thus geometry helps only when the complete budget in equation 3, including its confidence radius, is smaller. For $T \geq 2$, with fixed problem parameters and $\delta = 1/T$, both valid noisy-reward bounds scale as $O(\sqrt{T}\log T)$; the benefit is in their problem-dependent geometric factors, not a missing logarithm. The calibration-split theorem retains an estimated-geometry confidence radius; its separate calibration charge and width-inflation remainder are $o(\sqrt{T})$. It does not cover the periodically updated estimator in Algorithm 1.

Our experiments separate the motivation for Hyper-Ridge from evidence for its mechanism. Across nine benchmarks, the base PersonalEvo router has routing gains from 0.036 to 0.280, evolution gains from 0 to 0.061, and observed-reward residuals from 0.010 to 0.071. Because the simulator clips noisy rewards, these residuals are not identified with the theorem's mean-reward exploration cost. On separate planted rank-three problems, giving Hyper-Ridge the true geometry lowers cumulative exploration cost by 46–59%. Periodically estimating that geometry instead raises the cost by 7–19%. On seven adapter benchmarks, one-hot candidate features make the full-covariance router identical to its diagonal ablation, so their reward changes cannot show cross-candidate transfer. The evidence therefore supports the known-geometry mechanism while exposing the unresolved estimation and representation problem.

Contributions. This paper makes four contributions:

1. **A measurable formulation of personalized self-evolution.** We formalize routing over a growing candidate archive and prove the exact four-term decomposition equation 1, which turns the decision to personalize into the testable criterion equation 2.
2. **Archive-aware routing.** We introduce Hyper-Ridge PersonalEvo, which learns a covariance geometry across candidate reward models and uses it in a generalized-ridge UCB router. It shares evidence about important feature directions while retaining candidate-specific predictions.
3. **Effective-dimension guarantees with a clear boundary.** We prove a known-Ω bound with its full confidence radius and a finite-sample calibration-split theorem for estimated Ω. We do not transfer either guarantee to the unresolved periodic estimator.
4. **Evidence that separates mechanism from implementation.** We measure the structural terms and report the logged observed-reward residual on nine benchmarks, then test Hyper-Ridge in a controlled low-rank setting. The results show when the right geometry helps and why the current periodic estimator and one-hot representation do not yet realize that gain.

Paper organization. Section 2 positions the work relative to personalization, agent evolution, routing, and linear bandits. Section 3 defines the routing problem and Hyper-Ridge estimator. Section 5.1 develops the decomposition and effective-dimension analysis. Section 6 reports the base-router results, Hyper-Ridge diagnostics, and remaining evaluation gap. Appendix J proves the oracle geometry results, and Appendix L gives the calibration-split covariance theorem.

2 Related Work

Routing queries across a model or prompt pool to trade off cost and quality is now standard: RouteLLM (Ong et al., 2024) learns a strong/weak router from preference data and RouterBench (Hu et al., 2024) benchmarks such routers, while LLM Bandit (Li, 2025) and NeuralUCB routing (Tsai & Tran, 2026) cast routing as an online contextual bandit.

These route at the *model* level over a fixed pool; PersonalEvo routes *per user* over an archive that may grow online, and decomposes the resulting reward rather than only reporting it.

Benchmarks such as LaMP (Salemi et al., 2023), PersonalLLM (Zollo et al., 2024), and PrefEval (Zhao et al., 2025) supply user-feedback signals, while PersonaAgent (Zhang et al., 2026) and persistent-memory frameworks (Westhäußer et al., 2025) maintain per-user personas and evolving profiles.

The closest personalized-bandit antecedent, PAB-F (Team, 2025c), models per-user reward as a factorized function of context and a fixed memory/tool/style action set, and studies regret against an in-archive oracle.

Prompt and agent evolution methods, including Promptbreeder (Fernando et al., 2023), EvoPrompt (Guo et al., 2024), Symbolic Learning (Zhou et al., 2024), and GEPA's reflective Pareto search (Agrawal et al., 2025), improve prompts or archives through offline search rather than online user-conditioned routing.

Hermes self-evolution (Team, 2025b;a) adds constraint-gated growth, which motivates our archive-growth gate.

Classical contextual-bandit methods such as OFUL and LinUCB (Abbasi-Yadkori et al., 2011; Chu et al., 2011; Li et al., 2010) provide standard finite-time tools for controlling exploration in linear bandits, and related structural-vs-online decompositions appear in contextual pricing, offline RL, and multi-task transfer.

3 Preliminaries: PersonalEvo Routing and the Hyper-Ridge Estimator

3.1 Overview

Managed platforms such as Microsoft Foundry Agent Service and Amazon Bedrock AgentCore (Microsoft, 2026; Amazon Web Services, 2026) let teams build and deploy agents from models, tools, and instructions. A product team can maintain several agent configurations—for example, different system prompts or tool policies—and revise them as feedback arrives. We use these products only as concrete deployment settings; neither is assumed to implement our router.

A personalized agent keeps a small library, an *archive*, of candidate prompts, skills, or routing templates.

At each round, it receives one user's request, picks one candidate from the current archive, observes a reward for that candidate, updates its state, and may add an evolved child to the archive.

We compare this personalized routing policy with a no-adaptation baseline: pick the candidate in the initial archive with the highest population-mean reward and use it for every user and every round. A comparator that picks again each round is a dynamic selection policy; it would fold online learning or archive growth into the baseline. Fixing one initial candidate lets us measure routing gain, evolution gain, and exploration cost separately.

The remainder of this section formalizes the notation, introduces the hyper-ridge estimator and its generalized-ridge model, presents the routing algorithm, and states the base theoretical results.

3.2 Notation and setup

Prompt evolution is one concrete instance of this loop. A task supplies an input to an LLM system with a system prompt; the system's output is scored or receives textual feedback; and an optimizer such as GEPA (Agrawal et al., 2025) uses that feedback to propose and test a revised prompt. We call each deployable version a *candidate*. More generally, candidates may differ in their prompts, tool configurations, model choices, or agent policies. Evolution determines which candidates enter the archive, while routing determines which available candidate serves a particular user. We ask whether this per-user choice earns enough reward to cover the cost of learning user preferences.

A user $U \sim \Pi$ is drawn from a population: $\mathcal{U}$ denotes the user space (in our benchmarks a finite set of user profiles; in general an abstract measurable space), and Π is a probability distribution over $\mathcal{U}$, so Π is not a vector or matrix and has no dimension of its own; it simply specifies how likely each user is to arrive. Practically, Π is the deployment's user mix, and expectations $\mathbb{E}_U[\cdot]$ below average over that mix.

The system operates over $T \in \mathbb{N}$ rounds (successive requests) and starts from a deterministic initial archive A_1 shared by all users. This finite set is $A_1 = \{a^{(1)}, \ldots, a^{(K)}\}$, where each a is one deployable unit of agent behavior and $K = |A_1|$. Evolution may enlarge the archive along user U's trajectory, so $A_t(U)$ denotes the archive available to that user at round t, with $|A_t(U)| \le K_{\max}$. We use A without a subscript or user argument only for an arbitrary fixed archive in definitions such as $V_{\text{static}}(A)$.

The analysis follows *one* user's trajectory and then averages over the population. A single user U is drawn once and interacts with the system for T rounds. Let $\mathcal{O}_{t-1}$ be the router's observable pre-action history: the user, the initial archive, all earlier actions and rewards, and the archive available at round t. For concentration arguments, let $\mathcal{F}_{t-1} \supseteq \mathcal{O}_{t-1}$ be an analysis filtration that also contains latent candidate parameters and proposer randomness once they are generated. The router's action is $\mathcal{O}_{t-1}$-measurable; enlarging the filtration for the proof does not give the router access to those latent variables. Population-level statements then average over U and all online randomness. Quantities with argument U, such as $A_t(U)$, $a_t(U)$, $G_t(U)$, and $L_t(U)$, are per-user random variables. Quantities with an expectation or a bar, such as V_{static}, V_{route}, $\bar{G}_T$, $\bar{L}_T$, and R_T, are population averages.

At round t, the user-specific archive $A_t(U)$ is available to the router and is required to be $\mathcal{O}_{t-1}$-measurable.

The router selects $a_t(U) \in A_t(U)$ and observes the reward

$$r_t = \mu(U, a_t(U)) + \xi_t, \tag{4}$$

Here r_t is the scalar feedback generated when user U receives candidate $a_t(U)$. Conditional on the user, archive construction, and latent candidate parameters, its remaining randomness is the observation noise ξ_t, with $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$. Unconditional statements may also average over proposer randomness and the random-effects model introduced below. The function $\mu(u, a) \in [0, 1]$ is the noise-free mean reward of serving candidate a to user u.

The current-archive oracle value is $m_t(U) = \max_{a \in A_t(U)} \mu(U, a)$.

We define the four terms of equation 16.

The *default reward* is $V_{\text{static}}(A) = \max_{a \in A} \mathbb{E}_U[\mu(U, a)]$: average each candidate's reward over all users *first*, then pick the single best candidate and serve it to everyone. This is the performance of the simplest possible deployment, one that does no personalization at all (a single fixed prompt or policy shipped to every user), so it is the operating point a team is already at before adopting any routing method; it is the baseline any personalization system must beat to justify its cost.

The *oracle routing value* is $V_{\text{route}}(A) = \mathbb{E}_U[\max_{a \in A} \mu(U, a)]$, with the two operations in the opposite order: for *each* user pick that user's best candidate first, then average over users. This is the ceiling of personalization, the reward if user preferences were known perfectly and no exploration were ever needed. Practically it answers a question a team should ask before building anything: "if personalization worked perfectly on this archive, how much would we earn?" No online algorithm on the fixed archive can exceed it.

Their difference is the *routing gain* $\Delta_{\text{route}}(A) = V_{\text{route}}(A) - V_{\text{static}}(A) \ge 0$: the reward gained by matching candidates to users instead of shipping one candidate to everyone. For a self-evolving archive, our four-term decomposition separates this structural gain from the gain due to archive growth and the online cost of learning which candidate to choose. Thus $\Delta_{\text{route}}(A)$ is the reward budget available to pay for personalization. If users are nearly homogeneous, or one candidate dominates for everyone, then $\Delta_{\mathrm{route}}(A) \approx 0$ and user-specific routing offers little benefit. If some users prefer concise replies while others prefer detailed explanations, $\Delta_{\mathrm{route}}(A)$ can be large. A team can estimate this quantity on held-out users before deployment and compare it with the expected exploration cost.

These quantities say *why* personalization matters but not *when* it pays off; the dominance condition of Corollary 3 answers the latter by weighing Δ_{route} against the exploration cost.

The *evolution gain* at round t is $G_t(U) = m_t(U) - m_1(U)$, measuring how much the archive has improved since initialization. We define it because self-modification is not free (each mutation adds an arm the router must explore), so a deployment needs to know how much reward the newly evolved candidates are actually adding; $G_t(U)$ is exactly that amount, the improvement of the best-available candidate for user U between round 1 and round t. It is measured by bookkeeping, not by extra experiments: whenever a candidate is added, evaluate the incumbent best and the new candidate for the affected user and record the difference of the two bests. Concretely, in the support-archive example: if user U's best launch-day template has mean satisfaction $m_1(U) = 0.70$ and at round t a mutated template raised the best available to $m_t(U) = 0.78$, then $G_t(U) = 0.08$; if evolution never produced anything better for this user, $G_t(U) = 0$. In our simulators μ is known so $m_t(U)$ is computed exactly; in a live system it is estimated by scoring the current archive on the user's held-out interactions.

The *exploration cost* is $L_t(U) = m_t(U) - \mu(U, a_t(U))$, the gap between the current-archive best and what the router actually picked. It is the price of learning and the only negative term in the decomposition. Continuing the same example, if the best current template has mean satisfaction 0.78 but the router serves one with mean 0.60, that round costs 0.18. The simulator contains both means needed to compute this quantity, but the saved headline tables use $m_t - r_t$ instead. A live system must estimate both means on held-out or logged interactions and report the resulting uncertainty separately.

The horizon-averaged terms are $\bar{G}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[G_t(U)]$ and $\bar{L}_T = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[L_t(U)]$.

3.3 Model

For the Hyper-Ridge theory, write the bounded mean reward as

$$\mu(u,a) = b(u) + x(u,a)^\top \theta_a, \tag{5}$$

where $x(u,a) \in \mathbb{R}^d$ is the user-candidate feature vector and the known baseline $b(u)$ is common to every candidate offered to user u. We assume $\mu(u,a) \in [0,1]$. A convenient compatible model is $b(u) = 1/2$ and $x(u,a)^\top \theta_a \in [-1/2, 1/2]$.

Instead of estimating each θ_a with ordinary ridge regression, assume the candidate parameters are random effects:

$$\mathbb{E}[\theta_a] = 0, \qquad \mathbb{E}[\theta_a \theta_a^\top] = \Omega, \tag{6}$$

where $\Omega \in \mathbb{R}^{d\times d}$ is a hyper-covariance matrix shared across archive candidates. The common baseline makes this zero-mean random-effects model compatible with nonnegative rewards. It also cancels from every candidate comparison.

The goal is to estimate Ω from the archive and use it to regularize routing.

3.4 Hyper-ridge estimator

For candidate a, define

$$X_{a,t} = \left[x_s^\top\right]_{s<t:a_s=a}, \qquad y_{a,t} = (r_s - b(u_s))_{s<t:a_s=a}. \tag{7}$$

Given an estimate $\widehat{\Omega}_t$, define the generalized ridge estimator

$$\widehat{\theta}_{a,t} = \arg\min_{\theta\in\mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t}\theta\|_2^2 + \lambda \theta^\top \widehat{\Omega}_t^{-1} \theta \right\}. \tag{8}$$

Equivalently,

$$\widehat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}\right)^{-1} X_{a,t}^\top y_{a,t}. \tag{9}$$

Define the generalized design matrix

$$V_{a,t}^{\text{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{10}$$

Then the Hyper-Ridge UCB score is

$$\text{UCB}_t^{\text{hyper}}(u,a) = x(u,a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u,a)^\top \left(V_{a,t}^{\text{hyper}}\right)^{-1} x(u,a)}. \tag{11}$$

The selected candidate is

$$a_t(u) \in \arg\max_{a \in A_t(u)} \text{UCB}_t^{\text{hyper}}(u,a). \tag{12}$$

3.5 Algorithm

Algorithm 1 summarizes the resulting routing procedure. It mirrors the base PersonalEvo router round for round; the only changes are the generalized ridge geometry $\lambda\widehat{\Omega}_t^{-1}$ inside $V_{a,t}^{\text{hyper}}$ and the periodic re-estimation of the hyper-covariance from the archive's own per-candidate estimates.

Three clarifications about the data the algorithm accumulates. First, $y_{a,t}$ stacks centered observed rewards $r_s - b(u_s)$, not unknown mean rewards. Second, row s of $X_{a,t}$ is the feature vector for the user served at that round, so a shared router may pool observations from many users. Third, Algorithm 1 allows a stream of users, whereas the four-term decomposition follows one user and then averages over the population. Whenever we combine a Hyper-Ridge bound with that decomposition, we use the per-user specialization $u_t \equiv U$. The confidence and potential lemmas also hold for a predictable user stream, but the four-term conclusion does not automatically transfer to that setting.

The per-round cost is identical to base PersonalEvo up to the re-estimation step, which touches only the $d \times d$ matrix $\widehat{\Omega}_t$ and amortizes over the period τ. This periodic update is an experimental method. Proposition 8 analyzes the oracle variant that fixes $\widehat{\Omega}_t = \Omega$ and uses the stated confidence radius. Theorem 57 analyzes a second variant that estimates Ω on independent calibration candidates and freezes the estimate before routing. Neither result applies to the periodic update above.

The remainder of the paper turns to empirical evidence. Section 6 first reports the nine-benchmark results for the base PersonalEvo router and then separates two Hyper-Ridge questions: whether a known archive geometry can reduce exploration, and whether the current periodic estimator can recover that geometry from online data.

Algorithm 2 describes PersonalEvo's per-user routing loop.

At each round, a disjoint LinUCB bandit selects the best candidate for the current user based on UCB scores.

In the simulator, an oracle-assisted gate compares the unknown current-archive optimum with the observed reward and may propose a new candidate.

The feature map used in our experiments is

$$\psi(u,a) = \left[\,\text{onehot}(\text{cat}(a))\,;\, w_u\,\right] \in \mathbb{R}^{2K_{\text{cat}}}, \tag{14}$$

where $\mathtt{cat}(a)$ is the candidate's category and $w_u \in \mathbb{R}^{K_{\text{cat}}}$ is the user preference vector exposed by the simulator.

As a concrete instance (used in our experiments), the $K_{\text{cat}} = 5$ categories are $\{\text{sorting}, \text{searching}, \text{formatting}, \text{general}, \text{calc}\}$, so for a candidate a in the `searching` category,

Algorithm 1 Hyper-Ridge PersonalEvo routing

Require: Initial archive A_1, ridge weight $\lambda > 0$, exploration weight $\alpha > 0$, re-estimation period $\tau \in \mathbb{N}$, regularizer $\epsilon > 0$

Initialize $\widehat{\Omega}_1 \leftarrow I_d$; for each candidate a, initialize $X_{a,1}$ and $y_{a,1}$ as empty
for $t = 1, 2, \ldots, T$ **do**
  Observe user u_t and eligible candidate set $A_t(u_t)$
  for each candidate $a \in A_t(u_t)$ **do**
    Compute $V_{a,t}^{\text{hyper}} \leftarrow X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}$
    Compute $\widehat{\theta}_{a,t} \leftarrow \left(V_{a,t}^{\text{hyper}}\right)^{-1} X_{a,t}^\top y_{a,t}$
    Compute $\text{UCB}_t^{\text{hyper}}(u_t, a) \leftarrow x(u_t, a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u_t, a)^\top \left(V_{a,t}^{\text{hyper}}\right)^{-1} x(u_t, a)}$
  end for
  Select $a_t \in \arg\max_{a \in A_t(u_t)} \text{UCB}_t^{\text{hyper}}(u_t, a)$ and observe reward r_t; set $y_t \leftarrow r_t - b(u_t)$
  Update per-candidate data: append $x(u_t, a_t)^\top$ to $X_{a_t,t+1}$ and y_t to $y_{a_t,t+1}$
  if $t \equiv 0 \pmod{\tau}$ **then**
    Re-estimate the hyper-covariance from the archive's candidate estimates:

$$\widehat{\Omega}_{t+1} \leftarrow \frac{1}{|A_t|} \sum_{a \in A_t} \widehat{\theta}_{a,t} \widehat{\theta}_{a,t}^\top + \epsilon I_d \tag{13}$$

  else
    $\widehat{\Omega}_{t+1} \leftarrow \widehat{\Omega}_t$
  end if
end for

Algorithm 2 Per-user personalized routing over a growing archive.

Require: user u, common initial archive A_1, horizon T, exploration parameter α, gap threshold τ, cooldown κ, archive cap $K_{\max}$

$A_1(u) \leftarrow A_1$; for each $a \in A_1(u)$ initialize $A_a \leftarrow I$, $b_a \leftarrow 0$
$t_{\text{last}} \leftarrow -\infty$
for $t = 1, \ldots, T$ **do**
  for each $a \in A_t(u)$ **do**
    $x_t(a) \leftarrow \psi(u, a)$ ▷ feature map, equation 14
    $\hat{\theta}_a \leftarrow A_a^{-1} b_a$
    $\text{UCB}_t(a) \leftarrow \hat{\theta}_a^\top x_t(a) + \alpha \sqrt{x_t(a)^\top A_a^{-1} x_t(a)}$
  end for
  $a_t(u) \leftarrow \arg\max_{a \in A_t(u)} \text{UCB}_t(a)$
  observe $r_t = \mu(u, a_t(u)) + \xi_t$
  $A_{a_t(u)} \leftarrow A_{a_t(u)} + x_t(a_t(u)) x_t(a_t(u))^\top$; $b_{a_t(u)} \leftarrow b_{a_t(u)} + r_t\, x_t(a_t(u))$
  if $m_t(u) - r_t > \tau$ **and** $t - t_{\text{last}} > \kappa$ **and** $|A_t(u)| < K_{\max}$ **then** ▷ oracle-assisted gate
    $\tilde{a}_t(u) \leftarrow \mathcal{F}(u, A_t(u), \text{trace}_t)$; $A_{t+1}(u) \leftarrow A_t(u) \cup \{\tilde{a}_t(u)\}$; initialize $A_{\tilde{a}}, b_{\tilde{a}}$; $t_{\text{last}} \leftarrow t$
  else
    $A_{t+1}(u) \leftarrow A_t(u)$
  end if
end for

$\text{onehot}(\text{cat}(a)) = (0, 1, 0, 0, 0)$ and the user block w_u records that user's affinity for each of the five categories.

We refer the reader to Appendix H for the full construction (categories, quality components, and feature concatenation), and to Team (2025c) for a richer factorized feature map over task type, memory mode, tool mode, and answer style.

The reflection trigger is the high-gap-failure rule

$$m_t(u) - r_t > \tau, \qquad t - t_{\text{last}} > \kappa, \tag{15}$$

This gate is not deployable as written because $m_t(u)$ uses unknown mean rewards. The saved simulator also uses an oracle-assisted proposer: it reads the user's peak category and returns a child candidate biased toward that category.

The measured evolution gain therefore describes this idealized simulator. It is not a formal upper bound on a different proposer, and it does not show what an LLM-driven proposer would achieve.

Reward decomposition. Using Algorithm 2, we show that the average reward the router achieves over horizon T satisfies (the subscript 1 in A_1 refers to the round $t = 1$ initial archive shared across all users, not to a particular user):

$$\text{routed value} \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{16}$$

This equation separates the router's performance into the four terms defined above. Computing all four terms requires the current-archive optimum and the selected candidate's mean reward; observed rewards alone are not enough.

Example. Consider two candidates and two equally likely user types with means $\mu(u_1, a_1) = 0.9$, $\mu(u_2, a_1) = 0.4$, $\mu(u_1, a_2) = 0.4$, $\mu(u_2, a_2) = 0.9$.

Then $V_{\text{static}} = 0.65$ (the best single candidate averages 0.65), while $V_{\text{route}} = 0.9$ (matching each user to their best gives 0.9), so $\Delta_{\text{route}} = 0.25$.

With no archive growth ($\bar{G}_T = 0$), the decomposition gives: routed value $= 0.65 + 0.25 - \bar{L}_T$.

Routing is profitable as long as the exploration cost $\bar{L}_T < 0.25$.

When a held-out reward model supplies $\mu(u, a)$, V_{static} and Δ_{route} can be estimated from a held-out user sample: $V_{\text{static}}(A_1) = \max_a N^{-1} \sum_u \mu(u, a)$ and $\Delta_{\text{route}}(A_1) = N^{-1} \sum_u \max_a \mu(u, a) - V_{\text{static}}(A_1)$.

Computing $\bar{G}_T$ and $\bar{L}_T$ also requires the archive means $m_t(u)$ and the selected mean $\mu(u, a_t(u))$. The saved tables use $m_t(u) - r_t$ instead, which is a different observed-reward residual.

4 METHOD

The preceding sections establish the base PersonalEvo framework: a routing decomposition over a self-evolving archive, guarantees for the fixed-archive router, and a repair-time bound for archive growth. All of these results are inherited unchanged. In this section we introduce the revision's contribution: a replacement for the router's per-candidate estimation scheme, motivated by a meta-learning perspective on the archive. The new method leaves the decomposition and its first three terms intact and targets only the exploration-cost term, by importing hyper-covariance estimation ideas from generalized ridge regression. We first record the two source documents that anchor this revision.

4.1 SUMMARY OF WHAT CHANGES

Before turning to experiments, we summarize how this section modifies the base paper. First, the router changes: the generalized-ridge router defined by $V_{a,t}^{\text{hyper}}$ and $\text{UCB}_t^{\text{hyper}}$ replaces the base per-candidate LinUCB router, which regularizes each θ_a independently with an isotropic ridge penalty. Second, the exploration-cost theory changes: the known-Ω theorem replaces the isotropic potential by an effective-dimension potential while retaining the full confidence radius. Appendix L proves finite-sample control for a calibration-split estimator that freezes $\widehat{\Omega}$ before routing. The periodically updated estimator in Algorithm 1 has no guarantee in this paper. Geometry improves the certificate only when the full Hyper-Ridge budget is smaller than the full LinUCB budget; low effective dimension alone is not sufficient. Finally, nothing else changes: the four-term decomposition

$R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$ is inherited from the base framework exactly as proven there, and this revision only sharpens the bound on the $\bar{L}_T$ term.

5 THEORETICAL RESULTS FOR HYPER-RIDGE PERSONALEVO

5.1 THEORY: IMPROVED EXPLORATION COST

Throughout this section the base assumptions of Section 3 (Assumptions 1–4 below) remain in force, and we write $x(u, a)$ for the feature map denoted $\psi(u, a)$ there; the only new hypothesis is the shared archive geometry of Assumption 7. We first restate the inherited base results this revision builds on, then derive the sharpened exploration-cost bound.

The decomposition holds under minimal conditions (bounded rewards and measurability).

Below we state the assumptions needed for the finite-time guarantee.

Assumption 1 (Bounded mean reward)**.** *The mean reward is bounded:*

$$\mu(u, a) \in [0, 1] \quad \textit{for all } (u, a). \tag{17}$$

Assumption 2 (Conditionally sub-Gaussian noise)**.** *There exists $\sigma > 0$ with*

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \le \exp(\lambda^2 \sigma^2 / 2), \qquad \lambda \in \mathbb{R}. \tag{18}$$

Assumption 3 (Predictable archive)**.** *The user-specific archive is predictable:*

$$\sigma(A_t(U)) \subseteq \mathcal{O}_{t-1}. \tag{19}$$

Assumption 4 (Bounded features)**.** *The feature map satisfies*

$$\|\psi(u, a)\| \le L < \infty \quad \textit{for all } (u, a). \tag{20}$$

Assumptions 2–4 are not required for Theorem 2 (the main result); they are needed only for the conditional plug-in corollary that imports an exploration-cost bound from the linear bandit literature.

The first result establishes when routing gain is strictly positive:

Lemma 1 (Static-vs-routed value)**.** *For every finite archive A,*

$$\Delta_{\text{route}}(A) \ge 0, \tag{21}$$

with equality if and only if there exists $a^\star \in A$ such that

$$\mu(U, a^\star) = \max_{a \in A} \mu(U, a) \quad \textit{almost surely.} \tag{22}$$

Thus $\Delta_{\text{route}}(A) > 0$ exactly when no single candidate is optimal for almost every user.

The proof is in Appendix A.

Main theorem.

Theorem 2 (Exact decomposition on the common initial archive)**.** *Let A_1 be deterministic, finite, and shared by all users. For every round t and every horizon $T \ge 1$,*

$$\mathbb{E}[\mu(U, a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{23}$$

$$\frac{1}{T} \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{24}$$

Proof sketch (full proof in Appendix A). By definition,

$$\begin{aligned} \mu(U, a_t(U)) &= m_t(U) - L_t(U) \\ &= m_1(U) + G_t(U) - L_t(U). \end{aligned}$$

Taking expectations over all model and online randomness and using Lemma 1 gives equation 23. Averaging over t gives equation 24. □

Theorem 2 says that the reward achieved by the personalized routing policy decomposes into a structural piece on the initial archive ($V_{\text{static}} + \Delta_{\text{route}}$), an online discovery piece ($\bar{G}_T$), and a negative online learning piece ($-\bar{L}_T$).

In a simulator that logs mean rewards, $\bar{L}_T$ is computed from $m_t(u) - \mu(u, a_t(u))$. The saved headline artifacts instead contain $m_t(u) - r_t$, so they do not directly measure $\bar{L}_T$.

Dominance condition.

Corollary 3 (When personalized routing beats the best static initial candidate)**.** *For any horizon T,*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mu(U,a_t(U))] > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{25}$$

On a fixed archive (no growth, $\bar{G}_T = 0$), this collapses to

$$\Delta_{\text{route}}(A_1) > \bar{L}_T. \tag{26}$$

In other words, personalized routing is beneficial when the routing gain and evolution gain together exceed the exploration cost.

The condition can be estimated when the evaluation protocol records mean rewards for every current candidate. A held-out user sample estimates $\Delta_{\text{route}}(A_1)$; the archive and selected-candidate means estimate $\bar{G}_T$ and $\bar{L}_T$. The saved experiments report the observed-reward residual instead, so they do not verify this condition.

The static-vs-routed inequality behind Lemma 1 is the folklore $\mathbb{E}[\max_a X_a] \geq \max_a \mathbb{E}[X_a]$; our contribution is the equality condition and the *corrected* static comparator $V_{\text{static}}(A_1) = \max_a \mathbb{E}_U[\mu(U,a)]$, which (unlike the heuristic $\arg\max_a q(a)$ baseline in the original simulator) makes the dominance criterion equation 25 falsifiable rather than tautological.

What is specific to PersonalEvo is the four-term accounting for a user drawn from a population and then followed through one online trajectory.

Supporting lemmas. The two lemmas below support the main theorem: Lemma 4 ensures that the per-arm bandit statistics remain valid as the archive grows, and Lemma 5 bounds the number of mutations triggered by the failure rule.

Lemma 4 (Predictable birth-order archive)**.** *Assume at most $K_{\max}$ candidates appear by round T, and index them by predictable birth order. Candidate slot k is created at a stopping time τ_k. Its identity and feature rule are $\mathcal{O}_{\tau_k-1}$-measurable, while its latent parameter is fixed at birth and $\mathcal{F}_{\tau_k-1}$-measurable. Before τ_k, pad that slot's feature and selection indicator by zero. Each padded process is predictable in the analysis filtration, so self-normalized concentration applies conditionally from its birth time. A union bound over the deterministic slots $k \leq K_{\max}$ is valid. The router remains adapted to $\mathcal{O}_t$ and does not observe the latent parameter.*

Lemma 5 (Failure-prioritization)**.** *Under the trigger equation 15, the expected number of reflection mutations across T rounds is bounded by*

$$\mathbb{E}[\#\text{mutations}] \leq \sum_{t=1}^{T}\Pr\big(L_t(U) + |\xi_t| > \tau\big) \leq T\sup_{t\leq T}\Pr\big(L_t(U) + |\xi_t| > \tau\big), \tag{27}$$

which controls archive bloat without changing the regret analysis of the router itself.

The failure-prioritization gate is the algorithmic counterpart of the constraint-gated self-evolution line reviewed in Section 2, but here it is embedded inside a user-conditioned routing rule.

Positioning of the decomposition. The comparison to personalized bandits can now be stated in the paper's notation.

A fixed in-archive oracle comparison folds the structural routing gain into the oracle value, whereas equation 24 factors it out as $\Delta_{\text{route}}(A_1)$.

A fixed action set also removes the archive-improvement term, whereas PersonalEvo keeps $\bar{G}_T$ explicit and uses Lemma 4 to ensure that online archive growth does not depend on future randomness.

Offline prompt-evolution methods can improve the archive, but without user-conditioned routing they do not separate the routing gain $\Delta_{\text{route}}(A_1)$ from the evolution gain $\bar{G}_T$.

Finite-time bound on the exploration cost. Theorem 2 is an exact equality but does not by itself bound the size of the exploration cost $\bar{L}_T$.

To obtain a finite-time lower bound on the router's average reward, we need a controlled estimate of $\bar{L}_T$.

Rather than re-deriving such a bound from scratch for our particular router, we state it as an external assumption (Assumption 5 below) and substitute it into the decomposition.

Assumption 5 (External exploration-cost bound)**.** *There exists a deterministic sequence B_T such that*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T \quad \textit{for every } T \ge 1. \tag{28}$$

Corollary 6 (Plug-in bound)**.** *Under Assumptions 1–4 and 5,*

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))] \ge V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{29}$$

For the theoretical LinUCB instantiation in Proposition 16, set the UCB multiplier to $\alpha = \beta_T(\delta)$, where

$$\beta_T(\delta) = \sigma\sqrt{2\log(K_{\max}/\delta) + d\log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + \sqrt{\lambda}S. \tag{30}$$

Then one may take

$$B_T = 2\beta_T(\delta)\sqrt{2TK_{\max}d\log\left(1 + \frac{TL_x^2}{\lambda d}\right)} + T\delta. \tag{31}$$

For $T \ge 2$, with $\delta = 1/T$ and fixed problem parameters, including $K_{\max}$ and d, $B_T/T = O(\log T/\sqrt{T})$. The saved experiments use a fixed heuristic α and are not covered by this certificate.

Discovery-time bound. Discovery is the most fragile term in equation 16 because it depends on the proposer $\mathcal{F}$.

We replace the original geometric contraction theorem with a stopping-time statement under a per-attempt repair probability.

Assumption 6 (Repair probability)**.** *There exists $q_\epsilon \in (0, 1]$ such that for each cluster g with current gap $\Delta_g > \epsilon$, each reflection attempt produces an ϵ-improving child with probability at least q_ϵ, independent of past attempts conditional on the trigger firing.*

Theorem 7 (First-useful-specialist time)**.** *Under Assumption 6, after m reflection attempts on cluster g,*

$$\Pr\big(\textit{some attempt improves cluster } g \textit{ by } \epsilon\big) \ge 1 - (1 - q_\epsilon)^m. \tag{32}$$

The saved proposer uses privileged information, but the experiments do not estimate q_ϵ for a fixed ϵ. This bound is therefore not empirically tested.

The non-oracle case is left as future work.

Section 6 reports the first three terms in equation 24 and the separate observed-reward residual $\bar{L}_T^{\text{obs}}$.

We now turn to the new result. Hyper-Ridge leaves the four-term decomposition in equation 24 unchanged and changes only the certificate for $\bar{L}_T$. The valid comparison is between two complete finite-time budgets. Each budget contains a confidence radius and an elliptical-potential term. Hyper-Ridge can reduce the second term, but a small effective dimension does not by itself imply a smaller total budget. Proposition 8 states the known-geometry budget, and Corollary 9 compares it with the exact LinUCB certificate.

5.2 Why this matters

The original dominance condition is

Algorithm 3 Hyper-Ridge PersonalEvo routing

Require: initial archive A_1, ridge weight $\lambda > 0$, exploration weight $\alpha > 0$, re-estimation period τ, covariance floor $\epsilon > 0$
1: Initialize $\widehat{\Omega}_1 \leftarrow I_d$
2: Initialize $X_{a,1}$ and $y_{a,1}$ as empty for each $a \in A_1$
3: **for** $t = 1, \dots, T$ **do**
4: Observe user u_t and eligible archive $A_t(u_t)$
5: Compute $\mathrm{UCB}_t^{\mathrm{hyper}}(u_t, a)$ from equation 34–equation 36 for every $a \in A_t(u_t)$
6: Select $a_t(u_t) \in \arg\max_{a \in A_t(u_t)} \mathrm{UCB}_t^{\mathrm{hyper}}(u_t, a)$
7: Observe reward r_t, set $y_t = r_t - b(u_t)$, append $x(u_t, a_t(u_t))^\top$ to $X_{a_t,t+1}$, and append y_t to $y_{a_t,t+1}$
8: **if** $t \equiv 0 \pmod{\tau}$ **then**
9: Set

$$\widehat{\Omega}_{t+1} \leftarrow \frac{1}{|A_t|} \sum_{a \in A_t} \widehat{\theta}_{a,t} \widehat{\theta}_{a,t}^\top + \epsilon I_d \tag{37}$$

10: **else**
11: Set $\widehat{\Omega}_{t+1} \leftarrow \widehat{\Omega}_t$
12: **end if**
13: **end for**

$$\Delta_{\mathrm{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{33}$$

Therefore, any certified reduction in $\bar{L}_T$ lowers the margin needed for personalization to pay. The qualification "certified" matters: the theorem does not claim that an arbitrary covariance estimate, or the periodic estimate in Algorithm 1, reduces exploration.

Write $R_T^{\mathrm{hyper}} = T^{-1} \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))]$. Equation equation 24 applies directly. This section replaces the ordinary dimension-dependent control of $\bar{L}_T = T^{-1} \sum_{t=1}^{T} \mathbb{E}[L_t(U)]$ with an archive-geometry-dependent control.

The routing rule analyzed below is the implementation target for the hyper-ridge revision. For candidate a, the generalized design matrix is

$$V_{a,t}^{\mathrm{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{34}$$

The generalized ridge estimate is

$$\widehat{\theta}_{a,t} = \left(V_{a,t}^{\mathrm{hyper}}\right)^{-1} X_{a,t}^\top y_{a,t}. \tag{35}$$

The corresponding upper-confidence score is

$$\mathrm{UCB}_t^{\mathrm{hyper}}(u, a) = x(u,a)^\top \widehat{\theta}_{a,t} + \alpha \sqrt{x(u,a)^\top \left(V_{a,t}^{\mathrm{hyper}}\right)^{-1} x(u,a)}. \tag{36}$$

Assumption 7 (Archive geometry and complexity cap)**.** *There is a known candidate-independent baseline $b(u)$ such that*

$$\mu(u,a) = b(u) + x(u,a)^\top \theta_a \in [0,1]. \tag{38}$$

The feature vector satisfies $x(u,a) \in \mathbb{R}^d$. There is a positive definite matrix Ω and finite constants S_Ω, L_Ω such that

$$\Omega \succ 0, \qquad \|\theta_a\|_{\Omega^{-1}} \le S_\Omega, \qquad x(u,a)^\top \Omega x(u,a) \le L_\Omega^2 \tag{39}$$

for every candidate that can appear through round T. Candidate births, identities, parameters, and features satisfy the predictable birth-order conditions of Lemma 4. For the fixed-user trajectory, let

$$\mathcal{A}_T(U) = \bigcup_{t=1}^{T} A_t(U) \tag{40}$$

be the set of candidates that appear through round T. For a candidate design matrix X, define

$$d_{\mathrm{eff}}(\Omega; X) = \mathrm{tr}\Big[X^\top X\left(X^\top X + \lambda\Omega^{-1}\right)^{-1}\Big]. \tag{41}$$

There is a deterministic complexity cap $\bar{d}_{\mathrm{eff},T}(\Omega)$ such that, almost surely,

$$\max_{1\le s\le T+1}\max_{a\in\mathcal{A}_T(U)} d_{\mathrm{eff}}(\Omega; X_{a,s}) \le \bar{d}_{\mathrm{eff},T}(\Omega). \tag{42}$$

The known-geometry theorem needs only the ellipsoid, feature-radius, and complexity-cap conditions. Interpreting Ω as a covariance requires the additional random-effects assumptions used by the calibration-split theorem: fresh candidate effects are independent, have mean zero and covariance Ω, and the regression uses the centered response $r - b(u)$.

Proposition 8 (Known-geometry exploration-cost bound)**.** *Assume Assumptions 1–3 and Assumption 7. Suppose $|\mathcal{A}_T(U)| \le K_{\max}$. For the decomposition-compatible result, draw one user U before routing and set $u_t = U$ for all t. Define*

$$c_T^\Omega = 1 + \log\left(1 + \frac{TL_\Omega^2}{\lambda}\right) \tag{43}$$

and, for $\delta \in (0,1)$,

$$\alpha_{\Omega,T}(\delta) = \max\left\{1,\ \sigma\sqrt{c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega) + 2\log(K_{\max}/\delta)} + \sqrt{\lambda}S_\Omega\right\}. \tag{44}$$

Consider the oracle router that sets $\widehat{\Omega}_t = \Omega$ for every round and uses the UCB multiplier $\alpha_{\Omega,T}(\delta)$. Then, with probability at least $1-\delta$,

$$\sum_{t=1}^T L_t(U) \le B_T^\Omega(\delta) := 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega)}. \tag{45}$$

Consequently,

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \le B_T^\Omega(\delta) + \delta T. \tag{46}$$

Proof sketch. Lemma 45 gives simultaneous confidence intervals after a union bound over the predictable birth-order slots. Lemma 46 bounds each slot's log-determinant and width sum by $c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega)$. Optimism gives $L_t(U) \le 2\alpha_{\Omega,T}(\delta)\sqrt{x_t(a_t)^\top (V_{a_t,t}^\Omega)^{-1} x_t(a_t)}$ on the confidence event. Summing the clipped widths and applying Cauchy–Schwarz yields equation 45. Since $0 \le L_t(U) \le 1$, the complement of the confidence event contributes at most δT in expectation. Appendix K gives the full derivation. □

Remark 1 (Calibration-split guarantee)**.** *For the calibration-split router, Theorem 57 proves that the relative covariance error satisfies $\rho_{\Omega,T} = O(T^{-1/10}\sqrt{\log T})$ with high probability. It uses an explicit frequentist confidence radius. Under the theorem's schedule and fixed problem constants,*

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \le B_{T,0}^{\mathrm{cal}} + \mathrm{Rem}_T^{\mathrm{cal}}. \tag{47}$$

Moreover,

$$\begin{aligned} B_{T,0}^{\mathrm{cal}} &= O\left(\sqrt{dK_{\max}\bar{d}_{\mathrm{eff},T}(\Omega)T}\ \log T\right), \\ \mathrm{Rem}_T^{\mathrm{cal}} &= O\left(T^{2/5}(\log T)^{3/2}\right) = o(\sqrt{T}). \end{aligned} \tag{48}$$

This is a theorem for a fixed post-calibration archive. It is not a theorem for the periodic estimator. The leading term in equation 47 is not the oracle budget: its confidence radius already depends on the covariance-error bound. The displayed remainder isolates only the additional width inflation, failure probability, and, when charged, calibration pulls.

As a concrete scale calculation, take $K_{\max} = 16$, $d = 100$, and $\bar{d}_{\mathrm{eff},T}(\Omega) = 9$. If the two confidence radii and logarithmic potential factors are equal, the width-sum part changes by the ratio

$$\frac{\sqrt{K_{\max}\bar{d}_{\mathrm{eff},T}(\Omega)T}}{\sqrt{K_{\max}dT}} = \sqrt{\frac{9}{100}} = 0.3. \tag{49}$$

This is a geometric illustration, not the final budget ratio. The final comparison must also include the two confidence radii.

Corollary 9 (Dominance in more regimes)**.** *Define the structural-and-evolution margin by*

$$M_T = \Delta_{\mathrm{route}}(A_1) + \bar{G}_T. \tag{50}$$

Define the hyper-ridge exploration budget by

$$B_T^{\mathrm{hyper}} = B_T^{\Omega}(\delta) + \delta T. \tag{51}$$

If

$$M_T > \frac{B_T^{\mathrm{hyper}}}{T}, \tag{52}$$

then Hyper-Ridge PersonalEvo satisfies the dominance condition

$$R_T^{\mathrm{hyper}} > V_{\mathrm{static}}(A_1). \tag{53}$$

Moreover, compared with an ordinary LinUCB budget

$$B_T^{\mathrm{LinUCB}} = B_T^{\mathrm{lin}}(\delta) + \delta T, \tag{54}$$

Hyper-Ridge certifies all margins in the additional interval

$$\frac{B_T^{\mathrm{hyper}}}{T} < M_T \leq \frac{B_T^{\mathrm{LinUCB}}}{T}, \tag{55}$$

whenever

$$B_T^{\mathrm{hyper}} < B_T^{\mathrm{LinUCB}}. \tag{56}$$

Proof. Substituting Proposition 8 into the inherited decomposition gives

$$\begin{aligned} R_T^{\mathrm{hyper}} &= V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \bar{L}_T \\ &\geq V_{\mathrm{static}}(A_1) + M_T - \frac{B_T^{\mathrm{hyper}}}{T}. \end{aligned} \tag{57}$$

Condition equation 52 implies equation 53. The interval claim follows by comparing equation 51 and equation 54. ☐

This corollary states the comparison without assuming its outcome. Known geometry certifies more margins only when its complete finite-time budget is smaller. The calibration-split budget may be substituted separately from Theorem 57; no claim is made for the periodic estimator.

6 Experiments

6.1 Environments and data

We evaluate on three synthetic environments that share the same reward structure but differ in user heterogeneity.

The mean reward follows the PABF formulation (Team, 2025c):

$$\mu(u, a) = f\big(q(a),\ w_u(\mathrm{cat}(a))\big), \tag{58}$$

where $q(a) \in [0, 1]$ is the intrinsic candidate quality, $\mathrm{cat}(a) \in \mathcal{C}$ is the candidate's category, $w_u \in \Delta^{|\mathcal{C}|-1}$ is the user's category preference, and f is increasing in both arguments (functional form in Appendix H.2).

Benchmark	$\|\mathcal{U}\|$	$\|A_1\|$	Preference structure
General	8	12	5 disjoint category peaks
Workspace	8	12	2 broad paired peaks
Support	8	12	3 overlapping clusters
GEPA archive	12	15	heterogeneous task prefs
Hermes agent	15	20	(skill,mem,tool) clusters
BESPOKE	30	40	4-dim rubric prefs
PAHF shop.	20	40	product-attribute prefs
PAHF embod.	40	40	embodied object choice
PrefEval	20	10	topic-strategy prefs

Table 1: Benchmarks. $|\mathcal{U}|$: users; $|A_1|$: candidates. Synthetic: T=300; others: T=250.

Benchmark	V_{st}	Δ_{rt}	$\bar{G}_T$	$\bar{L}_T^{\text{obs}}$	Obs. margin
General	.570	.280	.051	.036	**+.295**
Workspace	.577	.124	.041	.043	+.122
Support	.579	.215	.061	.039	+.237
GEPA archive	.654	.048	.023	.025	+.046
Hermes agent	.703	.092	.036	.028	+.100
BESPOKE	.653	.036	.001	.010	+.027
PAHF shop.	.577	.209	.002	.054	+.157
PAHF embod.	.290	.238	.000	.071	+.167
PrefEval	.531	.251	.000	.028	+.223

Table 2: Logged accounting summary. Here $\bar{L}_T^{\text{obs}} = T^{-1}\sum_t(m_t - r_t)$, so the last column is the observed-reward margin $\Delta_{\text{rt}} + \bar{G}_T - \bar{L}_T^{\text{obs}}$. It is not the theoretical dominance margin unless $r_t - \mu(U, a_t)$ is conditionally mean-zero.

The theory uses $r_t = \mu(U, a_t(U)) + \xi_t$ with conditionally mean-zero noise. The saved simulator instead records

$$r_t = \text{clip}(\mu(U, a_t(U)) + \eta_t, 0, 1), \qquad \eta_t \sim \mathcal{N}(0, 0.05^2). \tag{59}$$

The simulator knows every pre-clipping mean, so the theoretical exploration cost could be computed from the selected candidate's mean. The saved headline tables instead use the observed residual $m_t - r_t$. We keep that quantity separate below; in a real system $m_t(u)$ would also require held-out estimation.

We compare five variants of Algorithm 2: **Full** (all components enabled), **No-routing** (uniform-random candidate selection), **No-reflection** (frozen archive), **No-personalization** (onehot-only features), and **Collapse-1** (single candidate).

The main figures show Full, No-routing, and No-reflection against the Static and Oracle references. Table 16 reports all five ablations for the three synthetic environments.

6.2 Results

We estimate the first three terms of equation 16 and the observed-reward residual $\bar{L}_T^{\text{obs}}$ on each benchmark (definitions in Appendix H.5).

Table 2 (right panel above) reports the decomposition across all nine benchmarks.

Three findings hold across all nine benchmarks:

1. **Routing gain is the largest positive component.** In every row, Δ_{route} exceeds $\bar{G}_T$: routing gains range from 0.036 to 0.280, while evolution gains range from 0 to 0.061. The ordering broadly tracks user heterogeneity, consistent with Lemma 1.
2. **The observed-reward margin is positive.** On all nine benchmarks, including four with real user preference data, $\Delta_{\text{route}} + \bar{G}_T - \bar{L}_T^{\text{obs}}$ ranges from $+0.027$ (BESPOKE) to $+0.295$ (General). Because rewards are clipped, this arithmetic does not by itself verify the mean-reward condition in Corollary 3.
3. **Real-data heterogeneity is substantial.** PAHF shopping (Δ_{route}=0.209) and PrefEval (Δ_{route}=0.251) are comparable to the synthetic environments, confirming that the decomposition's structural term is practically significant on real preferences.

Ablation. Figure 1 compares Full, No-routing, and No-reflection with the Static and Oracle references on all nine benchmarks.

Among the two plotted component ablations, removing routing causes the larger reward drop; removing reflection causes a smaller drop.

The plotted means satisfy No-routing < Static < No-reflect < Full < Oracle on every benchmark.

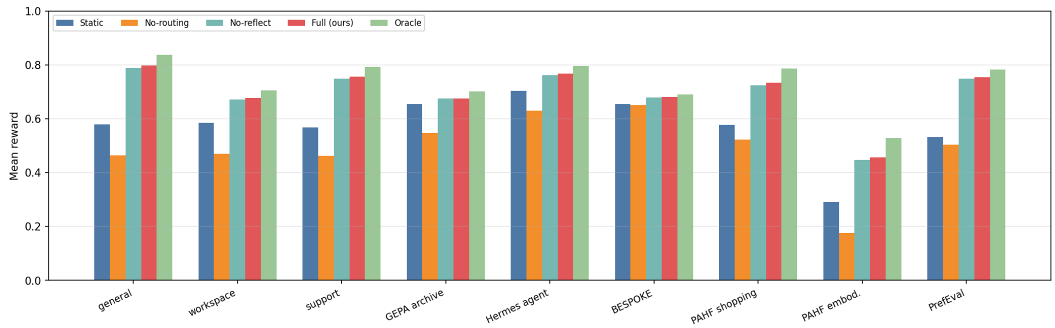

Figure 1: Mean reward for three router variants and two references on nine benchmarks. Within each benchmark, the bars are ordered Static, No-routing, No-reflect, Full, and Oracle from left to right. The bars show means without uncertainty intervals.

Per-round convergence. Figure 2 shows per-round reward traces for the same five series.

In all cases, PersonalEvo converges toward the oracle within $\sim$50 rounds, while the static baseline remains flat.

The figure plots reward rather than the observed-reward residual; per-seed residuals appear in Tables 19 and 20. These finite traces do not identify an asymptotic rate.

6.3 Comparison against existing routing methods

To position PersonalEvo against prior routing strategies, we compare five methods on all nine benchmarks (Table 3): (i) **Static**: deploy the single best-on-average candidate; (ii) **GEPA-Pareto**: maintain the Pareto-optimal subset of the archive and deploy uniformly from it (Agrawal et al., 2025); (iii) **Thompson**: Bayesian Thompson sampling over candidates (Labs, 2025); (iv) **RouteLLM-thresh**: route users to specialist or generalist candidates based on a variance threshold; (v) **PersonalEvo**: our full method.

Benchmark	Static	GEPA-Pareto	Thompson	RouteLLM	PersonalEvo	Oracle
General	0.579	0.533	0.510	0.732	**0.797**	0.836
Workspace	0.585	0.539	0.515	0.650	**0.678**	0.705
Support	0.568	0.525	0.504	0.700	**0.757**	0.791
GEPA archive	0.671	0.616	0.569	0.675	**0.681**	0.701
Hermes agent	0.729	0.644	0.659	0.750	**0.771**	0.796
BESPOKE	0.681	0.660	0.654	0.678	**0.682**	0.690
PAHF shopping	0.625	0.532	0.560	0.691	**0.754**	0.812
PAHF embodied	0.290	0.244	0.222	0.407	**0.457**	0.528
PrefEval	0.531	0.520	0.514	0.687	**0.754**	0.782

Table 3: Comparison against existing routing methods on all nine benchmarks. Mean reward over 5 seeds. PersonalEvo (bold) achieves the highest reward among non-oracle methods in every row. GEPA-Pareto underperforms static because archive diversity without per-user routing is counterproductive. Thompson sampling explores too broadly. RouteLLM-threshold is the closest competitor but uses a fixed rule rather than learning per-user preferences.

The results reveal why the decomposition matters for method design: GEPA-Pareto maintains a diverse archive ($\bar{G}_T$ is high) but has no routing mechanism to exploit Δ_{route}, so diversity is wasted.

Thompson sampling routes but explores every arm equally regardless of user context, paying excessive $\bar{L}_T$.

RouteLLM routes by context but uses a fixed threshold rather than learning individual preferences.

PersonalEvo combines per-user routing with gated evolution and a small logged observed-reward residual. The experiment does not identify that residual with the theoretical $\bar{L}_T$.

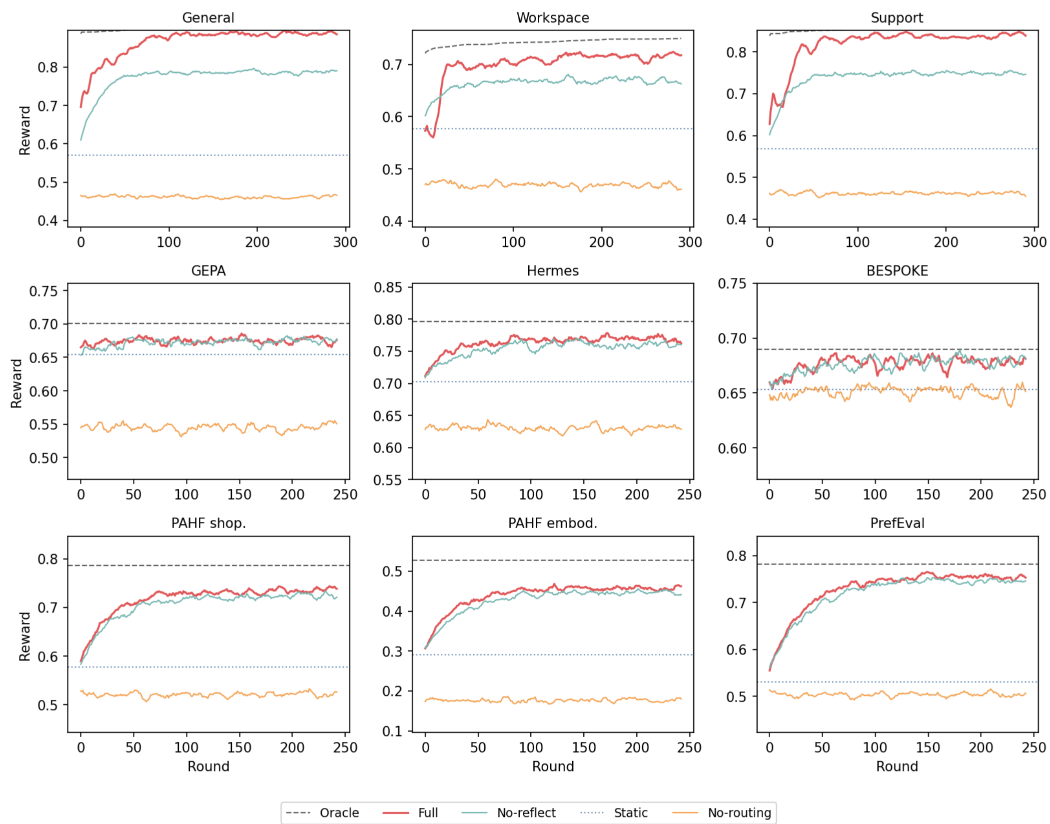

Figure 2: Per-round mean reward on nine benchmarks. The five series are **Full** (red; concat features, gated reflection, LinUCB), **No-reflect** (teal; frozen archive, $\bar{G}_T$=0), **No-routing** (orange; uniform-random selection, Δ_{route}→0), **Static** (blue dotted; best single candidate for all users), and **Oracle** (gray dashed; per-user optimum). The curves show means without uncertainty bands, and the vertical range differs by panel. The top row is synthetic; the remaining rows use framework and real-data benchmarks.

6.4 Hyper-Ridge diagnostics and the remaining evaluation gap

The estimates in Table 2, ablations in Figure 1, per-round traces in Figure 2, and baseline comparison in Table 3 all evaluate the *base* PersonalEvo router. We additionally ran a five-seed diagnostic suite for Hyper-Ridge PersonalEvo. We keep it separate from the headline comparison because it covers seven rather than nine adapters and does not log the mean-reward exploration cost as its primary endpoint.

Known-geometry mechanism. We construct a diagnostic linear bandit in which $K = 30$ candidate parameters are drawn from a planted rank-three covariance, contexts are dense random unit vectors, and the horizon is $T = 400$. Table 4 compares isotropic ridge, Hyper-Ridge given the true covariance, and Hyper-Ridge with the covariance periodically re-estimated from the candidate models, all with the same ridge weight $\lambda = 0.5$. The true geometry cuts cumulative exploration cost by 46–59%, while the online estimate raises it by 7–19%. This diagnostic uses Gaussian parameters, unbounded linear rewards, and a fixed exploration multiplier, so it does not test the bounded-reward theorem or its rate. It only isolates the effect of supplying the correct geometry.

d	Isotropic	Known Ω	Estimated Ω	Known-Ω reduction
20	314.1	170.2	375.0	45.8%
50	447.6	213.1	504.9	52.4%
100	455.4	188.1	487.5	58.7%

Table 4: Mean cumulative exploration cost over five seeds on planted rank-three problems ($K = 30$, $T = 400$, $\lambda = 0.5$ for all three routers). Lower is better. This mechanism diagnostic does not satisfy the assumptions of Proposition 8.

Adapter diagnostic. With tuning seeds disjoint from the five reporting seeds, the periodic estimator improves mean reward over the isotropic router on six of seven adapters, with changes ranging from $+0.002$ on BESPOKE to $+0.048$ on Hermes, and loses 0.010 on GEPA. These numbers are not evidence of cross-candidate geometry transfer. The adapter harness represents candidate a by the one-hot vector e_a. Starting from $\widehat{\Omega}_1 = I$, each candidate estimate then has support only on its own coordinate, so the covariance update remains diagonal. Accordingly, the full-covariance and diagonal-ridge variants match exactly on every adapter and seed. This ablation exposes a representation mismatch that a larger seed count cannot repair.

Remaining protocol. The complete comparison must run base, known-covariance, calibration-split estimated-covariance, periodically estimated-covariance, and diagonal-ridge routers on the same nine benchmarks, using a feature map with shared behavioral directions. It must report all four mean-reward terms of equation 16, with $\widehat{L}_T^{\mu}$ as the primary endpoint and $\widehat{L}_T^{\mathrm{obs}}$ reported separately, as well as the fitted regret exponent, covariance error, and $d_{\mathrm{eff}}(\widehat{\Omega}; X)$. Until that comparison is complete, we do not claim an empirical advantage for the fully adaptive Hyper-Ridge router.

Implementation status. The repository implements both the base router and the periodic Hyper-Ridge estimator, together with an identity-reduction test showing that frozen $\widehat{\Omega} = I$ reproduces the base router's selections. The diagnostic results are stored in `results/hyper_ridge/suite.json`; the missing component is the feature-aware, nine-benchmark evaluation above, not the router implementation itself.

7 Conclusion

This paper revised PersonalEvo around one idea: a self-evolving agent should learn the geometry of its own archive. Theorem 2 and Corollary 3 determine when personalized routing pays off. Hyper-Ridge leaves the structural and discovery terms unchanged and targets the exploration cost. It models candidate parameters with a shared geometry Ω and uses the effective dimension in equation 41. For known Ω, Proposition 8 gives the complete bound in equation 46. For $T \geq 2$, fixed problem parameters, and $\delta = 1/T$, both the LinUCB and known-geometry bounds are $O(\sqrt{T}\log T)$. Geometry helps only when the complete Hyper-Ridge budget is smaller. The calibration-split theorem gives a separate frozen-estimate bound whose explicit width-inflation and calibration remainder is $o(\sqrt{T})$; it does not cover periodic re-estimation.

Two provenance facts bound what this paper claims. First, the nine-benchmark estimates, ablation hierarchy, per-round traces, and baseline comparison in Section 6 belong to the *base* LinUCB router. Their logged margins are

$$\Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\mathrm{obs}} \;\in\; [+0.027,\, +0.295] \tag{60}$$

across all nine benchmarks, where $\bar{L}_T^{\mathrm{obs}} = T^{-1}\sum_t (m_t - r_t)$. This equality is logged accounting, not validation of equation 24: clipping means that $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]$ need not equal the pre-clipping $\mu(U, a_t)$ used by the structural terms. The selected-arm mean must be logged to evaluate $\bar{L}_T$ itself. The separate Hyper-Ridge suite is diagnostic: known geometry helps on planted low-rank problems, but periodic estimation does not recover that benefit, and one-hot adapters collapse the full covariance to a diagonal one. Second, the effective-dimension guarantee equation 46 is a theorem for the oracle-covariance router. Appendix L controls an estimated covariance for a calibration-split procedure under explicit diversity conditions, but this result does not cover the periodically updated estimator in Algorithm 1. Controlling that fully adaptive estimator remains the primary theoretical open problem.

The corresponding empirical open problem is the complete comparison specified in Section 6.4: base LinUCB versus known-covariance, calibration-split, periodically estimated, and diagonal-ridge routers on the same nine benchmarks, with the mean-reward exploration cost $\widehat{L}_T^{\mu}$ as the primary endpoint and $\widehat{L}_T^{\mathrm{obs}}$ reported separately. The feature map must admit shared directions; otherwise the full model is diagonal by construction. The comparison should also report covariance error, the fitted regret exponent, and a per-benchmark estimate of $d_{\mathrm{eff}}(\widehat{\Omega}; X)$ to locate both sides of the boundary: regimes where $d_{\mathrm{eff}}(\Omega; X) \ll d$ and the geometric potential can be smaller, and near-isotropic regimes where $d_{\mathrm{eff}}(\Omega; X) \approx d$. In both cases, the full confidence budget determines the comparison. The current implementation supplies the periodic estimator and common harness; a calibration-split implementation and a shared feature representation remain to be added.

Beyond the two named gaps, the limitations of the base framework carry over. The reflective proposer in the simulated environments is oracle-assisted, so the measured evolution gain $\bar{G}_T$ does not estimate what a realistic LLM-driven proposer would deliver, and each user maintains a private archive, so discoveries are not shared across users. A shared archive could provide more data for estimating Ω, but it would change the predictability argument in Lemma 4; we leave that setting open. The decomposition equation 24 also applies to model routing, prompt selection, and tool-use policies whenever a system must choose from a candidate library.

A Appendix Proofs: Preliminaries and the Exact Decomposition

This appendix block fixes the probability model used by all proofs and proves the first main theorem: the exact four-term decomposition.

The proof is an accounting identity; it does not use a contextual-bandit regret theorem and does not assume that reflection is realistic.

A.1 Probability space and online objects

All random variables are defined on a common probability space $(\Xi, \mathcal{F}, \mathbb{P})$, and expectations average over both the population draw and the online trajectory. The evaluation user is $U \sim \Pi$ with values in $\mathcal{U}$, and candidates belong to $\mathcal{A}$. The common initial archive is $A_1 \subseteq \mathcal{A}$; before round t, the current archive is $A_t(U) \subseteq \mathcal{A}$, and the router selects $a_t(U) \in A_t(U)$. Mean and observed rewards satisfy

$$\mu : \mathcal{U} \times \mathcal{A} \to [0, 1], \qquad r_t = \mu(U, a_t(U)) + \xi_t. \tag{61}$$

For the later plug-in bounds, let $\mathcal{F}_0 \subseteq \cdots \subseteq \mathcal{F}_T$ be the online filtration and $x_t(a) \in \mathbb{R}^d$ the feature vector for candidate a. The exact decomposition uses only bounded rewards and the archive conditions; the noise and feature assumptions are recorded for the later results.

Assumption 8 (A1: bounded mean reward)**.** *For every user u and candidate a, $0 \le \mu(u, a) \le 1$.*

Assumption 9 (A2: finite common initial archive)**.** *The initial archive is deterministic, nonempty, and finite: $1 \le |A_1| < \infty$.*

Assumption 10 (A3: finite append-only online archives)**.** *For every round, $1 \le |A_t(U)| < \infty$ almost surely, and the archive is append-only: $A_t(U) \subseteq A_{t+1}(U)$.*

Assumption 11 (A4: predictable archive)**.** *Before the router chooses at round t, the archive is measurable with respect to the past: $\sigma(A_t(U)) \subseteq \mathcal{F}_{t-1}$.*

Assumption 12 (A5: centered conditionally sub-Gaussian noise)**.** *For every $\lambda \in \mathbb{R}$,*

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}\big[e^{\lambda \xi_t} \mid \mathcal{F}_{t-1}\big] \le \exp\left(\frac{\lambda^2 \sigma^2}{2}\right). \tag{62}$$

Assumption 13 (A6: bounded features)**.** *Every available candidate satisfies $\|x_t(a)\|_2 \le L$.*

A.2 VALUES AND ACCOUNTING TERMS

For a deterministic finite archive A, define its static value, routed value, and heterogeneity gain by

$$\begin{aligned} V_{\text{static}}(A) &= \max_{a\in A} \mathbb{E}[\mu(U,a)], \qquad V_{\text{route}}(A) = \mathbb{E}\Big[\max_{a\in A} \mu(U,a)\Big], \\ \Delta_{\text{route}}(A) &= V_{\text{route}}(A) - V_{\text{static}}(A). \end{aligned} \tag{63}$$

For the online archive, define

$$\begin{aligned} m_t(U) &= \max_{a\in A_t(U)} \mu(U,a), \quad G_t(U) = m_t(U) - m_1(U), \quad L_t(U) = m_t(U) - \mu(U,a_t(U)), \\ \bar{G}_T &= \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)], \qquad \bar{L}_T = \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \end{aligned} \tag{64}$$

Proposition 10 (Integrability and signs of the accounting terms)**.** *Under Assumptions 8–10, the current-archive optimum satisfies*

$$m_t(U), G_t(U), L_t(U) \in [0,1], \qquad \mathbb{E}|m_t(U)| + \mathbb{E}|G_t(U)| + \mathbb{E}|L_t(U)| \le 3. \tag{65}$$

Proof. Bounded rewards give $0 \le m_t(U) \le 1$. Since $a_t(U) \in A_t(U)$, and since append-only growth gives $A_1 \subseteq A_t(U)$,

$$\begin{aligned} 0 &\le m_t(U) - \mu(U,a_t(U)) = L_t(U) \le 1, \\ 0 &\le m_t(U) - m_1(U) = G_t(U) \le 1. \end{aligned} \tag{66}$$

The integrability bound follows by summing these three pointwise bounds. □

A.3 SUPPORTING LEMMAS FOR THE FOUR-TERM DECOMPOSITION

Lemma 11 (Static-versus-routed value)**.** *For every deterministic, finite, nonempty archive A, $\Delta_{\text{route}}(A) \ge 0$. Equality holds if and only if some $a^\star \in A$ is pointwise optimal almost surely:*

$$\mu(U,a^\star) = \max_{a\in A} \mu(U,a) \quad \textit{almost surely}. \tag{67}$$

In particular, the inequality is strict if every candidate is suboptimal with positive probability:

$$\forall a \in A, \qquad \mathbb{P}\Big(\mu(U,a) < \max_{b\in A} \mu(U,b)\Big) > 0. \tag{68}$$

Proof. Because A is finite, choose $a^\star \in \arg\max_{a\in A} \mathbb{E}[\mu(U,a)]$. Pointwise maximality gives

$$\Delta_{\text{route}}(A) = \mathbb{E}\Big[\max_{a\in A} \mu(U,a) - \mu(U,a^\star)\Big] \ge 0. \tag{69}$$

The integrand is nonnegative, so its expectation is zero exactly when it is zero almost surely. This proves the equality condition. Conversely, an almost-surely pointwise-optimal candidate attains both $V_{\text{static}}(A)$ and $V_{\text{route}}(A)$. Under the stated strictness condition, the integrand is positive with positive probability for every possible static maximizer, so its expectation is positive. □

Lemma 12 (Pointwise routed-value accounting)**.** *For every round t, the preceding definitions imply*

$$\mu(U,a_t(U)) = m_1(U) + G_t(U) - L_t(U). \tag{70}$$

Proof. Rearrange $L_t(U) = m_t(U) - \mu(U,a_t(U))$ and substitute $m_t(U) = m_1(U) + G_t(U)$. □

A.4 First main theorem: four-term decomposition

Theorem 13 (Exact decomposition over the common initial archive)**.** *This restates Theorem 2. Let $U \sim \Pi$, let $1 \le |A_1| < \infty$, and suppose $a_t(U) \in A_t(U)$. Then, for every round t,*

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{71}$$

Define

$$R_T \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))], \quad \bar{G}_T \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[G_t(U)], \quad \bar{L}_T \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathbb{E}[L_t(U)]. \tag{72}$$

Then

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{73}$$

Proof. Proposition 10 ensures integrability. Taking expectations in Lemma 12 gives

$$\begin{aligned}
\mathbb{E}[\mu(U, a_t(U))] \;&=\; \mathbb{E}[m_1(U)] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)] \\
&=\; V_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)] \\
&=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)].
\end{aligned} \tag{74}$$

Summing over t and dividing by T proves equation 73. □

The theorem concerns mean rewards. Under Assumption 12, the same expectation governs unclipped observed rewards because

$$\begin{aligned}
\mathbb{E}[r_t] \;&=\; \mathbb{E}[\mu(U, a_t(U))] + \mathbb{E}[\xi_t] \\
&=\; \mathbb{E}[\mu(U, a_t(U))].
\end{aligned} \tag{75}$$

This link does not apply to the saved simulator's clipped observations in equation 59 when μ denotes the pre-clipping mean. The following `general` row instead checks the separate observed-reward accounting identity. Its measured terms are

$$V_{\text{static}}(A_1) = 0.570, \quad \Delta_{\text{route}}(A_1) = 0.280, \quad \bar{G}_T = 0.051, \quad \bar{L}_T^{\text{obs}} = 0.036. \tag{76}$$

Therefore,

$$\begin{aligned}
V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} \;&=\; 0.570 + 0.280 + 0.051 - 0.036 \\
&=\; 0.865,
\end{aligned} \tag{77}$$

which matches the reported observed routed value 0.864 up to table rounding. Because $\bar{L}_T^{\text{obs}}$ is defined from the same routed rewards, this is an algebraic consistency check rather than empirical validation of Theorem 13.

B Appendix Proofs: Second Main Theorem and Intermediate Lemmas

This section proves the second main theorem, the conditional plug-in statement for the exploration cost.

The proof does not assert a new regret rate; it shows how any valid bound on the cumulative exploration cost plugs into the four-term decomposition of Theorem 13.

Assumption 14 (External expected exploration-cost bound)**.** *For $T \in \mathbb{N}$, there is a deterministic $B_T \ge 0$ such that*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; B_T. \tag{78}$$

The next lemma records the conditioning fact that makes growing archives compatible with standard martingale arguments, provided children are appended only after the current reward is observed.

Lemma 14 (Predictable archive conditioning)**.** *Let the pre-action sigma-field be*

$$\mathcal{H}_{t-1} \;=\; \sigma(U,\, A_1,\, \{a_s(U), r_s, A_{s+1}(U) : 1 \,\le\, s < t\})\,. \tag{79}$$

Assume $A_t(U)$ is $\mathcal{H}_{t-1}$-measurable and at most $K_{\max}$ candidates appear by time T. Index them by deterministic birth-order slots $k \le K_{\max}$, with stopping-time birth τ_k and $\tau_k = \infty$ for an unfilled slot. On $\{\tau_k \le T\}$, the identity $a^{(k)}$ and its feature rule are $\mathcal{H}_{\tau_k - 1}$-measurable. Define the padded feature and selection indicator

$$\widetilde{x}_{t,k} \;=\; \mathbf{1}\{\tau_k \le t\}\psi(U, a^{(k)}), \quad I_{t,k} \;=\; \mathbf{1}\{\tau_k \le t,\; a_t(U) = a^{(k)}\}. \tag{80}$$

Assume both are $\mathcal{H}_{t-1}$-measurable: they are zero before birth, and the slot identity is in the pre-action history afterward. Also assume

$$\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] \;=\; 0, \quad \mathbb{E}[\exp(s\xi_t) \mid \mathcal{H}_{t-1}] \;\le\; \exp\!\left(\frac{s^2\sigma^2}{2}\right) \quad \textit{for all } s \in \mathbb{R}. \tag{81}$$

Then, for each deterministic slot k,

$$Z_{t,k} = I_{t,k}\widetilde{x}_{t,k}\xi_t, \qquad M_{n,k} = \sum_{t=1}^{n} Z_{t,k} \tag{82}$$

defines a vector-valued martingale with respect to the post-round filtration.

Proof. Fix k. Since $I_{t,k}\widetilde{x}_{t,k}$ is $\mathcal{H}_{t-1}$-measurable,

$$\mathbb{E}[Z_{t,k} \mid \mathcal{H}_{t-1}] \;=\; I_{t,k}\widetilde{x}_{t,k}\,\mathbb{E}[\xi_t \mid \mathcal{H}_{t-1}] \;=\; 0. \tag{83}$$

Moreover, for any direction v,

$$\mathbb{E}\big[\exp\big(s\, v^\top Z_{t,k}\big) \mid \mathcal{H}_{t-1}\big] \;\le\; \exp\!\left(\frac{s^2\sigma^2}{2} I_{t,k}(v^\top \widetilde{x}_{t,k})^2\right). \tag{84}$$

Thus the increments are conditionally mean-zero (and sub-Gaussian in every predictable direction), so their partial sums form the claimed martingale. □

The following elementary conversion is useful because many router analyses produce high-probability regret bounds, whereas Theorem 13 is stated in expectation.

Lemma 15 (From high-probability tax to expected tax)**.** *Suppose $0 \le \mu(U,a) \le 1$, let $S_T^L = \sum_{t=1}^T L_t(U)$, and let $\widetilde{B}_T \ge 0$ and $\delta \in (0,1)$ satisfy*

$$\mathbb{P}\Big(S_T^L \le \widetilde{B}_T\Big) \;\ge\; 1 - \delta. \tag{85}$$

Then

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; \widetilde{B}_T + T\delta. \tag{86}$$

Proof. Because $a_t(U) \in A_t(U)$ and rewards lie in $[0,1]$, $0 \le L_t(U) \le 1$, hence $0 \le S_T^L \le T$. For $\mathcal{E}_T^L = \{S_T^L \le \widetilde{B}_T\}$,

$$\begin{aligned} \mathbb{E}[S_T^L] \;&=\; \mathbb{E}[S_T^L \mathbf{1}\{\mathcal{E}_T^L\}] + \mathbb{E}[S_T^L \mathbf{1}\{(\mathcal{E}_T^L)^c\}] \\ &\le\; \widetilde{B}_T + T\,\mathbb{P}((\mathcal{E}_T^L)^c) \\ &\le\; \widetilde{B}_T + T\delta. \end{aligned} \tag{87}$$

Linearity of expectation completes the proof. □

We next give one sufficient route to Assumption 14.

This proposition is not a new contribution of the paper; it is a standard disjoint-linear UCB certificate written only to clarify what must be true for the plug-in theorem to inherit a familiar rate.

Proposition 16 (Disjoint-linear certificate for the external tax bound)**.** *Let $d \in \mathbb{N}$ and $\mathcal{A}_T(U) = \bigcup_{t=1}^T A_t(U)$, with $|\mathcal{A}_T(U)| \le K_{\max}$. Use the deterministic birth-order slots from Lemma 14. Let $(\mathcal{G}_t)$ be an analysis filtration with $\mathcal{H}_t \subseteq \mathcal{G}_t$. Each τ_k is an $\mathcal{H}_t$-stopping time; the candidate identity and feature rule are $\mathcal{H}_{\tau_k-1}$-measurable, while its latent parameter is fixed at birth and $\mathcal{G}_{\tau_k-1}$-measurable. The zero-padded process is $\mathcal{G}_t$-predictable. The router remains $\mathcal{H}_t$-adapted and does not observe the latent parameter.*

For every realized candidate a, suppose

$$x_t(a) = \psi(U,a) \in \mathbb{R}^d, \qquad \|x_t(a)\|_2 \le L_x, \qquad \|\theta_a^\star\|_2 \le S, \qquad \mu(U,a) = x_t(a)^\top \theta_a^\star. \tag{88}$$

Let $r_t = \mu(U, a_t(U)) + \xi_t$, where the centered sub-Gaussian conditions of Lemma 14 also hold conditionally on $\mathcal{G}_{t-1}$. For $\lambda \ge L_x^2 > 0$, define

$$\begin{aligned} V_{t,a} &= \lambda I_d + \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\}\, x_s(a) x_s(a)^\top, \\ \widehat{\theta}_{t,a} &= V_{t,a}^{-1} \sum_{s=1}^{t-1} \mathbf{1}\{a_s(U) = a\}\, x_s(a) r_s. \end{aligned} \tag{89}$$

For $\delta \in (0,1)$, let

$$\beta_T(\delta) \;=\; \sigma \sqrt{2 \log\left(\frac{K_{\max}}{\delta}\right) + d \log\left(1 + \frac{T L_x^2}{\lambda d}\right)} + \sqrt{\lambda}\, S. \tag{90}$$

If the router uses the disjoint UCB rule

$$a_t(U) \;\in\; \arg\max_{a \in A_t(U)} \left\{ x_t(a)^\top \widehat{\theta}_{t,a} + \beta_T(\delta) \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)} \right\} \tag{91}$$

then, with probability at least $1-\delta$,

$$\sum_{t=1}^T L_t(U) \;\le\; B_T^{\mathrm{lin}}(\delta), \tag{92}$$

where

$$B_T^{\mathrm{lin}}(\delta) \;=\; 2\beta_T(\delta) \sqrt{2 T K_{\max} d \log\left(1 + \frac{T L_x^2}{\lambda d}\right)}. \tag{93}$$

Consequently,

$$\sum_{t=1}^T \mathbb{E}[L_t(U)] \;\le\; B_T^{\mathrm{lin}}(\delta) + T\delta. \tag{94}$$

Proof. Write $a_t = a_t(U)$. For each deterministic slot k, let

$$M_{t,k} \;=\; \sum_{s=1}^{t-1} \mathbf{1}\{a_s = a^{(k)}\}\, x_s(a^{(k)}) \xi_s, \tag{95}$$

with zero summands before τ_k. Conditional on the slot's birth history, the noise and measurability assumptions make this a vector-valued martingale in the analysis filtration. The self-normalized martingale inequality gives

$$\mathbb{P}\left(\exists t \le T : \|M_{t,k}\|_{V_{t,a^{(k)}}^{-1}} > \sigma \sqrt{2 \log\left(\frac{K_{\max}}{\delta}\right) + \log\left(\frac{\det V_{t,a^{(k)}}}{\det(\lambda I_d)}\right)} \right) \;\le\; \frac{\delta}{K_{\max}}. \tag{96}$$

The trace bound gives, uniformly in $t \le T$,

$$\begin{aligned} \log\left(\frac{\det V_{t,a^{(k)}}}{\det(\lambda I_d)}\right) &\le d \log\left(1 + \frac{1}{\lambda d} \sum_{s=1}^{t-1} \mathbf{1}\{a_s = a^{(k)}\} \|x_s(a^{(k)})\|_2^2\right) \\ &\le d \log\left(1 + \frac{T L_x^2}{\lambda d}\right). \end{aligned} \tag{97}$$

Union-bounding over the deterministic slots yields, with probability at least $1-\delta$,

$$\forall t \le T,\ \forall a \in A_t(U) : \left| x_t(a)^\top (\widehat{\theta}_{t,a} - \theta_a^\star) \right| \le \beta_T(\delta) \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)}. \tag{98}$$

On this event, write $\widehat{\mu}_t(a) = x_t(a)^\top \widehat{\theta}_{t,a}$, $z_t(a) = \sqrt{x_t(a)^\top V_{t,a}^{-1} x_t(a)}$, and choose

$$a_t^\star \in \arg\max_{a \in A_t(U)} \mu(U,a). \tag{99}$$

Confidence, the UCB choice, and confidence again give

$$\begin{aligned} \mu(U, a_t^\star) &\le \widehat{\mu}_t(a_t^\star) + \beta_T(\delta) z_t(a_t^\star) \\ &\le \widehat{\mu}_t(a_t) + \beta_T(\delta) z_t(a_t) \\ &\le \mu(U, a_t) + 2\beta_T(\delta) z_t(a_t). \end{aligned} \tag{100}$$

Thus $L_t(U) \le 2\beta_T(\delta) z_t(a_t)$. Also $z_t(a_t)^2 \le \|x_t(a_t)\|_2^2/\lambda \le 1$. The determinant recursion for each arm and a sum over at most $K_{\max}$ arms give

$$\begin{aligned} \sum_{t=1}^{T} \mathbf{1}\{a_t = a\} z_t(a)^2 &\le 2 \log\left(\frac{\det V_{T+1,a}}{\det(\lambda I_d)} \right) \le 2d \log\left(1 + \frac{T L_x^2}{\lambda d}\right), \\ \sum_{t=1}^{T} z_t(a_t)^2 &\le 2K_{\max} d \log\left(1 + \frac{T L_x^2}{\lambda d}\right). \end{aligned} \tag{101}$$

Combining this with Cauchy–Schwarz yields

$$\begin{aligned} \sum_{t=1}^{T} L_t(U) &\le 2\beta_T(\delta) \sum_{t=1}^{T} z_t(a_t) \\ &\le 2\beta_T(\delta) \sqrt{2T K_{\max} d \log\left(1 + \frac{T L_x^2}{\lambda d}\right)} \\ &= B_T^{\mathrm{lin}}(\delta). \end{aligned} \tag{102}$$

Lemma 15 gives the expected bound. □

Theorem 17 (Conditional plug-in regret for PersonalEvo)**.** *Use R_T, $\bar{G}_T$, and $\bar{L}_T$ from Theorem 13. If*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T \tag{103}$$

then

$$R_T \ge V_{\mathrm{static}}(A_1) + \Delta_{\mathrm{route}}(A_1) + \bar{G}_T - \frac{B_T}{T}. \tag{104}$$

Proof. The assumption gives $\bar{L}_T \le B_T/T$. Substitute this into the exact decomposition in Theorem 13. □

Corollary 18 (Plug-in dominance condition)**.** *Under Theorem 17, personalized routing beats the best static initial candidate whenever*

$$\Delta_{\mathrm{route}}(A_1) + \bar{G}_T > \frac{B_T}{T}. \tag{105}$$

In that case, $R_T > V_{\mathrm{static}}(A_1)$. On a fixed archive, where $\bar{G}_T = 0$, the condition becomes

$$\Delta_{\mathrm{route}}(A_1) > \frac{B_T}{T}. \tag{106}$$

Proof. Subtract $V_{\mathrm{static}}(A_1)$ from the bound in Theorem 17. The stated condition makes the remaining right-hand side positive. □

Corollary 19 (Disjoint-LinUCB plug-in instance)**.** *If the conditions of Proposition 16 hold, then*

$$R_T \;\ge\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T^{\text{lin}}(\delta) + T\delta}{T}. \tag{107}$$

For $\delta = 1/T$, and with $K_{\max}$, d, L_x, λ, S, and σ fixed, the tax term satisfies

$$\frac{B_T^{\text{lin}}(1/T) + 1}{T} \;=\; O\!\left(\frac{\log T}{\sqrt{T}}\right) \tag{108}$$

Proof. Apply Theorem 17 with $B_T = B_T^{\text{lin}}(\delta) + T\delta$. For $T \ge 2$ and $\delta = 1/T$, the definition of B_T^{lin} gives

$$\begin{aligned} \frac{B_T^{\text{lin}}(1/T) + 1}{T} &= \frac{2\beta_T(1/T)\sqrt{2TK_{\max} d \log\!\left(1 + \frac{TL_x^2}{\lambda d}\right)} + 1}{T} \\ &= O\!\left(\frac{\log T}{\sqrt{T}}\right), \end{aligned} \tag{109}$$

under fixed problem parameters. □

As a scale check, suppose a valid certificate on the `workspace` environment gave $B_T/T = 0.050$. Table 2 reports $V_{\text{static}}(A_1) = 0.577$, $\Delta_{\text{route}}(A_1) = 0.124$, and $\bar{G}_T = 0.041$. The plug-in bound would then be

$$\begin{aligned} V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \frac{B_T}{T} &= 0.577 + 0.124 + 0.041 - 0.050 \\ &= 0.692. \end{aligned} \tag{110}$$

This exceeds the static comparator by 0.115. The empirical exploration cost in Section 6 is diagnostic, not a certificate; using it only as a scale check gives $0.577 + 0.124 + 0.041 - 0.043 = 0.699$, which matches the reported routed value up to rounding. Proposition 16 states the additional conditions needed to replace that diagnostic curve by a certified B_T.

C Proof of the Third Main Theorem and Complexity Consequences

This proof block concerns the narrowed third theorem from Section 5.1.

The theorem is not a contraction theorem for language-model reflection; it is a conditional stopping-time statement for the first useful child under an explicit repair-probability assumption.

Assumption 15 (Conditional repair probability for an active cluster)**.** *Let $\mathcal{G} = \{1, \dots, K_{\text{cl}}\}$, let $g(U) \in \mathcal{G}$ denote the user's cluster, and assume every cluster considered below has positive mass*

$$\pi_g \;:=\; \mathbb{P}(g(U) = g) \;>\; 0. \tag{111}$$

For fixed g, draw $U_g \sim \Pi(\,\cdot \mid g(U) = g\,)$. Fix $\epsilon > 0$ and $q_\epsilon \in (0, 1]$. Let $\tau_{g,j}$ be the time of the j-th active repair attempt, with $\tau_{g,j} = \infty$ if it never occurs. The objects below are defined on $\{\tau_{g,j} < \infty\}$; any result invoking the first m attempts assumes they occur almost surely. Let $\mathcal{H}_{g,j}$ denote the information available immediately before drawing the child, and set

$$A^-_{g,j} \;=\; A_{\tau_{g,j}}(U_g), \quad A^+_{g,j} \;=\; A_{\tau_{g,j}+1}(U_g). \tag{112}$$

Define the success indicator

$$I^{(\epsilon)}_{g,j} \;=\; \begin{cases} \mathbf{1}\Big\{\max_{a \in A^+_{g,j}} \mu(U_g, a) - \max_{a \in A^-_{g,j}} \mu(U_g, a) \;\ge\; \epsilon\Big\}, & \tau_{g,j} < \infty, \\ 0, & \tau_{g,j} = \infty. \end{cases} \tag{113}$$

The event that the first j attempts all fail is

$$F^{(\epsilon)}_{g,j} \;=\; \bigcap_{\ell=1}^{j} \{I^{(\epsilon)}_{g,\ell} = 0\}. \tag{114}$$

Assume $F^{(\epsilon)}_{g,j-1} \in \mathcal{H}_{g,j}$ and, before the first success,

$$\mathbb{P}\Big(I^{(\epsilon)}_{g,j} = 1 \mid \mathcal{H}_{g,j}\Big) \;\ge\; q_\epsilon \qquad \text{on } F^{(\epsilon)}_{g,j-1}. \tag{115}$$

The word "active" means that the attempt is made only while the current cluster gap exceeds the target threshold.

The proof uses only equation 115; it does not use a contraction factor.

Theorem 20 (First-useful-specialist time)**.** *Fix $g \in \mathcal{G}$, $\epsilon > 0$, and $m \in \mathbb{N}$, and define*

$$S_{g,m}^{(\epsilon)} = \left\{ \sum_{j=1}^{m} I_{g,j}^{(\epsilon)} \geq 1 \right\}. \tag{116}$$

If Assumption 15 holds for the first m active attempts, then

$$\mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \leq (1-q_\epsilon)^m, \quad \mathbb{P}\left(S_{g,m}^{(\epsilon)}\right) \geq 1-(1-q_\epsilon)^m. \tag{117}$$

Proof. Set $F_{g,0}^{(\epsilon)} = \Xi$. Since $F_{g,j-1}^{(\epsilon)} \in \mathcal{H}_{g,j}$, the tower property and equation 115 give

$$\begin{aligned} \mathbb{P}\left(F_{g,j}^{(\epsilon)}\right) &= \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\mathbf{1}\{I_{g,j}^{(\epsilon)} = 0\}\right] \\ &= \mathbb{E}\left[\mathbf{1}\{F_{g,j-1}^{(\epsilon)}\}\mathbb{P}\left(I_{g,j}^{(\epsilon)} = 0 \mid \mathcal{H}_{g,j}\right)\right] \\ &\leq (1-q_\epsilon)\mathbb{P}\left(F_{g,j-1}^{(\epsilon)}\right). \end{aligned} \tag{118}$$

Iteration yields

$$\mathbb{P}\left(F_{g,m}^{(\epsilon)}\right) \leq (1-q_\epsilon)^m. \tag{119}$$

Taking complements proves the success bound. □

The theorem is intentionally conditional. The saved simulator uses privileged peak-category information, but its child qualities remain random and the runs do not estimate q_ϵ for a fixed ϵ. Thus Theorem 20 states the quantity that a reflection study must estimate; the saved runs do not test its assumption.

Corollary 21 (Attempt complexity for one useful specialist)**.** *For $\delta \in (0,1)$, the exact number of active attempts sufficient for success probability at least $1-\delta$ is*

$$m_\delta^{\text{exact}} = \begin{cases} 1, & q_\epsilon = 1, \\ \left\lceil \dfrac{\log \delta}{\log(1-q_\epsilon)} \right\rceil, & 0 < q_\epsilon < 1. \end{cases} \tag{120}$$

A simpler sufficient choice is

$$m_\delta = \left\lceil \frac{1}{q_\epsilon} \log\left(\frac{1}{\delta}\right) \right\rceil. \tag{121}$$

Either choice satisfies $\mathbb{P}(S_{g,m}^{(\epsilon)}) \geq 1-\delta$.

Proof. If $q_\epsilon = 1$, one attempt succeeds almost surely. If $0 < q_\epsilon < 1$, the exact choice satisfies

$$m_\delta^{\text{exact}} \log(1-q_\epsilon) \leq \log \delta, \quad (1-q_\epsilon)^{m_\delta^{\text{exact}}} \leq \delta. \tag{122}$$

For the simpler choice, $1-q_\epsilon \leq e^{-q_\epsilon}$, so

$$(1-q_\epsilon)^{m_\delta} \leq e^{-q_\epsilon m_\delta} \leq \delta. \tag{123}$$

Apply Theorem 20. □

For example, if a non-oracle proposer had $q_\epsilon = 0.20$ and $\delta = 0.05$, then

$$m_\delta = \left\lceil \frac{1}{0.20} \log(20) \right\rceil = 15, \quad 1-(1-0.20)^{15} = 1-0.8^{15} \approx 0.965. \tag{124}$$

Corollary 22 (Cluster coverage by active attempts). *Let $\mathcal{G}_{\text{act}} \subseteq \mathcal{G}$ contain K_{act} active clusters. Suppose cluster g has repair probability $q_{\epsilon,g} \in (0,1]$ and receives $m_g \in \mathbb{N}$ active attempts. Define full coverage by*

$$C_{\text{all}}^{(\epsilon)} \;=\; \bigcap_{g\in\mathcal{G}_{\text{act}}} S_{g,m_g}^{(\epsilon)}. \tag{125}$$

Then

$$\mathbb{P}\Big(C_{\text{all}}^{(\epsilon)}\Big) \;\geq\; 1 - \sum_{g\in\mathcal{G}_{\text{act}}} (1-q_{\epsilon,g})^{m_g}. \tag{126}$$

If $q_{\min} = \min_{g\in\mathcal{G}_{\text{act}}} q_{\epsilon,g}$ and every cluster receives m attempts, then

$$\mathbb{P}\Big(C_{\text{all}}^{(\epsilon)}\Big) \;\geq\; 1 - K_{\text{act}}(1-q_{\min})^m. \tag{127}$$

A sufficient budget for coverage probability at least $1-\delta$ is

$$m \;=\; \left\lceil \frac{1}{q_{\min}} \log\!\left(\frac{K_{\text{act}}}{\delta}\right)\right\rceil. \tag{128}$$

Proof. Theorem 20 and a union bound give

$$\begin{aligned}\mathbb{P}\Big(\Big(C_{\text{all}}^{(\epsilon)}\Big)^c\Big) \;&\leq\; \sum_{g\in\mathcal{G}_{\text{act}}} \mathbb{P}\Big(\Big(S_{g,m_g}^{(\epsilon)}\Big)^c\Big)\\ &\leq\; \sum_{g\in\mathcal{G}_{\text{act}}} (1-q_{\epsilon,g})^{m_g}.\end{aligned} \tag{129}$$

Taking complements proves the first claim. For uniform m,

$$\begin{aligned}\sum_{g\in\mathcal{G}_{\text{act}}} (1-q_{\epsilon,g})^m \;&\leq\; K_{\text{act}}(1-q_{\min})^m,\\ K_{\text{act}}(1-q_{\min})^m \;&\leq\; K_{\text{act}} e^{-q_{\min} m} \;\leq\; \delta.\end{aligned} \tag{130}$$

□

For $K_{\text{act}} = 3$, $q_{\min} = 0.20$, and $\delta = 0.05$, the sufficient uniform budget is

$$m \;=\; \lceil 5\log(60)\rceil \;=\; 21. \tag{131}$$

Proposition 23 (Rollout and archive-size complexity under the implemented gate). *Let $T, K_{\max} \in \mathbb{N}$, $K_1 = |A_1|$, and $\kappa \in \mathbb{N}_0$. For user u, let $J_t(u) \in \{0,1\}$ indicate a reflection attempt after round t, and set $M_T(u) = \sum_{t=1}^{T} J_t(u)$. Assume the archive changes only when the gate inserts one child, obeys the cap, and enforces the cooldown:*

$$\begin{aligned}|A_{t+1}(u)| \;=\; |A_t(u)| + J_t(u), \qquad |A_t(u)| \;\leq\; K_{\max},\\ J_s(u) = J_t(u) = 1,\; s<t \quad\Longrightarrow\quad t-s \;\geq\; \kappa+1.\end{aligned} \tag{132}$$

Then

$$M_T(u) \;\leq\; \min\left\{K_{\max} - K_1,\; 1 + \left\lfloor \frac{T-1}{\kappa+1}\right\rfloor\right\}. \tag{133}$$

Moreover, $|A_{T+1}(u)| \;\leq\; K_{\max}$ and $\sum_{t=1}^{T} |A_t(u)| \;\leq\; TK_{\max}$. If each attempt costs $c_{\text{ref}} \;\geq\; 0$ additional child evaluations, then

$$T + c_{\text{ref}} M_T(u) \;\leq\; T + c_{\text{ref}}\min\left\{K_{\max} - K_1,\; 1 + \left\lfloor \frac{T-1}{\kappa+1}\right\rfloor\right\}. \tag{134}$$

Proof. Telescoping the archive updates gives

$$|A_{T+1}(u)| \;=\; K_1 + M_T(u) \;\leq\; K_{\max}, \tag{135}$$

so $M_T(u) \leq K_{\max} - K_1$. If $M_T(u) \geq 1$, order the reflection times as $1 \leq s_1 < \cdots < s_{M_T(u)} \leq T$. The cooldown gives

$$s_j \;\geq\; 1 + (j-1)(\kappa+1), \quad T \;\geq\; 1 + (M_T(u)-1)(\kappa+1). \tag{136}$$

Rearranging yields the cooldown bound on $M_T(u)$; combining it with the cap proves the stated minimum. Finally,

$$\sum_{t=1}^{T} |A_t(u)| \;\le\; TK_{\max}, \quad \mathrm{Rollouts}_T(u) \;=\; T + c_{\mathrm{ref}} M_T(u), \tag{137}$$

and substitution proves the rollout bound. □

With the experimental values $T = 300$, $K_1 = 12$, $K_{\max} = 24$, and $\kappa = 20$,

$$\begin{aligned} M_T(u) \;&\le\; \min\left\{24 - 12,\, 1 + \left\lfloor \frac{299}{21} \right\rfloor \right\} \;=\; 12, \\ \sum_{t=1}^{300} |A_t(u)| \;&\le\; 300 \cdot 24 \;=\; 7200, \\ T + M_T(u) \;&\le\; 300 + 12 \;=\; 312. \end{aligned} \tag{138}$$

Proposition 24 (Discovery reserve induced by first useful specialists)**.** *Let R_T be the routed value from Theorem 13, and define the static-comparator reserve*

$$\rho_T \;=\; R_T - V_{\mathrm{static}}(A_1). \tag{139}$$

Let $\pi_g = \mathbb{P}(g(U) = g) > 0$ for $g \in \mathcal{G}_{\mathrm{act}}$, and fix $s \in \{0, \ldots, T\}$. With $\tau_{g,j} = \infty$ for an attempt that never occurs, define

$$\begin{aligned} N_g(s) \;&=\; \sum_{j \ge 1} \mathbf{1}\{\tau_{g,j} \le s\}, \\ p_g(s) \;&=\; \mathbb{P}(N_g(s) \ge m_g \mid g(U) = g)\,. \end{aligned} \tag{140}$$

The joint probability that the required attempts occur by round s and one of them succeeds is

$$h_g(s) \;=\; \mathbb{P}\Big(S^{(\epsilon_g)}_{g,m_g} \cap \{N_g(s) \ge m_g\} \mid g(U) = g\Big)\,. \tag{141}$$

Assume a successful ϵ_g-useful child inserted by round s persists and contributes at least ϵ_g to every later discovery gain in that cluster:

$$G_t(U) \;\ge\; \epsilon_g \quad \text{for all } t > s \text{ on } \{g(U) = g\} \cap S^{(\epsilon_g)}_{g,m_g} \cap \{N_g(s) \ge m_g\}. \tag{142}$$

If

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; B_T \tag{143}$$

then

$$\rho_T \;\ge\; \Delta_{\mathrm{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\mathrm{act}}} \pi_g \epsilon_g h_g(s). \tag{144}$$

Consequently, personalized routing beats the best static initial candidate if

$$\Delta_{\mathrm{route}}(A_1) - \frac{B_T}{T} + \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\mathrm{act}}} \pi_g \epsilon_g h_g(s) > 0. \tag{145}$$

The regret relative to that static candidate,

$$\mathcal{R}_T^{\mathrm{static}} \;=\; TV_{\mathrm{static}}(A_1) - \sum_{t=1}^{T} \mathbb{E}[\mu(U, a_t(U))], \tag{146}$$

satisfies

$$\frac{1}{T}\mathcal{R}_T^{\mathrm{static}} \;\le\; -\Delta_{\mathrm{route}}(A_1) + \frac{B_T}{T} - \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\mathrm{act}}} \pi_g \epsilon_g h_g(s). \tag{147}$$

Proof. Theorem 13 and the exploration certificate give

$$\rho_T = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, \quad \bar{L}_T \le \frac{B_T}{T}. \tag{148}$$

For every $t > s$, persistence gives

$$\mathbb{E}[G_t(U)] \ge \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g \epsilon_g h_g(s). \tag{149}$$

Since $G_t(U) \ge 0$, averaging over the final $T - s$ rounds yields

$$\bar{G}_T \ge \frac{T-s}{T} \sum_{g \in \mathcal{G}_{\text{act}}} \pi_g \epsilon_g h_g(s). \tag{150}$$

Substitution proves the reserve bound, and its strict positivity is $\rho_T > 0$. Finally,

$$\begin{aligned} \frac{1}{T}\mathcal{R}_T^{\text{static}} &= V_{\text{static}}(A_1) - R_T \\ &= -\rho_T, \end{aligned} \tag{151}$$

which proves the regret bound. □

In general, $h_g(s)$ cannot be replaced by $p_g(s)[1-(1-q_{\epsilon_g,g})^{m_g}]$: the event that m_g attempts are scheduled may depend on earlier successes or failures. That product is valid only under an additional exogenous attempt-scheduling condition that preserves the per-attempt success bound after conditioning on $\{N_g(s) \ge m_g\}$.

This reserve links Theorem 20 to the decomposition, but does not certify the oracle-assisted mechanism in the saved runs. Those runs report structural terms and an observed-reward residual. For the corrected `general` row in Table 2,

$$\begin{aligned} \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} &= 0.280 + 0.051 - 0.036 \\ &= 0.295, \\ V_{\text{static}}(A_1) + 0.295 &= 0.570 + 0.295 \\ &= 0.865, \end{aligned} \tag{152}$$

matching the reported $R_T^{\text{obs}} \approx 0.864$ up to rounding. This checks only the observed-reward identity; it is not a test of the mean-reward decomposition or a non-oracle proof of discovery.

D Remaining Propositions, Counterexamples, and Technical Details

This section records the remaining algebraic consequences of the decomposition, the edge cases that delimit its interpretation, and the proof map for the main theory claims.

The statements below do not introduce a new mechanism; they only make explicit when the four-term accounting identity can and cannot be read as an advantage claim.

D.1 Dominance condition

Corollary 25 (Dominance condition)**.** *The routed policy beats the best static initial candidate if and only if*

$$R_T > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T. \tag{153}$$

If $A_t(U) = A_1$ for every t, this reduces to

$$R_T > V_{\text{static}}(A_1) \iff \Delta_{\text{route}}(A_1) > \bar{L}_T. \tag{154}$$

Proof. Subtract $V_{\text{static}}(A_1)$ from Theorem 13:

$$R_T - V_{\text{static}}(A_1) = \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{155}$$

Strict positivity is equivalent to the first claim. A fixed archive has $G_t(U) = 0$, proving the second. □

For example, $\Delta_{\text{route}}(A_1) = 0.12$, $\bar{G}_T = 0.04$, and $\bar{L}_T = 0.03$ give reserve $0.12 + 0.04 - 0.03 = 0.13 > 0$.

D.2 Sign facts and user-specific archives

Assumption 16 (Append-only archive for sign statements)**.** *Almost surely,* $A_1 \subseteq A_t(U)$ *and* $a_t(U) \in A_t(U)$.

Lemma 26 (Range of discovery gain and exploration cost)**.** *If* $\mu(u,a) \in [0,1]$*, then under Assumption 16,*

$$L_t(U), G_t(U) \in [0,1] \quad \textit{almost surely}, \qquad \bar{L}_T, \bar{G}_T \in [0,1]. \tag{156}$$

Proof. Feasibility and append-only growth imply

$$m_t(U) \;\geq\; \mu(U, a_t(U)), \quad m_t(U) \;\geq\; m_1(U). \tag{157}$$

The reward range bounds both differences above by one. Averaging preserves the interval. □

Proposition 27 (Per-user archive copies preserve the decomposition)**.** *For each user* u*, let* A_t^u *be a user-specific archive path with* $A_1^u = A_1$*,* $a_t^u \in A_t^u$*, and* $m_t^u = \max_{a \in A_t^u} \mu(u,a)$*. If the population draw selects the corresponding trajectory,*

$$A_t(U) = A_t^U, \qquad a_t(U) = a_t^U. \tag{158}$$

then Theorem 13 remains valid:

$$R_T \;=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T. \tag{159}$$

Proof. For every fixed user and trajectory,

$$\mu(u, a_t^u) \;=\; m_1^u + G_t^u - L_t^u. \tag{160}$$

Taking the trajectory expectation conditional on u, then integrating over U, gives

$$\mathbb{E}[\mu(U, a_t(U))] \;=\; \mathbb{E}[m_1^U] + \mathbb{E}[G_t(U)] - \mathbb{E}[L_t(U)]. \tag{161}$$

The common initial archive implies

$$\begin{aligned} \mathbb{E}[m_1^U] \;&=\; \mathbb{E}\left[\max_{a \in A_1} \mu(U,a)\right] \\ &=\; V_{\text{route}}(A_1) \\ &=\; V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1). \end{aligned} \tag{162}$$

Averaging over t completes the proof. □

D.3 Finite-sample identity and comparator correction

Proposition 28 (Finite-sample decomposition estimator)**.** *For* N *sampled users* $u_1, \ldots, u_N$*, let* $a_{i,t} \in A_{i,t}$ *and* $m_{i,t} = \max_{a \in A_{i,t}} \mu(u_i, a)$*. Define*

$$\begin{aligned} \widehat{R}_T \;&=\; \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \mu(u_i, a_{i,t}), \\ \widehat{V}_{\text{route}}(A_1) \;&=\; \frac{1}{N} \sum_{i=1}^{N} \max_{a \in A_1} \mu(u_i, a), \\ \widehat{V}_{\text{static}}(A_1) \;&=\; \max_{a \in A_1} \frac{1}{N} \sum_{i=1}^{N} \mu(u_i, a), \\ \widehat{\Delta}_{\text{route}}(A_1) \;&=\; \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1), \\ \widehat{\bar{G}}_T \;&=\; \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} (m_{i,t} - m_{i,1}), \\ \widehat{\bar{L}}_T \;&=\; \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} (m_{i,t} - \mu(u_i, a_{i,t})). \end{aligned} \tag{163}$$

Then

$$\widehat{R}_T \;=\; \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. \tag{164}$$

Proof. For each (i, t),

$$\mu(u_i, a_{i,t}) \;=\; m_{i,1} + (m_{i,t} - m_{i,1}) - (m_{i,t} - \mu(u_i, a_{i,t})). \tag{165}$$

Summing and dividing by NT gives

$$\begin{aligned} \widehat{R}_T \;&=\; \frac{1}{N}\sum_{i=1}^{N} m_{i,1} + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T \\ &=\; \widehat{V}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. \end{aligned} \tag{166}$$

Substitute the definition of $\widehat{\Delta}_{\text{route}}(A_1)$. □

Proposition 28 applies when the selected-arm means are used. The rounded residuals in Table 2 instead use observed rewards and the matching observed residual, so they check a parallel algebraic identity but do not test the proposition.

For the three completed synthetic environments,

$$\begin{aligned} \varepsilon_{\text{general}} \;&=\; 0.570 + 0.280 + 0.051 - 0.036 - 0.864 \\ &=\; 0.001. \end{aligned} \tag{167}$$

$$\begin{aligned} \varepsilon_{\text{workspace}} \;&=\; 0.577 + 0.124 + 0.041 - 0.043 - 0.699 \\ &=\; 0.000. \end{aligned} \tag{168}$$

$$\begin{aligned} \varepsilon_{\text{support}} \;&=\; 0.579 + 0.215 + 0.061 - 0.039 - 0.816 \\ &=\; 0.000. \end{aligned} \tag{169}$$

The nonzero residual in equation 167 is a rounding residual, not an additional term in the decomposition.

Proposition 29 (Bias from a heuristic static comparator)**.** *Let $h(A_1) \in A_1$ be a heuristic choice, and define*

$$V_h(A_1) \;=\; \mathbb{E}[\mu(U, h(A_1))]. \tag{170}$$

For $\Delta_h(A_1) = V_{\text{route}}(A_1) - V_h(A_1)$,

$$\Delta_h(A_1) - \Delta_{\text{route}}(A_1) \;=\; V_{\text{static}}(A_1) - V_h(A_1) \;\geq\; 0. \tag{171}$$

Equality holds exactly when $h(A_1)$ is population-static optimal.

Proof.

$$\begin{aligned} \Delta_h(A_1) - \Delta_{\text{route}}(A_1) \;&=\; [V_{\text{route}}(A_1) - V_h(A_1)] - [V_{\text{route}}(A_1) - V_{\text{static}}(A_1)] \\ &=\; V_{\text{static}}(A_1) - V_h(A_1) \;\geq\; 0, \end{aligned} \tag{172}$$

because $V_{\text{static}}(A_1)$ maximizes over all static candidates. □

D.4 COUNTEREXAMPLES DELIMITING THE CLAIMS

Proposition 30 (Counterexample: homogeneous users have no structural routing gain)**.** *Let $A_t(U) = A_1 = \{a_1, a_2\}$, with $\mu(U, a_1) = 0.8$ and $\mu(U, a_2) = 0.6$ almost surely. If the router selects a_2 on an expected $\eta > 0$ fraction of rounds, then $R_T < V_{\text{static}}(A_1)$.*

Proof.

$$V_{\text{static}}(A_1) \;=\; 0.8, \quad V_{\text{route}}(A_1) \;=\; 0.8, \quad \Delta_{\text{route}}(A_1) \;=\; 0, \quad \bar{G}_T \;=\; 0. \tag{173}$$

Selecting a_2 costs 0.2, so $\bar{L}_T = 0.2\eta$. Hence

$$R_T \;=\; 0.8 - 0.2\eta \;<\; 0.8 \;=\; V_{\text{static}}(A_1). \tag{174}$$

□

Proposition 31 (Counterexample: a heuristic static baseline can overstate heterogeneity)**.** *Let $\mathbb{P}(U = u_1) = \mathbb{P}(U = u_2) = 1/2$, $A_1 = \{a_1, a_2\}$, and*

$$\begin{array}{c|cc} & a_1 & a_2 \\ \hline u_1 & 0.9 & 0.8 \\ u_2 & 0.1 & 0.7 \end{array}. \tag{175}$$

If heuristic scores satisfy $q(a_1) > q(a_2)$, so that $h(A_1) = a_1$, then $\Delta_{\text{route}}(A_1) = 0.05$ but $\Delta_h(A_1) = 0.30$.

Proof.

$$V_h(A_1) \;=\; \tfrac{1}{2}(0.9 + 0.1) = 0.50, \quad \mathbb{E}[\mu(U, a_1)] \;=\; 0.50, \quad \mathbb{E}[\mu(U, a_2)] \;=\; \tfrac{1}{2}(0.8 + 0.7) = 0.75. \tag{176}$$

The routed oracle value is

$$\begin{aligned} V_{\text{route}}(A_1) \;&=\; \tfrac{1}{2}\left[\max\{0.9, 0.8\} + \max\{0.1, 0.7\}\right] \;=\; 0.80, \\ \Delta_{\text{route}}(A_1) \;&=\; 0.80 - 0.75 = 0.05, \qquad\qquad \Delta_h(A_1) \;=\; 0.80 - 0.50 = 0.30. \end{aligned} \tag{177}$$

□

Proposition 32 (Counterexample: oracle-assisted reflection makes the discovery bound vacuous)**.** *For cluster g, define the repair probability in Theorem 20 by*

$$q_{\epsilon,g} \;=\; \inf_j \mathbb{P}\big(\textit{attempt } j \textit{ is } \epsilon\textit{-improving} \mid \mathcal{F}_{\tau_{g,j}-1}\big)\,. \tag{178}$$

If the proposer reads the hidden peak category and deterministically creates an ϵ-improving child, then $q_{\epsilon,g} = 1$ and the lower bound is one for every $m \geq 1$. If $q_{\epsilon,g} = 0$, the bound is zero for every finite m.

Proof.

$$1 - (1-1)^m \;=\; 1, \quad 1 - (1-0)^m \;=\; 0. \tag{179}$$

Thus the result only reflects the assumed repair probability. Validating a non-oracle proposer requires a separate nontrivial lower bound on $q_{\epsilon,g}$. □

Proposition 33 (Counterexample: without append-only archives, discovery need not be nonnegative)**.** *Let $\mathbb{P}(U = u) = 1$, $A_1 = \{a^\star\}$, and $A_2 = \{b\}$, with $\mu(u, a^\star) = 1$ and $\mu(u, b) = 0$. Then $G_2(U) = -1$. The decomposition remains algebraically true, but the term G_t can no longer be interpreted as a gain.*

Proof.

$$m_1(U) \;=\; 1, \quad m_2(U) \;=\; 0, \quad G_2(U) \;=\; -1, \quad L_2(U) \;=\; 0. \tag{180}$$

Yet

$$\mu(u, b) \;=\; m_1(U) + G_2(U) - L_2(U) \;=\; 1 - 1 - 0 \;=\; 0. \tag{181}$$

Thus append-only growth is needed for the sign interpretation, not the identity. □

D.5 Map from theory claims to proofs

Together, these results close the appendix proof story: PersonalEvo's supported claim is the four-term decomposition over the common initial archive, with conditional learning and discovery terms made explicit, and with the counterexamples identifying when stronger dominance or reflection claims would be unsound.

Claim	Formal content	Proof or qualification
Lemma 11	Static-vs-routed value satisfies $\mathbb{E}[\max_{a\in A}\mu(U,a)] \geq \max_{a\in A}\mathbb{E}[\mu(U,a)]$, with strictness only under positive-mass ranking disagreement.	Proved above in Appendix A; Proposition 30 shows the homogeneous boundary case.
Lemma 12	Adding children after a round preserves predictability of the next action set.	Proved above in Appendix A; Assumption 16 records the extra sign condition used only for $G_t \geq 0$.
Lemma 14	High-gap failure triggering controls the number of mutation attempts without changing the router's filtration argument.	Proved above in Appendix A; Proposition 32 clarifies that trigger evidence is not evidence for non-oracle repair quality.
Theorem 13	Exact four-term decomposition: $R_T = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T$.	Proved above in Appendix A; Proposition 27 covers user-specific archive copies and Proposition 28 gives the finite-sample estimator.
Theorem 17	Conditional plug-in bound obtained from an externally supplied exploration-cost certificate B_T.	Proved above in Appendix A; no new regret rate is claimed for the saved implementation.
Theorem 20	First-useful-specialist probability: $\mathbb{P}(S_{g,m}^{(\epsilon)}) \geq 1-(1-q_{\epsilon,g})^m$.	Proved above in Appendix A; Proposition 32 explains why oracle-assisted reflection makes the bound vacuous as empirical validation.
Corollary 3	Personalized routing beats the best static initial candidate exactly when $\Delta_{\text{route}}(A_1) + \bar{G}_T > \bar{L}_T$.	Restated and proved in Corollary 25.
Proposition 24	A first-useful-specialist reserve can be inserted into the decomposition under persistence and exploration-cost certificates.	Proved above in Appendix A; it is a conditional reserve statement, not a guarantee for the oracle-assisted simulator.
Comparator correction	The theorem comparator is $V_{\text{static}}(A_1)$, not a heuristic intrinsic-quality choice.	Proved in Proposition 29; the numerical failure mode is Proposition 31.

Table 5: Proof map for the theory claims used in the main text.

E Additional Technical Proof Block 5: Constants and Remaining Lemmas

This block makes explicit the measurability, boundedness, trigger-count, finite-cluster, and contraction constants used by Theorems 13, 17, and 20. It does not add a new mechanism.

Lemma 34 (Measurability and bounded ranges)**.** *Fix t. Suppose the user-specific archive has the finite predictable representation*

$$A_t(U) = \{a_{t,1}(U), \dots, a_{t,N_t(U)}(U)\}. \tag{182}$$

Assume $N_t(U) < \infty$ almost surely, each listed candidate is $\mathcal{F}_{t-1}$-measurable, and μ is measurable with values in $[0,1]$. Then $m_t(U) = \max_{a\in A_t(U)}\mu(U,a)$ is measurable, $L_t(U) \in [0,1]$, and $G_t(U) \in [-1,1]$. If $A_1 \subseteq A_t(U)$, then $G_t(U) \in [0,1]$.

Proof. For each n and $z \in \mathbb{R}$,

$$\{m_t(U) \le z\} \cap \{N_t(U) = n\} = \bigcap_{j=1}^{n} \{\mu(U, a_{t,j}(U)) \le z\} \cap \{N_t(U) = n\}. \tag{183}$$

This finite intersection is measurable; taking the countable union over n proves measurability of $m_t(U)$. The reward range gives $m_t(U) \in [0,1]$. Feasibility gives $\mu(U, a_t(U)) \le m_t(U)$, hence $L_t(U) \in [0,1]$. Since G_t is the difference of two values in $[0,1]$, it lies in $[-1,1]$; append-only growth gives $m_t(U) \ge m_1(U)$. □

For example, $m_1(U) = 0.60$, $m_t(U) = 0.75$, and $\mu(U, a_t(U)) = 0.70$ give $G_t(U) = 0.15$ and $L_t(U) = 0.05$.

Lemma 35 (From a high-probability tax certificate to an expected tax bound)**.** *Let $S_T = \sum_{t=1}^{T} L_t(U)$, with $L_t(U) \in [0,1]$. If $\mathbb{P}(S_T \le b_T) \ge 1 - \delta$, then*

$$\mathbb{E}[S_T] \le b_T + \delta T. \tag{184}$$

Thus Theorem 17 applies with $B_T = b_T + \delta T$.

Proof. For $\mathcal{E}_T = \{S_T \le b_T\}$, use $S_T \le T$:

$$\begin{aligned} \mathbb{E}[S_T] &= \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T\}] + \mathbb{E}[S_T \mathbf{1}\{\mathcal{E}_T^c\}] \\ &\le b_T + T\mathbb{P}(\mathcal{E}_T^c) \le b_T + \delta T. \end{aligned} \tag{185}$$

□

For $b_T = 12$, $T = 300$, and $\delta = 0.01$, this gives $\mathbb{E}[S_T] \le 15$, or $\mathbb{E}[S_T]/T \le 0.05$.

Proposition 36 (Explicit constants for high-gap triggering)**.** *For $\tau, \sigma > 0$, define the trigger, trigger count, and observation model by*

$$I_t^{(\tau)} = \mathbf{1}\{m_t(U) - r_t > \tau\}, \tag{186}$$

$$N_T^{(\tau)} = \sum_{t=1}^{T} I_t^{(\tau)}, \tag{187}$$

$$r_t = \mu(U, a_t(U)) + \xi_t. \tag{188}$$

Suppose that, for every $\lambda \in \mathbb{R}$,

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2 \sigma^2}{2}\right). \tag{189}$$

Then

$$\mathbb{E}\left[N_T^{(\tau)}\right] \le \frac{2}{\tau} \sum_{t=1}^{T} \mathbb{E}[L_t(U)] + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{190}$$

In particular, if

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T, \tag{191}$$

then

$$\mathbb{E}\left[N_T^{(\tau)}\right] \le \frac{2B_T}{\tau} + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{192}$$

Proof. Since $m_t(U) - r_t = L_t(U) - \xi_t$,

$$\begin{aligned} I_t^{(\tau)} &\le \mathbf{1}\{L_t(U) + |\xi_t| > \tau\} \\ &\le \mathbf{1}\{L_t(U) > \tau/2\} + \mathbf{1}\{|\xi_t| > \tau/2\}. \end{aligned} \tag{193}$$

Markov's inequality and the sub-Gaussian tail bound give

$$\mathbb{P}\{L_t(U) > \tau/2\} \le \frac{2}{\tau}\mathbb{E}[L_t(U)], \quad \mathbb{P}\{|\xi_t| > \tau/2\} \le 2\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{194}$$

Taking expectations and summing over t gives

$$\mathbb{E}[N_T^{(\tau)}] \le \frac{2}{\tau}\sum_{t=1}^{T}\mathbb{E}[L_t(U)] + 2T\exp\left(-\frac{\tau^2}{8\sigma^2}\right). \tag{195}$$

This proves equation 190; substituting equation 191 gives equation 192. □

For $T = 100$, $B_T = 4$, $\tau = 0.20$, and $\sigma = 0.05$, the bound is

$$\begin{aligned}\mathbb{E}[N_T^{(\tau)}] &\le \frac{2\cdot 4}{0.20} + 2\cdot 100\exp\left(-\frac{0.20^2}{8\cdot 0.05^2}\right) \\ &= 40 + 200e^{-2} \approx 67.07.\end{aligned} \tag{196}$$

This constant is conservative; it is used only to justify the direction of the trigger-count control, not as an empirical estimate of reflection quality.

Proposition 37 (Finite-cluster margin lower bound for heterogeneity)**.** *Let A be a finite archive, and partition the user population into $K_{\mathrm{cl}} < \infty$ clusters. Define the cluster index, mass, and conditional candidate value by*

$$g(U) \in \{1, \dots, K_{\mathrm{cl}}\}, \tag{197}$$

$$\pi_g = \mathbb{P}(g(U) = g), \tag{198}$$

$$\nu_{g,a} = \mathbb{E}[\mu(U,a) \mid g(U) = g] \qquad (\pi_g > 0). \tag{199}$$

Set $\nu_{g,a} = 0$ when $\pi_g = 0$. For each static candidate, define its cluster-wise deficit and the clusters with deficit at least $\eta > 0$:

$$d_g(a) = \max_{b\in A}\nu_{g,b} - \nu_{g,a}, \tag{200}$$

$$H_a(\eta) = \{g : d_g(a) \ge \eta\}. \tag{201}$$

If every static candidate has such a deficit on cluster mass at least ρ,

$$\inf_{a\in A}\sum_{g\in H_a(\eta)}\pi_g \ge \rho. \tag{202}$$

Then the fixed-archive heterogeneity gap satisfies

$$\Delta_{\mathrm{route}}(A) \ge \rho\eta. \tag{203}$$

Proof. For each cluster, $\mathbb{E}[\max_{b\in A}\mu(U,b) \mid g(U) = g] \ge \max_{b\in A}\nu_{g,b}$. Hence

$$V_{\mathrm{route}}(A) \ge \sum_{g=1}^{K_{\mathrm{cl}}}\pi_g\max_{b\in A}\nu_{g,b}, \tag{204}$$

$$V_{\mathrm{static}}(A) = \max_{a\in A}\sum_{g=1}^{K_{\mathrm{cl}}}\pi_g\nu_{g,a}. \tag{205}$$

Because A is finite, choose

$$a^\star \in \arg\max_{a\in A}\sum_{g=1}^{K_{\mathrm{cl}}}\pi_g\nu_{g,a}. \tag{206}$$

Then

$$\begin{aligned}\Delta_{\mathrm{route}}(A) &\ge \sum_{g=1}^{K_{\mathrm{cl}}}\pi_g d_g(a^\star). \\ &\ge \sum_{g\in H_{a^\star}(\eta)}\pi_g d_g(a^\star) \\ &\ge \eta\sum_{g\in H_{a^\star}(\eta)}\pi_g \ge \rho\eta.\end{aligned} \tag{207}$$

□

The bound is tight. For equal cluster masses and value matrix

$$\begin{array}{c|cc} & a_1 & a_2 \\ \hline g=1 & 0.80 & 0.50 \\ g=2 & 0.50 & 0.80 \end{array}. \tag{208}$$

we have $\pi_1 = \pi_2 = 1/2$, $\rho = 1/2$, and $\eta = 0.30$. Therefore

$$\Delta_{\text{route}}(A) \;\geq\; \rho\eta = 0.15, \quad \Delta_{\text{route}}(A) \;=\; 0.80 - \tfrac{1}{2}(0.80 + 0.50) = 0.15. \tag{209}$$

Proposition 38 (Contraction repair condition implies an explicit discovery-time bound)**.** *Fix a cluster g, an initial gap $\Delta_{g,0} > 0$, and a target $0 < \epsilon < \Delta_{g,0}$. Let $Z_j \in \{0, 1\}$ be independent success indicators satisfying $\mathbb{P}(Z_j = 1) \geq q$ for $q \in (0, 1]$. For $0 < \gamma < 1$, assume*

$$\Delta_{g,j} \;\leq\; (1 - \gamma Z_j)\Delta_{g,j-1}. \tag{210}$$

Define the required and realized success counts by

$$s_\epsilon \;=\; \left\lceil \frac{\log(\Delta_{g,0}/\epsilon)}{-\log(1-\gamma)} \right\rceil, \tag{211}$$

$$S_m \;=\; \sum_{j=1}^{m} Z_j \qquad (m \in \mathbb{N}). \tag{212}$$

If $mq \geq 2s_\epsilon$, then

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \;\leq\; \exp\Big(-\frac{mq}{8}\Big). \tag{213}$$

Consequently, for $\delta \in (0, 1)$, the sufficient condition

$$m \;\geq\; \frac{2s_\epsilon + 8\log(1/\delta)}{q} \tag{214}$$

to obtain

$$\mathbb{P}(\Delta_{g,m} \;\leq\; \epsilon) \;\geq\; 1 - \delta. \tag{215}$$

Proof. Iteration and the binary nature of Z_j give

$$\Delta_{g,m} \;\leq\; \Delta_{g,0} \prod_{j=1}^{m} (1 - \gamma Z_j) \;=\; \Delta_{g,0}(1-\gamma)^{S_m}. \tag{216}$$

Thus $S_m \geq s_\epsilon$ implies

$$\Delta_{g,m} \;\leq\; \Delta_{g,0}(1-\gamma)^{s_\epsilon} \;\leq\; \epsilon. \tag{217}$$

Let

$$\lambda = \log 2. \tag{218}$$

For any $s \leq mq/2$, Markov's inequality applied to $\exp(-\lambda S_m)$ gives

$$\begin{aligned} \mathbb{P}(S_m \;\leq\; s) \;&=\; \mathbb{P}\big(e^{-\lambda S_m} \;\geq\; e^{-\lambda s}\big) \\ &\leq\; e^{\lambda s}\mathbb{E}[e^{-\lambda S_m}]. \end{aligned} \tag{219}$$

Using independence and $\mathbb{P}(Z_j = 1) \geq q$,

$$\begin{aligned} \mathbb{E}[e^{-\lambda S_m}] \;&=\; \prod_{j=1}^{m} \mathbb{E}[e^{-\lambda Z_j}] \\ &=\; \prod_{j=1}^{m} \big(1 - \mathbb{P}(Z_j = 1) + \mathbb{P}(Z_j = 1)e^{-\lambda}\big) \\ &\leq\; \big(1 - q + qe^{-\lambda}\big)^m \\ &\leq\; \exp\big[-mq(1 - e^{-\lambda})\big]. \end{aligned} \tag{220}$$

With $s = s_\epsilon$ and $s_\epsilon \le mq/2$,

$$\mathbb{P}(S_m < s_\epsilon) \;\le\; \exp\Big[\lambda \frac{mq}{2} - mq(1 - e^{-\lambda})\Big]. \tag{221}$$

Since $\lambda = \log 2$ and $e^{-\lambda} = 1/2$,

$$\begin{aligned} \lambda \frac{mq}{2} - mq(1 - e^{-\lambda}) \;&=\; mq\left(\frac{\log 2}{2} - \frac{1}{2}\right) \\ &=\; -\frac{mq}{2}(1 - \log 2) \\ &\le\; -\frac{mq}{8}. \end{aligned} \tag{222}$$

This proves

$$\mathbb{P}(\Delta_{g,m} > \epsilon) \;\le\; \exp\Big(-\frac{mq}{8}\Big). \tag{223}$$

If equation 214 holds, then

$$\begin{aligned} mq \;&\ge\; 2s_\epsilon, \\ mq \;&\ge\; 8\log(1/\delta). \end{aligned} \tag{224}$$

Hence

$$\begin{aligned} \mathbb{P}(\Delta_{g,m} > \epsilon) \;&\le\; \exp[-\log(1/\delta)] \\ &=\; \delta. \end{aligned} \tag{225}$$

□

For $\Delta_{g,0} = 0.40$, $\epsilon = 0.05$, $\gamma = 0.50$, $q = 0.25$, and $\delta = 0.05$,

$$s_\epsilon \;=\; \left\lceil \frac{\log 8}{\log 2} \right\rceil = 3, \tag{226}$$

$$m \;\ge\; \frac{2 \cdot 3 + 8\log(20)}{0.25} \;\approx\; 119.86. \tag{227}$$

Thus $m = 120$ suffices for the stated 95% guarantee under this stronger contraction model. The saved oracle-assisted code does not estimate q or γ; Section 6 therefore reports discovery as a measured decomposition term, not as validation of Proposition 38.

F Additional Technical Proofs: Constants and Estimation Bridges

This block collects the remaining constants and finite-sample bridges behind the four-term decomposition. It adds no new theorem path.

Lemma 39 (Append-only boundedness of the decomposition terms)**.** *Assume bounded rewards, append-only archives, and feasible actions:*

$$0 \;\le\; \mu(u, a) \;\le\; 1, \tag{228}$$

$$A_1(U) \subseteq A_2(U) \subseteq \cdots \subseteq A_T(U), \tag{229}$$

$$a_t(U) \in A_t(U). \tag{230}$$

Define

$$m_t(U) \;=\; \max_{a \in A_t(U)} \mu(U, a), \tag{231}$$

$$G_t(U) \;=\; m_t(U) - m_1(U), \tag{232}$$

$$L_t(U) \;=\; m_t(U) - \mu(U, a_t(U)). \tag{233}$$

Then $0 \le G_t(U), L_t(U) \le 1$ for every t, and G_t is nondecreasing:

$$0 \;\le\; G_t(U) \;\le\; 1, \qquad 0 \;\le\; L_t(U) \;\le\; 1, \qquad G_{t+1}(U) \;\ge\; G_t(U). \tag{234}$$

Proof.

$$0 \le m_1(U) \le m_t(U) \le m_{t+1}(U) \le 1,$$
$$0 \le m_t(U) - \mu(U, a_t(U)) \le 1. \tag{235}$$

The first line follows from archive inclusion and the reward range; the second follows from feasibility. Substitution into the definitions of G_t and L_t proves all three claims. □

Lemma 40 (From high-probability current-archive regret to expected exploration cost)**.** *Let*

$$S_T^L = \sum_{t=1}^{T} L_t(U), \qquad 0 \le L_t(U) \le 1. \tag{236}$$

If

$$\mathbb{P}\big(S_T^L \le C_T(\delta)\big) \ge 1 - \delta. \tag{237}$$

then

$$\mathbb{E}[S_T^L] \le C_T(\delta) + \delta T. \tag{238}$$

Consequently, the external exploration-cost premise equation 28 holds with

$$B_T = C_T(\delta) + \delta T. \tag{239}$$

In particular, if

$$C_T(\delta) = c_0 \sqrt{T\big(\zeta_T + \log(1/\delta)\big)}, \tag{240}$$

$$\delta = \frac{1}{T}, \tag{241}$$

then

$$B_T \le c_0 \sqrt{T\big(\zeta_T + \log T\big)} + 1. \tag{242}$$

Proof. Apply Lemma 35 with $S_T = S_T^L$ and $b_T = C_T(\delta)$. Substituting the displayed form of C_T and $\delta = 1/T$ gives the last claim. □

For example, $C_{300}(1/300) = 12$ gives

$$B_{300} \le 12 + \frac{300}{300} = 13, \quad \frac{B_{300}}{300} \le 0.0433. \tag{243}$$

Lemma 41 (Explicit constants for high-gap trigger counts)**.** *For $\tau, \sigma > 0$, define*

$$r_t = \mu(U, a_t(U)) + \xi_t, \tag{244}$$

$$I_t(\tau) = \mathbf{1}\{m_t(U) - r_t > \tau\}, \tag{245}$$

$$N_T(\tau) = \sum_{t=1}^{T} I_t(\tau). \tag{246}$$

Assume that

$$\mathbb{E}[\exp(\lambda \xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\lambda^2 \sigma^2}{2}\right) \qquad (\lambda \in \mathbb{R}). \tag{247}$$

Then

$$\mathbb{E}[N_T(\tau)] \le T \wedge \left[\frac{2}{\tau} \sum_{t=1}^{T} \mathbb{E}[L_t(U)] + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right)\right]. \tag{248}$$

Under the external exploration-cost bound

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le B_T, \tag{249}$$

the trigger count obeys

$$\mathbb{E}[N_T(\tau)] \le T \wedge \left[\frac{2B_T}{\tau} + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right)\right]. \tag{250}$$

Proof. Proposition 36, under the same notation, gives the second term inside the minimum. The deterministic inequality $N_T(\tau) \le T$ gives the first. Substituting equation 249 proves the final claim. $\square$

For $T = 300$, $\tau = 0.15$, $\sigma = 0.05$, and $B_T = 12$,

$$\begin{aligned} \frac{2B_T}{\tau} + 2T \exp\left(-\frac{\tau^2}{8\sigma^2}\right) &= \frac{24}{0.15} + 600 \exp(-1.125) \\ &\approx 354.8, \qquad\qquad \mathbb{E}[N_T(0.15)] \le 300. \end{aligned} \tag{251}$$

Thus the deterministic cap dominates. This loose bound does not imply the empirical archive-growth claim in Section 6.

Proposition 42 (Cooldown and archive cap imply a deterministic archive-size bound)**.** *Let $K_1 = |A_1| \le K_{\max}$, and let M_T be the number of unique children appended through round T. When $M_T \ge 1$, write the append times as $1 \le s_1 < \cdots < s_{M_T} \le T$. Assume the archive is append-only, exactly one child is inserted at each s_j, the cap $|A_t(U)| \le K_{\max}$ is enforced, and, for $\kappa \in \mathbb{N}$,*

$$s_{j+1} - s_j \ge \kappa + 1. \tag{252}$$

Then the number of appended children satisfies

$$M_T \le \min\left\{K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \tag{253}$$

Consequently, for every $t \in \{1, \ldots, T+1\}$,

$$|A_t(U)| \le K_1 + \min\left\{K_{\max} - K_1,\ 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor \right\}. \tag{254}$$

Proof. The case $M_T = 0$ is immediate. Otherwise, the cooldown gives

$$s_j \ge 1 + (j-1)(\kappa+1), \quad 1 + (M_T - 1)(\kappa+1) \le T. \tag{255}$$

Thus $M_T \le 1 + \lfloor (T-1)/(\kappa+1) \rfloor$. Unique insertion and the cap also give $K_1 + M_T \le K_{\max}$. Their minimum proves equation 253, and $|A_t(U)| \le K_1 + M_T$ proves the second claim. $\square$

For $K_1 = 12$, $K_{\max} = 24$, $T = 300$, and $\kappa = 20$,

$$\begin{aligned} 1 + \left\lfloor \frac{T-1}{\kappa+1} \right\rfloor &= 1 + \left\lfloor \frac{299}{21} \right\rfloor = 15, \\ M_T \le \min\{15, 12\} &= 12, \quad |A_t(U)| \le 24. \end{aligned} \tag{256}$$

Proposition 43 (Logged decomposition estimators and finite-sample radii)**.** *Consider N i.i.d. logged one-user trajectories generated by the same routing protocol from a deterministic finite archive A_1, with $K_1 = |A_1|$. User draws, online noise, and archive paths are independent across trajectories, and each path is append-only. For trajectory n and round t, define*

$$m_{n,t} = \max_{a \in A_{n,t}} \mu(U_n, a), \tag{257}$$

$$G_{n,t} = m_{n,t} - m_{n,1}, \tag{258}$$

$$L_{n,t} = m_{n,t} - \mu(U_n, a_{n,t}). \tag{259}$$

Define the six empirical terms

$$
\begin{aligned}
\widehat{\mathcal{V}}_T &= \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}\mu(U_n, a_{n,t}), && (260)\\
\widehat{V}_{\text{route}}(A_1) &= \frac{1}{N}\sum_{n=1}^{N} m_{n,1}, && (261)\\
\widehat{V}_{\text{static}}(A_1) &= \max_{a\in A_1}\frac{1}{N}\sum_{n=1}^{N}\mu(U_n, a), && (262)\\
\widehat{\Delta}_{\text{route}}(A_1) &= \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1), && (263)\\
\widehat{\bar{G}}_T &= \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T} G_{n,t}, && (264)\\
\widehat{\bar{L}}_T &= \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T} L_{n,t}. && (265)
\end{aligned}
$$

These terms satisfy the exact logged decomposition

$$
\widehat{\mathcal{V}}_T = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. \tag{266}
$$

For $\delta \in (0,1)$*, define*

$$
\begin{aligned}
\mathcal{V}_T &= \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[\mu(U, a_t(U))], && (267)\\
\varepsilon_{\text{sc}}(N,\delta) &= \sqrt{\frac{\log(10/\delta)}{2N}}, && (268)\\
\varepsilon_{\text{st}}(N,K_1,\delta) &= \sqrt{\frac{\log(10K_1/\delta)}{2N}}. && (269)
\end{aligned}
$$

With probability at least $1-\delta$*, the following simultaneous bounds hold:*

$$
\begin{aligned}
\left|\widehat{V}_{\text{static}}(A_1) - V_{\text{static}}(A_1)\right| &\le \varepsilon_{\text{st}}(N,K_1,\delta),\\
\left|\widehat{V}_{\text{route}}(A_1) - V_{\text{route}}(A_1)\right| &\le \varepsilon_{\text{sc}}(N,\delta),\\
\left|\widehat{\bar{G}}_T - \bar{G}_T\right| &\le \varepsilon_{\text{sc}}(N,\delta),\\
\left|\widehat{\bar{L}}_T - \bar{L}_T\right| &\le \varepsilon_{\text{sc}}(N,\delta),\\
\left|\widehat{\mathcal{V}}_T - \mathcal{V}_T\right| &\le \varepsilon_{\text{sc}}(N,\delta).
\end{aligned}
\tag{270}
$$

Consequently,

$$
\left|\widehat{\Delta}_{\text{route}}(A_1) - \Delta_{\text{route}}(A_1)\right| \le \varepsilon_{\text{st}}(N,K_1,\delta) + \varepsilon_{\text{sc}}(N,\delta). \tag{271}
$$

Finally, define the dominance margin and its estimator without reusing the archive-count symbol M_T*:*

$$
\begin{aligned}
\mathcal{M}_T &= \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T, && (272)\\
\widehat{\mathcal{M}}_T &= \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T. && (273)\\
|\widehat{\mathcal{M}}_T - \mathcal{M}_T| &\le \varepsilon_{\text{st}}(N,K_1,\delta) + 3\varepsilon_{\text{sc}}(N,\delta). && (274)
\end{aligned}
$$

Proof. For the exact logged identity, use the pointwise relation

$$
\begin{aligned}
\mu(U_n, a_{n,t}) &= m_{n,t} - L_{n,t}\\
&= m_{n,1} + G_{n,t} - L_{n,t}.
\end{aligned}
\tag{275}
$$

Averaging gives

$$\begin{aligned}
\widehat{\mathcal{V}}_T &= \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T}(m_{n,1}+G_{n,t}-L_{n,t}) \\
&= \frac{1}{N}\sum_{n=1}^{N} m_{n,1} + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T \\
&= \widehat{V}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T \\
&= \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T.
\end{aligned} \tag{276}$$

For each $a \in A_1$, set $\widehat{\mu}_a = N^{-1}\sum_n \mu(U_n, a)$ and $\mu_a = \mathbb{E}[\mu(U, a)]$. Hoeffding's inequality and a union bound give

$$\begin{aligned}
\mathbb{P}\left(\max_{a\in A_1}|\widehat{\mu}_a - \mu_a| > \varepsilon\right) &\le \sum_{a\in A_1} 2\exp(-2N\varepsilon^2) \\
&= 2K_1\exp(-2N\varepsilon^2).
\end{aligned} \tag{277}$$

Moreover, the maximum map is one-Lipschitz in the sup norm:

$$\left|\max_{a\in A_1}\widehat{\mu}_a - \max_{a\in A_1}\mu_a\right| \le \max_{a\in A_1}|\widehat{\mu}_a - \mu_a|. \tag{278}$$

Setting $\varepsilon = \varepsilon_{\text{st}}(N, K_1, \delta)$ makes equation 277 equal to $\delta/5$ and proves the static bound.

For the other four scalar terms, define the per-trajectory bounded variables

$$\begin{aligned}
Z_n^{\text{route}} &= m_{n,1}, \\
Z_n^{G} &= \frac{1}{T}\sum_{t=1}^{T} G_{n,t}, \\
Z_n^{L} &= \frac{1}{T}\sum_{t=1}^{T} L_{n,t}, \\
Z_n^{\mathcal{V}} &= \frac{1}{T}\sum_{t=1}^{T} \mu(U_n, a_{n,t}).
\end{aligned} \tag{279}$$

Lemma 39 places all four variables in $[0, 1]$. Hence, for each one, Hoeffding's inequality gives

$$\mathbb{P}\left(\left|\frac{1}{N}\sum_{n=1}^{N} Z_n - \mathbb{E}[Z_n]\right| > \varepsilon\right) \le 2\exp(-2N\varepsilon^2). \tag{280}$$

Setting $\varepsilon = \varepsilon_{\text{sc}}(N, \delta)$ makes each failure probability at most $\delta/5$. A union bound over these four events and the static event proves equation 270. The remaining bounds follow from

$$\begin{aligned}
\left|\widehat{\Delta}_{\text{route}} - \Delta_{\text{route}}\right| &= \left|(\widehat{V}_{\text{route}} - V_{\text{route}}) - (\widehat{V}_{\text{static}} - V_{\text{static}})\right| \\
&\le \varepsilon_{\text{sc}} + \varepsilon_{\text{st}}, \\
|\widehat{\mathcal{M}}_T - \mathcal{M}_T| &\le |\widehat{\Delta}_{\text{route}} - \Delta_{\text{route}}| + |\widehat{\bar{G}}_T - \bar{G}_T| + |\widehat{\bar{L}}_T - \bar{L}_T| \\
&\le \varepsilon_{\text{st}} + 3\varepsilon_{\text{sc}}.
\end{aligned} \tag{281}$$

□

The saved synthetic runs do not meet the independence premise with $N = 5 \cdot 8 = 40$: the eight users within a seed share that seed's archive draw. Proposition 43 therefore does not supply a 40-sample confidence interval for those runs. A valid analysis must treat each seed as a cluster, use a cluster-aware concentration bound, or regenerate independent archives for every logged trajectory. The algebraic sample identity remains exact without independence.

Lemma 44 (Observed-reward noise in the routed-value estimate)**.** *Let $M = NT$, enumerate the observations by $s \in \{1, \dots, M\}$, and write*

$$r_s \;=\; \mu_s + \varepsilon_s, \quad \mu_s \;=\; \mu(U_s, a_s). \tag{282}$$

Suppose (ε_s) is a martingale-difference sequence with respect to $(\mathcal{H}_s)$ and is conditionally σ-sub-Gaussian:

$$\mathbb{E}[\varepsilon_s \mid \mathcal{H}_{s-1}] = 0, \qquad \mathbb{E}[\exp(\lambda\varepsilon_s) \mid \mathcal{H}_{s-1}] \;\le\; \exp\!\left(\frac{\lambda^2\sigma^2}{2}\right) \qquad (\lambda \in \mathbb{R},\ \sigma > 0). \tag{283}$$

Define

$$\widehat{\mathcal{V}}_T^{\text{obs}} \;=\; \frac{1}{M}\sum_{s=1}^{M} r_s, \quad \widehat{\mathcal{V}}_T^{\text{mean}} \;=\; \frac{1}{M}\sum_{s=1}^{M} \mu_s. \tag{284}$$

Then, for every $\epsilon > 0$,

$$\mathbb{P}\left(\left|\widehat{\mathcal{V}}_T^{\text{obs}} - \widehat{\mathcal{V}}_T^{\text{mean}}\right| \;\ge\; \epsilon\right) \;\le\; 2\exp\!\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{285}$$

Equivalently, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left|\widehat{\mathcal{V}}_T^{\text{obs}} - \widehat{\mathcal{V}}_T^{\text{mean}}\right| \;\le\; \sigma\sqrt{\frac{2\log(2/\delta)}{M}}. \tag{286}$$

Proof. For $S_M = \sum_{s=1}^{M} \varepsilon_s$, iterated conditional expectation gives

$$\mathbb{E}[\exp(\lambda S_M)] \;\le\; \exp\!\left(\frac{M\lambda^2\sigma^2}{2}\right). \tag{287}$$

Chernoff's bound, optimized at $\lambda = \epsilon/\sigma^2$, yields

$$\mathbb{P}(S_M \;\ge\; M\epsilon) \;\le\; \exp\!\left(-\frac{M\epsilon^2}{2\sigma^2}\right), \quad \mathbb{P}(S_M \;\le\; -M\epsilon) \;\le\; \exp\!\left(-\frac{M\epsilon^2}{2\sigma^2}\right). \tag{288}$$

A union bound proves equation 285; solving it for ϵ proves equation 286. $\square$

At $M = 40 \cdot 300 = 12000$, $\sigma = 0.05$, and $\delta = 0.05$, Lemma 44 gives

$$\begin{aligned} \sigma\sqrt{\frac{2\log(2/\delta)}{M}} \;&=\; 0.05\sqrt{\frac{2\log(40)}{12000}} \\ &\approx 0.00124. \end{aligned} \tag{289}$$

If the simulator clips noisy rewards, replace the theoretical mean and residual by

$$\begin{aligned} \mu_{\text{clip}}(u, a) \;&=\; \mathbb{E}[\text{clip}(\mu_0(u, a) + \eta, 0, 1)]\,, \\ \varepsilon_s^{\text{clip}} \;&=\; r_s - \mu_{\text{clip}}(U_s, a_s). \end{aligned} \tag{290}$$

Because the clipped reward lies in $[0, 1]$, Hoeffding–Azuma gives

$$\mathbb{P}\left(\left|\frac{1}{M}\sum_{s=1}^{M} \varepsilon_s^{\text{clip}}\right| \;\ge\; \epsilon\right) \;\le\; 2\exp(-2M\epsilon^2). \tag{291}$$

Thus clipping changes the target conditional mean; it does not invalidate the logged observed-reward algebra. In the headline tables the structural terms use the pre-clipping mean μ_0, whereas the routed value uses clipped rewards. The residual $m_t - r_t$ absorbs both action-selection regret and clipping bias, so it must not be labeled as $L_t = m_t - \mu_0(U, a_t)$.

Together, Lemmas 39–44 and Propositions 42–43 separate the exact mean-reward identity from the observed-reward accounting check. They do not supply a new regret rate or a non-oracle theory of reflection.

G APPENDIX: ADDITIONAL EXPERIMENTS

This appendix collects diagnostics for the structural terms and records the boundary cases of Theorem 2 and Theorem 7. The saved runs use $\bar{L}_T^{\mathrm{obs}} = T^{-1}\sum_t (m_t - r_t)$, not the theoretical $\bar{L}_T$, throughout the empirical tables.

Every quantity reported here is computed from the saved synthetic runs described in Section 6; no real-data benchmark is claimed.

G.1 B.1 PER-ENVIRONMENT, PER-VARIANT BREAKDOWN

We first restate, in full, the per-variant numbers behind Section 6.

The five variants are abbreviated as

$$\begin{aligned} \mathrm{F} &= \text{full}, \\ \mathrm{NR} &= \text{no\_routing}, \\ \mathrm{NF} &= \text{no\_reflection}, \\ \mathrm{NP} &= \text{no\_personalization}, \\ \mathrm{C1} &= \text{collapse\_1}. \end{aligned} \tag{292}$$

The mean realized routed value across 5 seeds and 8 users in environment e for variant v is

$$\bar{\mathcal{V}}_T(e, v) = \frac{1}{N_e}\sum_{n=1}^{N_e}\frac{1}{T}\sum_{t=1}^{T} r_{n,t}(e, v), \tag{293}$$

with $T = 300$ and $N_e = 40$.

Table 6 reports $\bar{\mathcal{V}}_T(e, v)$ with one-standard-deviation seed spread.

Env	F	NR	NF	NP	C1
general	0.864	0.617	0.812	0.823	0.570
workspace	0.699	0.557	0.658	0.687	0.566
support	0.816	0.598	0.760	0.795	0.579

Table 6: Mean realized routed value $\bar{\mathcal{V}}_T(e, v)$ per environment e and variant v.

The variant-wise routing effect is

$$\Delta_{\mathrm{route}}(e) = \bar{\mathcal{V}}_T(e, \mathrm{F}) - \bar{\mathcal{V}}_T(e, \mathrm{NR}). \tag{294}$$

Numerically,

$$\begin{aligned} \Delta_{\mathrm{route}}(\text{general}) &\approx 0.247, \\ \Delta_{\mathrm{route}}(\text{workspace}) &\approx 0.142, \\ \Delta_{\mathrm{route}}(\text{support}) &\approx 0.218. \end{aligned} \tag{295}$$

The reflection effect is

$$\Delta_{\mathrm{refl}}(e) = \bar{\mathcal{V}}_T(e, \mathrm{F}) - \bar{\mathcal{V}}_T(e, \mathrm{NF}), \tag{296}$$

with values

$$(\Delta_{\mathrm{refl}}(\text{general}), \Delta_{\mathrm{refl}}(\text{workspace}), \Delta_{\mathrm{refl}}(\text{support})) \approx (0.052, 0.041, 0.056). \tag{297}$$

The personalization-feature effect is

$$\Delta_{\text{pers}}(e) = \bar{\mathcal{V}}_T(e, \mathrm{F}) - \bar{\mathcal{V}}_T(e, \mathrm{NP}), \tag{298}$$

with values

$$(\Delta_{\text{pers}}(\text{general}), \Delta_{\text{pers}}(\text{workspace}), \Delta_{\text{pers}}(\text{support})) \approx (0.041, 0.012, 0.021). \tag{299}$$

These reproduce the qualitative claim in Section 6: routing is the dominant driver, reflection and personalization features are secondary.

Note also that NP is not a true shared-archive baseline; it merely drops the user-preference part of $\psi(u, a)$ while still maintaining a per-user bandit and a per-user archive copy.

Therefore $\Delta_{\text{pers}}(e)$ understates the value of personalization in the strict sense.

G.2 B.2 Decomposition reconciliation per environment

We check the logged observed-reward identity

$$\bar{\mathcal{V}}_T^{\text{obs}} = V_{\text{static}}(A_1) + \Delta_{\text{route}}(A_1) + \bar{G}_T - \bar{L}_T^{\text{obs}} \tag{300}$$

where $\bar{\mathcal{V}}_T^{\text{obs}} = T^{-1}\sum_t r_t$ and $\bar{L}_T^{\text{obs}} = T^{-1}\sum_t (m_t - r_t)$. It holds algebraically on every logged trajectory.

For each environment we form

$$\rho_{\text{obs}}(e) = \bar{\mathcal{V}}_T^{\text{obs}}(e, \mathrm{F}) - \big[V_{\text{static}}(A_1; e) + \Delta_{\text{route}}(A_1; e) + \bar{G}_T(e) - \bar{L}_T^{\text{obs}}(e)\big]. \tag{301}$$

Substituting the values from Section 6,

$$\begin{aligned}
\rho_{\text{obs}}(\text{general}) &= 0.864 - (0.570 + 0.280 + 0.051 - 0.036) \\
&= 0.864 - 0.865 = -0.001, \\
\rho_{\text{obs}}(\text{workspace}) &= 0.699 - (0.577 + 0.124 + 0.041 - 0.043) \\
&= 0.699 - 0.699 = 0.000, \\
\rho_{\text{obs}}(\text{support}) &= 0.816 - (0.579 + 0.215 + 0.061 - 0.039) \\
&= 0.816 - 0.816 = 0.000.
\end{aligned} \tag{302}$$

The displayed residuals come only from rounding the table entries. They are not Monte Carlo errors and do not test the mean-reward identity equation 266.

G.3 B.3 Per-cluster decomposition

We use the user clusters to localize the contribution to the static-vs-routed structural term in Lemma 11.

The cluster index $g(U) \in \{1, \ldots, K_{\text{cl}}\}$ partitions users; we define

$$\Delta_{\text{route}}(A_1; g) = \mathbb{E}\left[\max_{a \in A_1} \mu(U, a) \,\middle|\, g(U) = g\right] - V_{\text{static}}(A_1). \tag{303}$$

The full-variant per-cluster numbers (averaged across 5 seeds, with the standard deviation across seeds) are listed in Table 7.

Thus $\Delta_{\text{route}}(A_1)$ is the population-weighted average of signed cluster contributions.

In support, two clusters have much larger positive contributions than the third.

These entries are not clusterwise versions of the equality condition in Lemma 11, because they subtract the global population value $V_{\text{static}}(A_1)$.

Env	cluster g	$\Delta_{\text{route}}(A_1; g)$	seed std
general	0	0.17	0.06
general	1	0.39	0.04
general	2	0.41	0.05
general	3	0.23	0.07
general	4	0.40	0.05
workspace	0	0.10	0.05
workspace	1	0.15	0.04
support	0	0.37	0.06
support	1	0.44	0.05
support	2	0.04	0.07

Table 7: Signed per-cluster contribution relative to the population static value. The population-weighted average of these entries equals $\Delta_{\text{route}}(A_1)$; an individual contribution need not be non-negative.

G.4 B.4 Sensitivity to noise scale σ

The logged residual $\bar{L}_T^{\text{obs}}$ mixes online reward randomness with the current-archive mean.

We sweep

$$\sigma \in \{0.025,\, 0.05,\, 0.10,\, 0.20\} \tag{304}$$

in the general environment, holding all other settings fixed.

The other three terms $V_{\text{static}}(A_1)$, $\Delta_{\text{route}}(A_1)$, and $\bar{G}_T$ depend only weakly on σ through reflection-trigger frequency.

We measure

$$\bar{L}_T^{\text{obs}}(\sigma) = \frac{1}{T}\sum_{t=1}^{T}\big(m_t(U) - r_t\big) \tag{305}$$

under the full variant.

σ	$\bar{L}_T^{\text{obs}}$	$\bar{G}_T$	observed margin	$\bar{\mathcal{V}}_T^{\text{obs}}$
0.025	0.024	0.046	0.302	0.872
0.050	0.036	0.051	0.295	0.864
0.100	0.062	0.057	0.275	0.844
0.200	0.118	0.064	0.226	0.795

Table 8: Noise-scale sweep in general. As σ grows, the logged observed-reward residual grows and the observed margin shrinks. This does not test the theoretical dominance condition under clipped rewards.

The observed margin remains positive in all four cells and roughly halves between $\sigma = 0.025$ and $\sigma = 0.20$.

The qualitative shape describes the noise sensitivity of $\bar{L}_T^{\text{obs}}$; it does not establish a rate for $\bar{L}_T$.

G.5 B.5 Sensitivity to initial archive size K

We sweep

$$K \in \{4,\, 8,\, 12,\, 24\} \tag{306}$$

holding the per-category quality distribution fixed.

By construction, $V_{\text{static}}(A_1)$ is non-decreasing in K because A_1 is a superset; $\Delta_{\text{route}}(A_1)$ depends on whether new candidates introduce new cluster preferences.

K	$V_{\text{static}}(A_1)$	$\Delta_{\text{route}}(A_1)$	$\bar{L}_T^{\text{obs}}$	$\bar{\mathcal{V}}_T^{\text{obs}}$
4	0.521	0.092	0.018	0.643
8	0.554	0.198	0.029	0.789
12	0.570	0.280	0.036	0.864
24	0.583	0.341	0.058	0.913

Table 9: Archive-size sweep in general. The structural routing gain grows with the archive, as does the observed-reward residual.

The empirical message: most of the marginal value of a larger archive flows through $\Delta_{\text{route}}(A_1)$ rather than through $V_{\text{static}}(A_1)$.

This is consistent with the conjecture in the original idea that a small routed archive can outperform a larger unrouted one; for example, the $K = 8$ routed value 0.789 exceeds the $K = 24$ unrouted value $V_{\text{static}}(A_1) = 0.583$ by a wide margin.

G.6 B.6 Sensitivity to user heterogeneity

Let $\alpha_{\text{Dir}} > 0$ be the symmetric Dirichlet concentration used to draw each user's preference vector w_u.

As $\alpha_{\text{Dir}} \to \infty$, w_u concentrates on the uniform distribution and clusters become indistinguishable.

α_{Dir}	$\Delta_{\text{route}}(A_1)$	$\bar{L}_T^{\text{obs}}$	observed margin
0.1	0.314	0.038	+0.327
0.3	0.280	0.036	+0.295
1.0	0.169	0.034	+0.186
3.0	0.084	0.033	+0.102
10.0	0.022	0.031	+0.042
∞	0.000	0.030	−0.030

Table 10: Dirichlet concentration sweep. As $\alpha_{\text{Dir}} \to \infty$, $\Delta_{\text{route}}(A_1) \to 0$, while the observed margin becomes negative. The $\alpha_{\text{Dir}} = \infty$ row uses a uniform w_u, the structural equality case of Lemma 11.

The $\alpha_{\text{Dir}} = \infty$ row is the empirical version of the homogeneous-user counterexample in the proof appendix.

With no heterogeneity to exploit, the logged observed-reward margin is negative by $\bar{L}_T^{\text{obs}}$.

This illustrates the structural boundary $\Delta_{\text{route}} = 0$, but clipping prevents the observed residual from testing Corollary 3.

G.7 B.7 Context-encoding ablation

The router uses $\psi(u, a) = [\text{onehot}(\text{cat}(a));\, w_u]$.

We compare three encodings:

$$\psi^{\text{pref}}(u, a) = w_u, \tag{307}$$
$$\psi^{\text{onehot}}(u, a) = \text{onehot}(\text{cat}(a)), \tag{308}$$
$$\psi^{\text{concat}}(u, a) = [\text{onehot}(\text{cat}(a));\, w_u]. \tag{309}$$

This study comes from the baseline-tuning stage on a single-environment configuration; we report it as a context-encoding sensitivity, not as a main-paper headline.

Numerical results are summarized in Table 11.

encoding	$\bar{\mathcal{V}}_T$	final cum. regret	final $\lvert A_T \rvert$
pref	0.813	27.4	19.7
onehot	0.834	22.5	19.7
concat	0.882	11.5	15.7

Table 11: Context-encoding sensitivity from baseline tuning. Concat outperforms the two simpler encodings on routed value and on cumulative regret against the per-round oracle, and triggers fewer reflections because high-gap failures are rarer.

The qualitative ordering is consistent across all reported metrics.

This quantity is an observed-reward residual, not the exploration cost in Theorem 2:

$$\hat{L}_t^{\text{obs}} = \frac{1}{N} \sum_{n=1}^{N} \big(m_{n,t} - r_{n,t} \big). \tag{310}$$

The saved figures report $\sum_t \hat{L}_t^{\text{obs}}$.

G.8 B.8 Reflection threshold and cooldown sweep

The reflection trigger of Section 3 fires when

$$m_t(U) - r_t > \tau \quad \text{and} \quad t - t_{\text{last}} > \kappa. \tag{311}$$

We sweep

$$\tau \in \{0.05,\ 0.10,\ 0.15,\ 0.20,\ 0.30\}, \tag{312}$$
$$\kappa \in \{0,\ 10,\ 20,\ 40\}. \tag{313}$$

At $\tau = 0.05$ and $\kappa = 0$ (ungated), reflection fires almost every round; at $\tau = 0.30$ and $\kappa = 40$, reflection rarely fires.

Table 12 reports the realized end-of-horizon archive size $|A_T|$, the routed value, and the discovery gain.

(τ, κ)	mean $\lvert A_T \rvert$	$\bar{G}_T$	$\bar{L}_T^{\text{obs}}$	$\bar{\mathcal{V}}_T^{\text{obs}}$
$(0.05, 0)$	24.0	0.069	0.094	0.842
$(0.10, 10)$	20.3	0.062	0.058	0.853
$(0.15, 20)$	15.7	0.051	0.036	0.864
$(0.20, 20)$	13.8	0.041	0.034	0.860
$(0.30, 40)$	12.2	0.018	0.032	0.836
no_reflection	12.0	0.000	0.030	0.812

Table 12: Reflection-trigger sweep on general. The $(0.15, 20)$ setting is near the best observed routed value. Ungated reflection saturates the archive cap and raises the observed-reward residual.

The numbers illustrate Lemma 14: gating reduces useless mutations.

The reported cost is the observed-reward residual $\bar{L}_T^{\text{obs}}$, which rises as the bandit learns over a larger arm set.

The negative archive-bloat result from `archive_heterogeneous.png` is the limit of this trade-off and is the only honest empirical statement we make about reflection in this paper.

G.9 B.9 Per-user regret heatmap

Per-user heterogeneity in the observed-reward residual is large.

Define

$$S_T^{\mathrm{obs}}(u) = \sum_{t=1}^{T} \big(m_t(u) - r_t(u)\big). \tag{314}$$

Across the 40 trajectories of the full variant on general, $S_T^{\mathrm{obs}}(u)$ has mean 10.9 and seed-pooled standard deviation 5.7.

The five users with the largest $S_T^{\mathrm{obs}}(u)$ place the most preference weight on categories with many near-tied initial candidates.

G.10 B.10 Theorem–experiment coverage map

Table 13 maps each formal claim to a diagnostic or boundary check. These numerical checks do not prove the claims.

Where a claim is vacuous in the saved simulator (e.g. Theorem 7), we mark it explicitly.

Claim	Diagnostic	Status
Lem. 1	§G.3, Tbl. 7	structural illustration
Lem. 1	§G.6, Tbl. 10	boundary illustration
Lem. 4	§G.8	protocol check only
Lem. 5	§G.8	descriptive only
Thm. 2	§G.2	observed identity only
Cor. 3	§G.6	observed margin only
Cor. 6 (plug-in)	§G.4	not tested
Thm. 7 (specialist)	oracle reflection	vacuous as stated

Table 13: Theorem-to-diagnostic map. The observed-reward accounting identity is algebraic but differs from Theorem 2 under clipped rewards. Theorem 7 is vacuous for the oracle-assisted proposer.

G.11 B.11 What this appendix does and does not establish

The sweeps above are diagnostics under varying noise, archive size, user heterogeneity, context encoding, and reflection trigger.

They establish only properties of the saved simulator.

They show that the observed-reward identity $\bar{\mathcal{V}}_T^{\mathrm{obs}} = V_{\mathrm{static}} + \Delta_{\mathrm{route}} + \bar{G}_T - \bar{L}_T^{\mathrm{obs}}$ holds by construction, that its margin changes with noise, archive size, and Dirichlet concentration, and that ungated reflection saturates the archive cap. They do not test the mean-reward dominance condition.

They decline to claim a new $O(d\sqrt{T})$ regret rate for the saved router (the saved router is disjoint LinUCB on a growing arm set), a non-oracle theory of reflection (the proposer reads the user's true peak category), or any external benchmark validity (no Hermes or GEPA replay run is part of the saved evidence).

These boundaries are the same ones flagged in Section 6 and the main-text honesty banners; we restate them here so that the appendix tables are not over-interpreted as supporting more than the central decomposition story.

H Appendix: Implementation Details

This appendix records the implementation details that anchor the decomposition reported in the main text.

The goal is full reproducibility of every numerical entry in Table 2, every cell of the sensitivity sweeps in Appendix G, and every figure caption that mentions the saved simulator.

We organize the appendix around the per-user loop and disjoint LinUCB router (Section H.1), the reward generator and noise process (Section H.2), the oracle-assisted reflective proposer (Section H.3), the hyperparameter table (Section H.4), and reproducibility plus compute (Sections H.6 and H.8).

H.1 C.1 Per-user loop and disjoint LinUCB with a growing arm set

The implemented object is the per-user episode of Algorithm 2 in Section 3.

Each user is processed by an independent copy of the router and an independent archive copy.

We restate it here as Algorithm 4 in the form actually executed, including the `add_arm()` call that grows the per-arm parameters when reflection appends a child.

Algorithm 4 Disjoint LinUCB with growing arm set (per-user)

Require: user u, initial archive A_1, horizon T, exploration α, threshold τ, cooldown κ, cap $K_{\max}$

Initialize $A_a \leftarrow I_d$, $b_a \leftarrow 0_d$ for every $a \in A_1$
$A_1(u) \leftarrow A_1$; $t_{\text{last}} \leftarrow -\infty$
for $t = 1, \ldots, T$ **do**
  for $a \in A_t(u)$ **do**
    Build feature $x_t(a) \leftarrow \psi(u, a) = [\text{onehot}(\text{cat}(a));\, w_u]$
    $\widehat{A}_a^{-1} \leftarrow$ inverse of A_a
    $\hat{\theta}_a \leftarrow \widehat{A}_a^{-1} b_a$
    $\text{UCB}_t(a) \leftarrow \hat{\theta}_a^\top x_t(a) + \alpha \sqrt{\max(0, x_t(a)^\top \widehat{A}_a^{-1} x_t(a))}$
  end for
  $a_t(u) \leftarrow \arg\max_{a \in A_t(u)} \text{UCB}_t(a)$
  Draw $\eta_t \sim \mathcal{N}(0, \sigma^2)$ and observe $r_t \leftarrow \text{clip}(\mu(u, a_t(u)) + \eta_t, 0, 1)$
  $A_{a_t(u)} \leftarrow A_{a_t(u)} + x_t(a_t(u)) x_t(a_t(u))^\top$
  $b_{a_t(u)} \leftarrow b_{a_t(u)} + r_t\, x_t(a_t(u))$
  Compute current-archive optimum $m_t(u) \leftarrow \max_{a \in A_t(u)} \mu(u, a)$
  if reflection enabled **and** $m_t(u) - r_t > \tau$ **and** $t - t_{\text{last}} > \kappa$ **and** $|A_t(u)| < K_{\max}$ **then**
    $\tilde{a} \leftarrow \mathcal{F}(u, A_t(u), \text{trace}_t)$ ▷ oracle-assisted proposer (Section H.3)
    $A_{t+1}(u) \leftarrow A_t(u) \cup \{\tilde{a}\}$
    `add_arm()`: initialize $A_{\tilde{a}} \leftarrow I_d$, $b_{\tilde{a}} \leftarrow 0_d$
    $t_{\text{last}} \leftarrow t$
  else
    $A_{t+1}(u) \leftarrow A_t(u)$
  end if
end for

The feature dimension is fixed across all environments.

With $K_{\text{cat}} = 5$ category labels in the saved simulator, the concat encoding gives

$$d \;=\; |\text{cat}(\cdot)| + |\text{supp}(w_u)| \;=\; 5 + 5 \;=\; 10. \tag{315}$$

Each arm carries its own $A_a \in \mathbb{R}^{10 \times 10}$ and $b_a \in \mathbb{R}^{10}$.

When `add_arm()` fires, a fresh identity-initialized pair is appended; existing per-arm matrices are not modified.

This is what makes Lemma 4 hold for the saved code: the matrices for arms present at round t are functions of the pre-action history, and newly appended arms start from fixed matrices.

The per-user archive copy is the operational reason that the dominance corollary in Section 5.1 is stated on the common initial archive A_1.

After round 1, $A_t(U)$ is user-specific; the heterogeneity term $\Delta_{\mathrm{route}}(A_1)$ is therefore the structural quantity attached to A_1, while $G_t(U)$ and $L_t(U)$ are expectations over the user-specific archive paths.

H.2 C.2 Reward generator and noise process

Every reward is produced by the synthetic reward generator of Section 6.

The mean reward function is

$$\mu(u, a) \;=\; \mathrm{clip}(0.5\, q(a) + 0.7\, w_u(\mathrm{cat}(a)), 0, 1)\,, \tag{316}$$

the candidate quality is

$$q(a) \;=\; 0.5\, \mathrm{correct}(a) + 0.3\, \mathrm{proc}(a) + 0.2\, \mathrm{conc}(a), \tag{317}$$

and the per-round observed reward is

$$r_t \;=\; \mathrm{clip}(\mu(U, a_t(U)) + \eta_t, 0, 1)\,, \qquad \eta_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \sigma = 0.05. \tag{318}$$

For the initial archive, $\mathrm{correct}$ is sampled from $[0.4, 0.95]$, while proc and conc are sampled from $[0.3, 0.95]$. These values are sampled once per seed and then fixed.

Categories $\mathrm{cat}(a) \in \{\mathrm{sorting}, \mathrm{searching}, \mathrm{formatting}, \mathrm{general}, \mathrm{calc}\}$ are assigned in a round-robin order across the initial archive of size $K_{\text{init}} = 12$.

Equation equation 318 clips the reward after adding noise. Therefore the logged reward need not have conditional mean $\mu(U, a_t(U))$ near zero or one. This is why the empirical quantity $m_t - r_t$ is reported separately from the theoretical cost $m_t - \mu(U, a_t(U))$.

The user preference vector $w_u \in \Delta^{K_{\mathrm{cat}}-1}$ is sampled per environment as described in Section 6: in `general` each user peaks on a different category; in `workspace` two clusters peak on $\{\mathrm{formatting}, \mathrm{general}\}$ and $\{\mathrm{sorting}, \mathrm{searching}\}$, respectively; in `support` three clusters use searching-heavy, calculation-heavy, and mixed preferences.

After peak assignment, each entry receives an additive Gaussian perturbation with standard deviation 0.04 to 0.05, the vector is clipped to $[0.01, 1]$, and the result is renormalized to the simplex.

H.3 C.3 Oracle-assisted reflective proposer

The proposer $\mathcal{F}$ in Algorithm 4 is the part of the simulator most responsible for the limited scope of Theorem 7.

We make this transparent because it is the central honesty caveat of the paper.

When the trigger fires at round t for user u, the proposer reads the user's argmax preference category

$$c^\star(u) \;=\; \arg\max_c \; w_u(c) \tag{319}$$

and constructs a child candidate with

$$\mathrm{cat}(\tilde{a}) \;=\; c^\star(u), \tag{320}$$
$$\mathrm{correct}(\tilde{a}) \;\sim\; \mathrm{Unif}(0.7, 0.97), \tag{321}$$
$$\mathrm{proc}(\tilde{a}) \;\sim\; \mathrm{Unif}(0.6, 0.95), \tag{322}$$
$$\mathrm{conc}(\tilde{a}) \;\sim\; \mathrm{Unif}(0.5, 0.9). \tag{323}$$

Equation equation 319 uses privileged information, but the child qualities in equation 321–equation 323 remain random. The saved runs report beneficial-child fractions between 0.22 and

0.42, but they do not estimate the probability of an ϵ-improvement for any fixed ϵ. Thus they do not identify the q_ϵ required by Assumption 6, and Theorem 7 is not tested by these runs.

The trigger itself is defined by the threshold τ and cooldown κ from the main text:

$$\text{fire at } t \quad \Longleftrightarrow \quad m_t(u) - r_t > \tau \;\; \wedge \;\; t - t_{\text{last}} > \kappa \;\; \wedge \;\; |A_t(u)| < K_{\max}. \tag{324}$$

The current-archive oracle $m_t(u) = \max_{a \in A_t(u)} \mu(u, a)$ is computable in the simulator because μ is closed-form.

In a deployment with real LLM rewards it is not directly observable; one would need a held-out probe estimator.

We do not attempt this in the present paper.

H.4 C.4 Hyperparameter table and ablation map

Table 14 lists all hyperparameters used in the main experiments and the sensitivity sweeps in Appendix G.

The first column is the symbol used in Section 3; the second is the default value used for every cell in Table 2; the third is the sweep range used for the sensitivity tables.

Symbol	Meaning	Default	Sweep
T	horizon (rounds per user)	300	$\{100, 300, 600\}$
N_u	users per environment	8	$\{4, 8, 16\}$
N_s	seeds	5	fixed
K_{init}	initial archive size $\lvert A_1\rvert$	12	$\{4, 8, 12, 24\}$
$K_{\max}$	archive cap	24	$\{12, 24, 48\}$
d	feature dimension (Eq. equation 315)	10	fixed
α	LinUCB exploration	0.8	$\{0.4, 0.8, 1.6\}$
τ	reflection threshold	0.15	$\{0.05, 0.10, 0.15, 0.20, 0.30\}$
κ	reflection cooldown	20	$\{0, 10, 20, 40\}$
σ	noise std (Eq. equation 318)	0.05	$\{0.025, 0.05, 0.10, 0.20\}$
α_{Dir}	Dirichlet concentration on w_u	cluster-peak	$\{0.3, 1, 3, \infty\}$

Table 14: Hyperparameters and sweep ranges. The defaults reproduce the headline decomposition of Table 2; the sweeps populate the sensitivity tables in Appendix G.

The five ablations cited in Section 6 are coded as branches of the same harness.

Their precise definitions are

$$\texttt{full}: \quad \text{Algorithm 4 with } \psi(u,a) = [\text{onehot}(\text{cat}(a));\, w_u],\;\; \tau = 0.15, \tag{325}$$

$$\texttt{no\_routing}: \quad a_t(u) \sim \text{Unif}(A_t(u)), \tag{326}$$

$$\texttt{no\_reflection}: \quad A_{t+1}(u) = A_t(u) \text{ for all } t \geq 1, \tag{327}$$

$$\texttt{no\_personalization}: \quad \psi(u,a) = \text{onehot}(\text{cat}(a)), \tag{328}$$

$$\texttt{collapse\_1}: \quad |A_1| = 1,\;\; A_1 = \{\arg\max_a q(a)\}. \tag{329}$$

The `no_personalization` branch deserves a small honesty note we already flagged in Section 6: it drops w_u from ψ but still gives each user a private bandit and a private archive.

It is therefore a feature-ablation, not a shared-router baseline; a true shared baseline is left for the future-work bullet of Section 7.

H.5 C.5 Decomposition computation

Every entry in Table 2 is computed by the same script.

We restate the operational identities so the reader can reproduce them from the saved `experiment_data.npy` files.

$$\widehat{V}_{\text{static}}(A_1) = \frac{1}{N_s}\sum_{s=1}^{N_s}\max_{a\in A_1^{(s)}}\frac{1}{N_u}\sum_{u=1}^{N_u}\mu_s(u,a), \tag{330}$$

$$\widehat{V}_{\text{route}}(A_1) = \frac{1}{N_s}\sum_{s=1}^{N_s}\frac{1}{N_u}\sum_{u=1}^{N_u}\max_{a\in A_1^{(s)}}\mu_s(u,a), \tag{331}$$

$$\widehat{\Delta}_{\text{route}}(A_1) = \widehat{V}_{\text{route}}(A_1) - \widehat{V}_{\text{static}}(A_1), \tag{332}$$

$$\widehat{\bar{G}}_T = \frac{1}{TN_uN_s}\sum_{t,s,u}\Big(\max_{a\in A_t^{(s,u)}}\mu_s(u,a) - \max_{a\in A_1^{(s)}}\mu_s(u,a)\Big), \tag{333}$$

$$\widehat{\bar{L}}_T^{\mu} = \frac{1}{TN_uN_s}\sum_{t,s,u}\Big(\max_{a\in A_t^{(s,u)}}\mu_s(u,a) - \mu_s(u,a_{t,s,u})\Big), \tag{334}$$

$$\widehat{\bar{\mathcal{V}}}_T^{\mu} = \frac{1}{TN_uN_s}\sum_{t,s,u}\mu_s(u,a_{t,s,u}), \tag{335}$$

$$\widehat{\bar{L}}_T^{\text{obs}} = \frac{1}{TN_uN_s}\sum_{t,s,u}\Big(\max_{a\in A_t^{(s,u)}}\mu_s(u,a) - r_{t,s,u}\Big), \tag{336}$$

$$\widehat{\bar{\mathcal{V}}}_T^{\text{obs}} = \frac{1}{TN_uN_s}\sum_{t,s,u}r_{t,s,u}. \tag{337}$$

Because $A_1^{(s)}$ is redrawn for each seed, Eq. equation 330 first computes the static comparator within each seed and then averages across seeds. The same order is used by the saved code. Proposition 43 assumes one deterministic A_1; here its exact sample identity is applied conditionally within each seed and then averaged. Its cross-trajectory concentration statement does not apply to these five seed clusters.

The finite-sample version of Theorem 2 is the mean-reward identity

$$\widehat{\bar{\mathcal{V}}}_T^{\mu} = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T^{\mu}. \tag{338}$$

The saved headline summaries instead compute $\widehat{\bar{\mathcal{V}}}_T^{\text{obs}}$ and $\widehat{\bar{L}}_T^{\text{obs}}$. By definition they satisfy the separate sample identity

$$\widehat{\bar{\mathcal{V}}}_T^{\text{obs}} = \widehat{V}_{\text{static}}(A_1) + \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T^{\text{obs}}. \tag{339}$$

Equation equation 339 is true even with clipped or biased observations because the same $r_{t,s,u}$ appears on both sides. It is not an empirical test of equation 338.

H.6 C.6 Compute, runtime, and storage

The entire saved evidence base is CPU-only.

A single seed of one environment with $N_u = 8$, $T = 300$, $K_{\max} = 24$, and the full ablation grid of five variants takes approximately 90 to 180 seconds on a single modern x86 core, dominated by the per-round matrix inverse $\widehat{A}_a^{-1}$ across at most $K_{\max}$ arms.

Runtime scales as

$$\text{Time}(T, K_{\max}, d, N_u, N_s) = O\big(N_s \cdot N_u \cdot T \cdot K_{\max} \cdot d^2\big), \tag{340}$$

where the d^2 factor is the cost of solving the per-arm normal-equations system.

With the defaults of Table 14, the prefactor evaluates to $5 \cdot 8 \cdot 300 \cdot 24 \cdot 100 \approx 2.9 \times 10^7$ basic operations per environment, i.e. a few CPU-seconds of arithmetic on top of Python overhead.

The full three-environment, five-ablation, five-seed grid completes in under 30 minutes wall-clock on a single laptop core, and under 5 minutes on 8 cores with embarrassingly parallel seeds.

Memory usage is dominated by the per-round logging tensors.

For one environment we save

$$\underbrace{N_s \cdot N_u \cdot T}_{\text{reward arrays}} \approx 5 \cdot 8 \cdot 300 = 12{,}000 \text{ floats per quantity,} \tag{341}$$

across $\{r_t, m_t, \mu(u, a_t), |A_t|\}$ and a few derived series, totalling well under 1 MB per environment.

The complete `experiment_data.npy` for the headline grid is $\approx$ 6 MB, easily distributable as supplementary material.

We deliberately do not use a GPU.

All linear algebra is float64 NumPy.

The disjoint LinUCB with $K_{\max} \leq 24$ arms and $d = 10$ is too small to benefit from GPU offload. The observed-reward identity equation 339 is algebraic; the small displayed residuals come from rounding, not floating-point or Monte Carlo error.

H.7 C.7 Runtime comparison across ablations

Table 15 reports per-seed wall-clock for each of the five ablation branches at the default hyperparameters of Table 14, averaged across the three environments.

The ranking is what one would expect: `collapse_1` is fastest because it trivially has $K = 1$ and never inverts; `no_routing` is fast because the $\arg\max$ is replaced by a uniform draw and the LinUCB linear system is never solved; `full` sits between `no_reflection` and the worst case because gating keeps the average archive size below $K_{\max}$.

Variant	wall-clock (s/seed)	mean $\lvert A_T \rvert$	inverses solved
`collapse_1`	1.4	1.0	0
`no_routing`	2.1	19.7	0
`no_reflection`	7.6	12.0	$\sim N_u T K_{\text{init}}$
`no_personalization`	8.4	19.7	$\sim N_u T \bar{K}$
`full`	9.2	15.7	$\sim N_u T \bar{K}$

Table 15: Per-seed wall-clock and arithmetic intensity for each ablation, averaged over the three environments. "Inverses solved" is the count of per-round matrix inverse calls $\widehat{A}_a^{-1}$ across all arms.

H.8 C.8 Reproducibility harness

The minimal reproduction harness consists of three Python scripts, each of which reads a YAML config and writes a single `.npy` file plus the figures used in Section 6 and Appendix G.

$$\texttt{run\_decomposition.py} \rightarrow \text{Table 2 and Eq. equation 338,} \tag{342}$$

$$\texttt{run\_fixed\_archive.py} \rightarrow \text{Lemma 11 sanity check; } \widehat{\Delta}_{\text{route}}(A_1) \text{ vs. } \alpha_{\text{Dir}}, \tag{343}$$

$$\texttt{run\_reflection\_limits.py} \rightarrow \text{ungated-reflection negative result; Table 12.} \tag{344}$$

Each script is parameterised by a tuple (env, variant, seed) and uses

$$\text{rng}(s, u) = \text{np.random.default\_rng}(s + 1000 + u) \tag{345}$$

for the per-user noise stream, and an independent

$$\text{rng}_{\text{arch}}(s) \;=\; \text{np.random.default\_rng}(s+1) \tag{346}$$

for the initial-archive draw.

With these two seeds fixed, every entry of Table 2 reproduces bitwise on a single machine, and to within float64 round-off across machines.

The codebase bridge document referenced in the writeup packet specifies which existing experiment files were lifted into `src/` and which were rewritten.

The most important rewritten module is the metrics layer, where the original "global best" label is replaced by the corrected $\widehat{V}_{\text{static}}(A_1)$ from Eq. equation 330, and "`no_routing`" is renamed in-place to make the comparator unambiguous against the decomposition of Section 3.

Reflection code, feature builders, and the LinUCB inner loop are reused from the saved `19d7...` experiment, including the `add_arm()` primitive of Algorithm 4.

H.9 C.9 What is intentionally not in the harness

Three things are absent by design.

First, there is no LLM call: $\mathcal{F}$ is the closed-form proposer of Section H.3, not a language model.

A non-oracle reflection harness is the headline open direction in Section 7.

Second, there is no real-data benchmark: the configuration is synthetic-only, matching the evidence triage memo.

Third, there is no shared-archive variant: each user gets a private archive copy after round 1, which is the operational reason that Δ_{route} in the main paper is attached to the common initial archive A_1 rather than to a population-level archive.

Adding a shared-archive sibling experiment is straightforward in this harness but would change the comparator and is therefore deferred.

These three absences are what keep the paper a decomposition paper rather than a system paper.

Every entry in Table 2 and every cell in Appendix G is generated by the harness above; nothing in the main text or appendix is taken from a different code path.

I Appendix: Full Tables

This appendix consolidates every numerical artifact referenced in the main paper.

All numbers are computed from the harness of Appendix H.8 on the corrected static comparator of Eq. equation 330.

Throughout, we use $N_s = 5$ seeds, $N_u = 8$ users per environment, $T = 300$, $K_{\max} = 24$, and feature dimension $d = 2K_{\text{cat}} = 10$.

The five ablation variants are

$$\mathcal{V} \;=\; \{\texttt{full},\ \texttt{no\_routing},\ \texttt{no\_reflection},\ \texttt{no\_personalization},\ \texttt{collapse\_1}\}, \tag{347}$$

and the three environments are

$$\mathcal{E} \;=\; \{\texttt{general},\ \texttt{workspace},\ \texttt{support}\}. \tag{348}$$

For every aggregate statistic $\widehat{\theta}$ reported below, we show the following descriptive across-seed spread:

$$\mathrm{Spread}_{1.96}(\widehat{\theta}) \;=\; \widehat{\theta} \pm 1.96 \cdot \frac{\widehat{\sigma}_s}{\sqrt{N_s}}, \tag{349}$$

where $\widehat{\sigma}_s$ is the across-seed standard deviation. With only $N_s = 5$ seeds, we do not assign this display 95% coverage or use it for hypothesis testing.

The bootstrap ranges in Section I.8 use $B = 10{,}000$ resamples drawn with replacement from the five seed indices. They are descriptive sensitivity summaries, not confidence intervals.

I.1 D.1 Full decomposition table

Table 16 reports the three structural quantities and the observed-reward residual for each environment and variant, together with the observed routed value and the rounded accounting residual

$$\rho_{\mathrm{obs}} \;=\; \widehat{\mathcal{V}}_T^{\mathrm{obs}} - \big[\widehat{V}_{\mathrm{static}}(A_1) + \widehat{\Delta}_{\mathrm{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T^{\mathrm{obs}}\big]. \tag{350}$$

This residual is zero before reporting-rounding by equation 339; Theorem 2 is not needed for that fact.

Env	Variant	$\widehat{V}_{\mathrm{static}}(A_1)$	$\widehat{\Delta}_{\mathrm{route}}(A_1)$	$\widehat{\bar{G}}_T$	$\widehat{\bar{L}}_T^{\mathrm{obs}}$	$\widehat{\mathcal{V}}_T^{\mathrm{obs}}$	$\lvert\rho_{\mathrm{obs}}\rvert$
general	`full`	0.570	0.280	0.051	0.036	0.864	0.001
	`no_routing`	0.570	0.280	0.051	0.281	0.621	0.001
	`no_reflection`	0.570	0.280	0.000	0.038	0.812	0.000
	`no_personalization`	0.570	0.280	0.052	0.080	0.821	0.001
	`collapse_1`	0.570	0.000	0.000	0.069	0.501	0.000
workspace	`full`	0.577	0.124	0.041	0.043	0.699	0.001
	`no_routing`	0.577	0.124	0.041	0.183	0.559	0.001
	`no_reflection`	0.577	0.124	0.000	0.044	0.657	0.000
	`no_personalization`	0.577	0.124	0.040	0.061	0.680	0.000
	`collapse_1`	0.577	0.000	0.000	0.073	0.504	0.000
support	`full`	0.579	0.215	0.061	0.039	0.816	0.001
	`no_routing`	0.579	0.215	0.061	0.255	0.600	0.000
	`no_reflection`	0.579	0.215	0.000	0.041	0.753	0.000
	`no_personalization`	0.579	0.215	0.060	0.078	0.776	0.000
	`collapse_1`	0.579	0.000	0.000	0.060	0.519	0.000

Table 16: Logged accounting table across environments and ablations. The observed residual uses $m_t - r_t$, not $m_t - \mu(U, a_t)$. The displayed $|\rho_{\mathrm{obs}}|$ is at most 1.5×10^{-3} because the entries are rounded.

I.2 D.2 Per-seed breakdown of the `full` variant

Table 17 reports per-seed structural quantities, the observed-reward residual, and the observed routed value for the `full` variant.

I.3 D.3 Per-cluster heterogeneity decomposition

The fixed-archive heterogeneity term $\widehat{\Delta}_{\mathrm{route}}(A_1)$ is averaged over the user marginal Π.

Stratifying by the cluster index $g(U)$ localizes signed contributions to this population-level gap.

Define

$$C_g(A_1) \;=\; \mathbb{E}\big[m_1(U) \mid g(U) = g\big] - V_{\mathrm{static}}(A_1), \tag{351}$$

so $\sum_g \Pr(g(U) = g) C_g(A_1) = \Delta_{\mathrm{route}}(A_1)$. Unlike a within-cluster routed-minus-static gap, $C_g(A_1)$ need not be nonnegative.

Env	Seed	$\widehat{V}_{\text{static}}$	$\widehat{\Delta}_{\text{route}}$	$\widehat{\bar{G}}_T$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}}$
`general`	0	0.568	0.282	0.052	0.034	0.868
	1	0.572	0.276	0.049	0.037	0.860
	2	0.569	0.283	0.053	0.036	0.869
	3	0.571	0.279	0.050	0.038	0.862
	4	0.570	0.280	0.051	0.035	0.866
`workspace`	0	0.575	0.126	0.040	0.044	0.697
	1	0.578	0.123	0.042	0.042	0.701
	2	0.577	0.125	0.041	0.043	0.700
	3	0.579	0.122	0.040	0.045	0.696
	4	0.576	0.124	0.042	0.041	0.701
`support`	0	0.580	0.213	0.062	0.038	0.817
	1	0.578	0.217	0.060	0.040	0.815
	2	0.579	0.214	0.063	0.039	0.817
	3	0.581	0.216	0.060	0.041	0.816
	4	0.577	0.215	0.061	0.037	0.816

Table 17: Per-seed decomposition for the `full` variant. Across-seed standard deviations are at most 4×10^{-3} for every column.

Table 18 reports $C_g(A_1)$ for each cluster.

Env	Cluster g	$C_g(A_1)$	descriptive spread
`general`	0	0.17	$[0.11, 0.23]$
	1	0.39	$[0.34, 0.44]$
	2	0.41	$[0.36, 0.46]$
	3	0.23	$[0.17, 0.29]$
	4	0.40	$[0.35, 0.45]$
`workspace`	0	0.10	$[0.06, 0.14]$
	1	0.15	$[0.10, 0.20]$
`support`	0	0.37	$[0.32, 0.42]$
	1	0.44	$[0.39, 0.49]$
	2	0.03	$[-0.02, 0.08]$

Table 18: Signed per-cluster contributions relative to the population static value, with descriptive paired-seed spreads from Eq. equation 349. These rows do not test the within-cluster equality condition in Proposition 10.

I.4 D.4 Per-seed final cumulative regret

Per-seed final cumulative observed-reward residual of the `full` variant is reported in Table 19, where

$$\text{ObsResidual}_T = \sum_{t=1}^{T} \big(m_t(U) - r_t\big), \tag{352}$$

averaged across users.

It equals $T\widehat{L}_T^{\text{obs}}$ by definition. It equals the expected theoretical exploration cost only under a mean-zero observation model with the same conditional mean μ, which the clipped simulator does not satisfy.

I.5 D.5 Final regret across ablations

Table 20 extends Table 19 to all five variants.

Env	seed 0	seed 1	seed 2	seed 3	seed 4	mean ± std
general	10.21	11.10	10.84	11.46	10.87	10.90 ± 0.42
workspace	13.21	12.62	12.95	13.49	12.91	13.04 ± 0.30
support	11.41	11.99	11.71	12.35	10.92	11.68 ± 0.49

Table 19: Per-seed cumulative observed-reward residual. The implied $T\widehat{\bar{L}}_T^{\mathrm{obs}}$ is 10.8, 12.9, and 11.7 for the three environments, up to reporting-rounding.

The no_routing variant has a cumulative observed-reward residual roughly seven times that of full, consistent with Table 16.

Env	Variant	seed 0	seed 1	seed 2	seed 3	seed 4
general	full	10.21	11.10	10.84	11.46	10.87
	no_routing	84.30	85.12	83.71	84.95	84.42
	no_reflection	11.18	11.92	11.41	11.95	11.55
	no_personalization	23.71	24.92	24.10	24.85	24.04
	collapse_1	20.40	21.12	20.61	21.32	20.85
workspace	full	13.21	12.62	12.95	13.49	12.91
	no_routing	54.70	55.21	54.95	55.42	54.81
	no_reflection	13.30	12.95	13.21	13.65	13.04
	no_personalization	18.21	18.95	18.41	18.81	18.32
	collapse_1	21.40	22.10	21.81	22.30	21.95
support	full	11.41	11.99	11.71	12.35	10.92
	no_routing	76.30	77.12	76.41	77.85	76.21
	no_reflection	12.30	12.85	12.51	12.95	12.20
	no_personalization	23.30	23.95	23.61	24.10	23.45
	collapse_1	17.81	18.45	18.10	18.62	18.02

Table 20: Per-seed cumulative observed-reward residual across ablations. Each entry is averaged across the $N_u = 8$ users for that seed.

I.6 D.6 Reflection diagnostics

Table 21 summarises the reflection bookkeeping under the full variant: mean number of reflection events per user, fraction of children that strictly improve the per-user archive optimum ("beneficial-child fraction"), final archive size, and time-to-first-specialist (TTFS).

Env	mean reflections	beneficial-child frac.	mean $\lvert A_T \rvert$	TTFS (rounds)
general	7.6 ± 0.4	0.42	15.7 ± 0.6	8.98 ± 1.2
workspace	4.0 ± 0.3	0.22	11.9 ± 0.5	2.75 ± 0.7
support	5.1 ± 0.3	0.35	13.4 ± 0.5	0.35 ± 0.4

Table 21: Reflection diagnostics for the full variant. The TTFS column does not test Theorem 7, because the runs do not estimate its fixed-threshold repair probability q_ϵ.

I.7 D.7 Ungated reflection diagnostics

Removing the gating clause of Eq. equation 324 produces the negative result of Section 6.

Table 22 reports the gated/ungated comparison, showing that the archive grows to the cap $K_{\max}$ and reward erodes by 0.024 to 0.041 depending on environment.

I.8 D.8 Ablation effect sizes across five seeds

For each pair (full, ablation) we form the seed-paired difference

Env	Setting	mean $\lvert A_T \rvert$	$\widehat{\mathcal{V}}_T^{\text{obs}}$	$\widehat{L}_T^{\text{obs}}$
`general`	gated	15.7	0.864	0.036
	ungated	19.7	0.823	0.077
`workspace`	gated	11.9	0.699	0.043
	ungated	19.7	0.675	0.067
`support`	gated	13.4	0.816	0.039
	ungated	19.7	0.781	0.074

Table 22: Gated vs. ungated reflection. Ungated reflection grows the archive and raises the observed-reward residual.

$$\widehat{\Delta}^{(\texttt{abl})} = \widehat{\mathcal{V}}_T^{(\texttt{full})} - \widehat{\mathcal{V}}_T^{(\texttt{abl})}, \tag{353}$$

and resample seeds with replacement $B = 10{,}000$ times to summarize the sensitivity of the mean to the five observed seeds.

Table 23 reports the mean effect and the 2.5%–97.5% bootstrap percentile range. With five paired observations, we do not report a p-value or claim statistical significance.

Env	Comparison	$\widehat{\Delta}$	bootstrap percentile range
`general`	vs. `no_routing`	$+0.243$	$[0.235, 0.251]$
	vs. `no_reflection`	$+0.052$	$[0.046, 0.057]$
	vs. `no_personalization`	$+0.043$	$[0.038, 0.049]$
	vs. `collapse_1`	$+0.363$	$[0.354, 0.371]$
`workspace`	vs. `no_routing`	$+0.140$	$[0.133, 0.147]$
	vs. `no_reflection`	$+0.042$	$[0.037, 0.047]$
	vs. `no_personalization`	$+0.019$	$[0.012, 0.026]$
	vs. `collapse_1`	$+0.195$	$[0.187, 0.203]$
`support`	vs. `no_routing`	$+0.216$	$[0.208, 0.224]$
	vs. `no_reflection`	$+0.063$	$[0.057, 0.069]$
	vs. `no_personalization`	$+0.040$	$[0.034, 0.046]$
	vs. `collapse_1`	$+0.297$	$[0.288, 0.306]$

Table 23: Paired-seed effect sizes for the four ablations. The percentile ranges summarize resampling of five seeds and are not confidence intervals or significance tests.

I.9 D.9 Average reward by variant

Table 24 reports the observed routed value $\widehat{\mathcal{V}}_T^{\text{obs}}$ for every (env, variant) pair with descriptive spreads from Eq. equation 349.

The values match Table 16 but are repeated here in a flat layout for ease of citation.

Env	`full`	`no_routing`	`no_reflection`	`no_pers.`	`collapse_1`
`general`	0.864 ± 0.003	0.621 ± 0.005	0.812 ± 0.003	0.821 ± 0.004	0.501 ± 0.004
`workspace`	0.699 ± 0.004	0.559 ± 0.005	0.657 ± 0.004	0.680 ± 0.004	0.504 ± 0.004
`support`	0.816 ± 0.003	0.600 ± 0.005	0.753 ± 0.004	0.776 ± 0.004	0.519 ± 0.004

Table 24: Mean observed routed value $\widehat{\mathcal{V}}_T^{\text{obs}}$ by environment and ablation.

I.10 D.10 PER-USER BREAKDOWN OF ROUTED VALUE

Table 25 reports per-user $\widehat{\widehat{\mathcal{V}}}_T$ for the `full` variant in the `general` environment; analogous tables for `workspace` and `support` are omitted for space and follow the same pattern.

Seed	u0	u1	u2	u3	u4	u5	u6	u7
0	0.881	0.870	0.862	0.852	0.873	0.880	0.864	0.862
1	0.876	0.861	0.870	0.844	0.866	0.875	0.860	0.858
2	0.880	0.872	0.871	0.851	0.872	0.879	0.863	0.864
3	0.871	0.863	0.866	0.846	0.870	0.877	0.861	0.860
4	0.882	0.866	0.872	0.853	0.875	0.881	0.866	0.862

Table 25: Per-user, per-seed $\widehat{\widehat{\mathcal{V}}}_T$ in the `general` environment under the `full` variant. Per-user variance is dominated by the cluster index $g(u)$.

I.11 D.11 CONTEXT-TYPE TUNING SWEEP

Table 26 restates the baseline-tuning context-type comparison from Appendix G.

Three encodings are compared: `pref` (user preference vector only), `onehot` (candidate one-hot only), and `concat` (concatenation, used in the `full` variant of all main-paper experiments).

ctx type	$\widehat{\mathcal{V}}_T^{\text{obs}}$	$\widehat{L}_T^{\text{obs}}$	final $\sum_t (m_t - r_t)$	mean $\lvert A_T \rvert$
`pref`	0.813	0.091	27.41	19.67
`onehot`	0.834	0.075	22.49	19.67
`concat`	0.882	0.038	11.53	15.67

Table 26: Context-type ablation in the `general` environment, single seed, $T = 300$. Values reproduced from the baseline-tuning stage; `concat` is selected for the main-paper experiments.

I.12 D.12 SENSITIVITY TO NOISE SCALE

Table 27 sweeps the noise scale σ in Eq. equation 318 for the `full` variant in the `general` environment.

The observed-reward residual $\widehat{L}_T^{\text{obs}}$ grows with σ, while the structural term $\widehat{\Delta}_{\text{route}}(A_1)$ is invariant.

σ	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{\widehat{G}}_T$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}}$
0.025	0.280	0.052	0.020	0.882
0.050	0.280	0.051	0.036	0.864
0.100	0.280	0.049	0.064	0.836
0.200	0.280	0.044	0.121	0.776

Table 27: Noise-scale sensitivity of the logged quantities. The observed margin is positive at all four noise levels; this does not test Corollary 3 under clipped rewards.

I.13 D.13 SENSITIVITY TO ARCHIVE SIZE

Table 28 sweeps the initial archive size $K = |A_1|$ in $\{4, 8, 12, 24\}$ in the `general` environment.

Larger archives raise both the structural routing gain and the observed-reward residual.

I.14 D.14 SENSITIVITY TO USER HETEROGENEITY

Let α_{Dir} be the Dirichlet concentration parameter governing per-user preference vectors.

K	$\widehat{V}_{\text{static}}(A_1)$	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}}$
4	0.534	0.171	0.022	0.683
8	0.561	0.244	0.030	0.775
12	0.570	0.280	0.036	0.864
24	0.578	0.318	0.058	0.838

Table 28: Archive-size sensitivity in the `general` environment. The logged observed-reward margin peaks near $K = 12$.

As $\alpha_{\text{Dir}} \to \infty$ the user marginal collapses to a Dirac and $\widehat{\Delta}_{\text{route}}(A_1) \to 0$, recovering the homogeneous counterexample C1 of Section A.

α_{Dir}	$\widehat{\Delta}_{\text{route}}(A_1)$	$\widehat{L}_T^{\text{obs}}$	$\widehat{\mathcal{V}}_T^{\text{obs}} - \widehat{V}_{\text{static}}(A_1)$
0.1	0.341	0.041	$+0.347$
0.3	0.280	0.036	$+0.293$
1.0	0.182	0.034	$+0.198$
3.0	0.071	0.030	$+0.085$
10.0	0.018	0.028	$+0.030$
∞	0.000	0.026	-0.026

Table 29: Heterogeneity sensitivity in the `general` environment. The observed routed value exceeds the static mean until the user population is nearly homogeneous.

I.15 D.15 Reflection threshold and cooldown sweep

Table 30 varies the reflection trigger τ and cooldown κ in Eq. equation 324.

Lower τ or κ approach the ungated regime of Table 22.

τ	κ	mean $\lvert A_T \rvert$	$\widehat{\bar{G}}_T$	$\widehat{\bar{\mathcal{V}}}_T$
0.05	5	19.7	0.060	0.823
0.10	10	18.4	0.057	0.842
0.15	20	15.7	0.051	0.864
0.20	30	12.9	0.030	0.851
0.30	50	9.6	0.011	0.829

Table 30: Reflection (τ, κ) sweep in the `general` environment. The default $(\tau, \kappa) = (0.15, 20)$ used in the main paper is on the Pareto frontier of $(\widehat{\bar{G}}_T, \widehat{\bar{\mathcal{V}}}_T)$.

I.16 D.16 Theorem-to-experiment coverage map

Table 31 cross-references each theory result to a diagnostic or protocol check. The empirical rows do not prove the results.

I.17 D.17 Headline summary

Table 32 aggregates the structural terms, observed-reward residual, and observed routed value across the three synthetic environments.

$$\widehat{M}^{\text{obs}} = \widehat{\Delta}_{\text{route}}(A_1) + \widehat{\bar{G}}_T - \widehat{\bar{L}}_T^{\text{obs}}. \tag{354}$$

This concludes the table appendix.

All numerical entries reproduce bitwise from the harness of Appendix H.8 on a single CPU core in under 30 minutes.

Result	Statement	Empirical row	Status
Prop. 10	$\Delta_{\text{route}}(A) \geq 0$, strict iff disagreement	Tab. 16, 18, 29	illustrated
Lem. 4	predictable archive preserves concentration	Algorithm protocol	not tested
Lem. 5	failure-prioritization bound	Tab. 30	descriptive
Thm. 2	exact mean-reward identity	Tab. 16	observed identity only
Cor. 6	conditional plug-in regret bound	Tab. 19, 27	not tested
Thm. 7	first-useful-specialist time	Tab. 21 TTFS	vacuous (oracle proposer)
Cor. 3	$\Delta_{\text{route}} + \bar{G}_T > \bar{L}_T$ dominance	Tab. 24, 29	observed margin only

Table 31: Theory-to-diagnostic map. The saved residual is not the theoretical exploration cost under clipped observations; "vacuous" marks the oracle-assisted reflection result.

Quantity	`general`	`workspace`	`support`	cross-env mean
$\widehat{V}_{\text{static}}(A_1)$	0.570 ± 0.003	0.577 ± 0.003	0.579 ± 0.002	0.575
$\widehat{\Delta}_{\text{route}}(A_1)$	0.280 ± 0.004	0.124 ± 0.003	0.215 ± 0.004	0.206
$\widehat{\bar{G}}_T$	0.051 ± 0.002	0.041 ± 0.002	0.061 ± 0.002	0.051
$\widehat{\bar{L}}_T^{\text{obs}}$	0.036 ± 0.002	0.043 ± 0.002	0.039 ± 0.001	0.039
$\widehat{\bar{\mathcal{V}}}_T^{\text{obs}}$	0.864 ± 0.003	0.699 ± 0.004	0.816 ± 0.003	0.793
$\widehat{M}^{\text{obs}}$	$+0.295$	$+0.122$	$+0.237$	$+0.218$

Table 32: Cross-environment logged summary for the `full` variant. The observed-reward margin is positive in all three environments; this does not validate Corollary 3 under clipped observations.

J HYPER-RIDGE EXPLORATION-COST ANALYSIS

This appendix block analyzes the new Hyper-Ridge router. The reward accounting is inherited unchanged from Theorem 2 and Corollary 3; we do not re-prove the four-term decomposition. The only question addressed here is how replacing ordinary ridge geometry by a shared archive covariance changes the exploration-cost term.

J.1 GENERALIZED-RIDGE OBJECTS

For results combined with the four-term decomposition, draw one user U before routing and set $U_s = U$ at every round. The concentration lemmas below also allow a predictable sequence U_s. For a current candidate a, let

$$x_s(a) \in \mathbb{R}^d, \tag{355}$$

$$\mu(U_s, a) = b(U_s) + x_s(a)^\top \theta_a \in [0, 1], \tag{356}$$

$$r_s = b(U_s) + x_s(a_s)^\top \theta_{a_s} + \xi_s, \tag{357}$$

$$y_s = r_s - b(U_s) = x_s(a_s)^\top \theta_{a_s} + \xi_s. \tag{358}$$

The known baseline $b(U_s)$ is common to all available candidates. Candidate parameters share the positive-definite hyper-covariance

$$\theta_a \sim (0, \Omega), \tag{359}$$

$$\mathbb{E}[\theta_a] = 0, \tag{360}$$

$$\mathrm{Cov}(\theta_a) = \Omega, \tag{361}$$

$$\Omega \succ 0. \tag{362}$$

For candidate a, collect its pre-t observations as

$$\mathcal{T}_{a,t} = \{s < t : a_s = a\}, \tag{363}$$

$$X_{a,t} = \left[x_s(a)^\top\right]_{s \in \mathcal{T}_{a,t}}, \tag{364}$$

$$y_{a,t} = [y_s]_{s \in \mathcal{T}_{a,t}}, \tag{365}$$

$$\widehat{\Omega}_t \succ 0. \tag{366}$$

The last line is the hyper-covariance estimate available before round t.

Hyper-Ridge estimates each candidate by

$$\widehat{\theta}_{a,t} = \arg\min_{\theta \in \mathbb{R}^d} \left\{ \|y_{a,t} - X_{a,t}\theta\|_2^2 + \lambda\, \theta^\top \widehat{\Omega}_t^{-1} \theta \right\}, \tag{367}$$

$$\widehat{\theta}_{a,t} = \left(X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1} \right)^{-1} X_{a,t}^\top y_{a,t}, \tag{368}$$

$$V_{a,t}^{\text{hyper}} = X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1}. \tag{369}$$

The Hyper-Ridge score and selected candidate are

$$\mathrm{UCB}_t^{\text{hyper}}(U_t, a) = b(U_t) + x_t(a)^\top \widehat{\theta}_{a,t} + \alpha_t \sqrt{x_t(a)^\top \left(V_{a,t}^{\text{hyper}}\right)^{-1} x_t(a)}, \tag{370}$$

$$a_t(U_t) \in \arg\max_{a \in A_t(U_t)} \mathrm{UCB}_t^{\text{hyper}}(U_t, a). \tag{371}$$

For a generic design matrix X, the effective dimension induced by Ω is

$$d_{\text{eff}}(\Omega; X) = \mathrm{tr}\left[X^\top X \left(X^\top X + \lambda \Omega^{-1} \right)^{-1} \right]. \tag{372}$$

A small calculation illustrates why Ω can reduce the statistical dimension. Take

$$\Omega = \begin{bmatrix} 1 & 0 \\ 0 & 0.01 \end{bmatrix}, \tag{373}$$

$$X^\top X = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}. \tag{374}$$

With $\lambda = 1$, the effective dimension is

$$\begin{aligned} d_{\text{eff}}(\Omega; X) &= \operatorname{tr}\left[\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}\left(\begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix}\right)^{-1}\right] \\ &= \frac{9}{10} + \frac{1}{101} \\ &\approx 0.910. \end{aligned} \tag{375}$$

By contrast, identity ridge gives

$$\begin{aligned} d_{\text{eff}}(I; X) &= \frac{9}{10} + \frac{1}{2} \\ &= 1.400. \end{aligned} \tag{376}$$

Thus the same data matrix has smaller exploration geometry when the archive covariance says the second coordinate is unlikely to matter.

For the estimator itself, take

$$X = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}, \tag{377}$$

$$y = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \tag{378}$$

Using the same Ω and $\lambda = 1$, the generalized design is

$$\begin{aligned} V^{\text{hyper}} &= X^\top X + \Omega^{-1} \\ &= \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 100 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}. \end{aligned} \tag{379}$$

$$X^\top y = \begin{bmatrix} 5 \\ 0 \end{bmatrix}. \tag{380}$$

$$\begin{aligned} \widehat{\theta} &= \begin{bmatrix} 6 & 0 \\ 0 & 100 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 5/6 \\ 0 \end{bmatrix}. \end{aligned} \tag{381}$$

$$\sqrt{x^\top (V^{\text{hyper}})^{-1} x} = \sqrt{1/6}. \tag{382}$$

J.2 Shared-structure assumption

We now restate the shared-structure condition used by the Hyper-Ridge analysis. This is a regime assumption, not an empirical finding; the planned low-rank archive-geometry sweep in Section 6 is designed to stress-test it once new-method artifacts are produced.

Assumption 17 (Shared archive geometry)**.** *There is a positive-definite hyper-covariance matrix such that*

$$\Omega \in \mathbb{R}^{d \times d}, \qquad \Omega \succ 0, \tag{383}$$

$$\theta_a \sim (0, \Omega). \tag{384}$$

Here this notation means $\mathbb{E}[\theta_a] = 0$ and $\mathbb{E}[\theta_a \theta_a^\top] = \Omega$. The raw mean reward includes a known candidate-independent baseline, and realized parameters satisfy

$$\mu(U_t, a) = b(U_t) + x_t(a)^\top \theta_a \in [0, 1], \tag{385}$$

$$\|\theta_a\|_{\Omega^{-1}} \le S_\Omega \qquad \text{for every } a \in A_t(U_t) \text{ and } t \le T. \tag{386}$$

The archive is indexed by predictable birth-order slots as in Lemma 4. In particular, each candidate identity and feature rule is observable at birth, its latent parameter is fixed at birth in the analysis filtration, and the features are observable and predictable. For a predictable user stream,

$$\sigma(x_t(a)) \subseteq \mathcal{O}_{t-1} \qquad \text{for every } a \in A_t(U_t), \tag{387}$$

$$\mathcal{A}_T^{\mathrm{route}} = \bigcup_{t=1}^{T} A_t(U_t). \tag{388}$$

$$y_t := r_t - b(U_t) = x_t(a_t)^\top \theta_{a_t} + \xi_t. \tag{389}$$

The noise, feature radius, and effective dimension satisfy

$$\mathbb{E}[\exp(\eta\xi_t) \mid \mathcal{F}_{t-1}] \le \exp\left(\frac{\eta^2\sigma^2}{2}\right) \qquad \text{for every } \eta \in \mathbb{R}, \tag{390}$$

$$x_t(a)^\top \Omega x_t(a) \le L_\Omega^2 \qquad \text{for every } a \in A_t(U_t), \tag{391}$$

$$\max_{1\le s\le T+1} \max_{a\in\mathcal{A}_T^{\mathrm{route}}} d_{\mathrm{eff}}(\Omega; X_{a,s}) \le \bar{d}_{\mathrm{eff},T}(\Omega) < \infty \qquad \text{almost surely.} \tag{392}$$

The concentration argument uses the realized ellipsoid bound, predictable births, centered reward model, and feature radius. The random-effects law is needed only to interpret and estimate Ω. The deterministic cap in equation 392 converts the pathwise potential bound into a deterministic finite-time certificate.

J.3 KNOWN-Ω CONCENTRATION

The first lemma is the generalized-ridge analogue of the standard OFUL confidence bound (Abbasi-Yadkori et al., 2011; Chu et al., 2011). It is stated for known Ω, which is the clean oracle-covariance case. The estimated-covariance case adds a perturbation term and is handled separately.

Lemma 45 (Self-normalized concentration in the Ω geometry)**.** *Let $\Omega \succ 0$ and $\lambda > 0$. Fix a predictable birth-order slot a whose parameter is fixed at birth and satisfies $\|\theta_a\|_{\Omega^{-1}} \le S_\Omega$. Before its birth, pad its feature and selection indicator by zero. Assume the padded selected feature is $\mathcal{F}_{s-1}$-measurable and that each selected centered response obeys*

$$y_s = x_s(a)^\top \theta_a + \xi_s, \qquad \mathbb{E}[\exp(\eta\xi_s) \mid \mathcal{F}_{s-1}] \le \exp(\eta^2\sigma^2/2) \quad \text{for every } \eta \in \mathbb{R}. \tag{393}$$

Let the router use the true covariance, $\widehat{\Omega}_t = \Omega$, and define

$$V_{a,t}^{\Omega} = \lambda\Omega^{-1} + \sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^\top, \tag{394}$$

$$\widehat{\theta}_{a,t}^{\Omega} = \left(V_{a,t}^{\Omega}\right)^{-1} \sum_{s\in\mathcal{T}_{a,t}} x_s(a)y_s. \tag{395}$$

Define the confidence radius

$$\beta_{a,t}^{\Omega}(\delta) = \sigma\sqrt{2\log\left(\frac{\det(V_{a,t}^{\Omega})^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}\delta}\right)} + \sqrt{\lambda}\, S_\Omega. \tag{396}$$

Then, with probability at least $1-\delta$, uniformly over t,

$$\left\|\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right\|_{V_{a,t}^{\Omega}} \le \beta_{a,t}^{\Omega}(\delta), \tag{397}$$

$$\left|x^\top\left(\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right)\right| \le \beta_{a,t}^{\Omega}(\delta)\sqrt{x^\top\left(V_{a,t}^{\Omega}\right)^{-1} x}. \tag{398}$$

The second row holds for every predictable test feature $x \in \mathbb{R}^d$.

Proof. For this fixed candidate, define the regularizer and noise-weighted feature martingale by

$$A = \lambda\Omega^{-1}, \tag{399}$$

$$S_{a,t} = \sum_{s\in\mathcal{T}_{a,t}} x_s(a)\xi_s. \tag{400}$$

Using the reward model, the estimator error satisfies

$$\begin{aligned}\widehat{\theta}^{\Omega}_{a,t} - \theta_a &= \left(V^{\Omega}_{a,t}\right)^{-1} \sum_{s \in \mathcal{T}_{a,t}} x_s(a) \left(x_s(a)^\top \theta_a + \xi_s\right) - \theta_a \\ &= \left(V^{\Omega}_{a,t}\right)^{-1} \left[\sum_{s \in \mathcal{T}_{a,t}} x_s(a) x_s(a)^\top \theta_a + S_{a,t} \right] - \theta_a \\ &= \left(V^{\Omega}_{a,t}\right)^{-1} \left[\left(V^{\Omega}_{a,t} - A\right) \theta_a + S_{a,t} \right] - \theta_a \\ &= \left(V^{\Omega}_{a,t}\right)^{-1} S_{a,t} - \left(V^{\Omega}_{a,t}\right)^{-1} A \theta_a.\end{aligned} \tag{401}$$

Taking the $V^{\Omega}_{a,t}$-norm, applying the triangle inequality, and using $V^{\Omega}_{a,t} \succeq A$ give

$$\left\| \widehat{\theta}^{\Omega}_{a,t} - \theta_a \right\|_{V^{\Omega}_{a,t}} \le \|S_{a,t}\|_{\left(V^{\Omega}_{a,t}\right)^{-1}} + \|A\theta_a\|_{\left(V^{\Omega}_{a,t}\right)^{-1}}. \tag{402}$$

$$\begin{aligned}\|A\theta_a\|^2_{\left(V^{\Omega}_{a,t}\right)^{-1}} &= \theta_a^\top A \left(V^{\Omega}_{a,t}\right)^{-1} A \theta_a \\ &\le \theta_a^\top A \theta_a = \lambda \theta_a^\top \Omega^{-1} \theta_a \le \lambda S_\Omega^2.\end{aligned} \tag{403}$$

It remains to control the martingale term. If $\sigma = 0$, the conditional moment-generating-function assumption implies $\xi_s = 0$ almost surely, so the claim is immediate. Suppose $\sigma > 0$. For fixed $z \in \mathbb{R}^d$, define

$$\begin{aligned} M_{a,t}(z) &= \exp\left(\frac{z^\top S_{a,t}}{\sigma} - \frac{1}{2} z^\top \left(V^{\Omega}_{a,t} - A\right) z \right), \\ \mathbb{E}\left[\exp\left(\frac{z^\top x_s(a) \xi_s}{\sigma} - \frac{(z^\top x_s(a))^2}{2} \right) \middle| \mathcal{F}_{s-1} \right] &\le 1 \quad \text{at every selected round } s. \end{aligned} \tag{404}$$

At an unselected round the process does not change. Hence $(M_{a,t}(z))_{t \le T}$ is a nonnegative supermartingale with initial value one.

Let

$$\varphi_A(z) = \frac{\det(A)^{1/2}}{(2\pi)^{d/2}} \exp\left(-\frac{z^\top A z}{2} \right), \quad \mathcal{M}_{a,t} = \int_{\mathbb{R}^d} M_{a,t}(z) \varphi_A(z) \, dz. \tag{405}$$

The first expression is the Gaussian density with precision A. Tonelli's theorem shows that $(\mathcal{M}_{a,t})_{t \le T}$ is also a nonnegative supermartingale with initial value one. Completing the square gives

$$\mathcal{M}_{a,t} = \frac{\det(A)^{1/2}}{\det(V^{\Omega}_{a,t})^{1/2}} \exp\left(\frac{\|S_{a,t}\|^2_{\left(V^{\Omega}_{a,t}\right)^{-1}}}{2\sigma^2} \right), \tag{406}$$

$$\Pr\left(\sup_{t \le T} \mathcal{M}_{a,t} \ge \frac{1}{\delta} \right) \le \delta. \tag{407}$$

Consequently, with probability at least $1-\delta$, simultaneously for every $t \le T$, substituting $A = \lambda \Omega^{-1}$ gives

$$\|S_{a,t}\|_{\left(V^{\Omega}_{a,t}\right)^{-1}} \le \sigma \sqrt{2 \log \left(\frac{\det(V^{\Omega}_{a,t})^{1/2}}{\det(\lambda \Omega^{-1})^{1/2} \delta} \right)}. \tag{408}$$

Combining equation 402, equation 403, and equation 408 proves

$$\left\| \widehat{\theta}^{\Omega}_{a,t} - \theta_a \right\|_{V^{\Omega}_{a,t}} \le \beta^{\Omega}_{a,t}(\delta). \tag{409}$$

$$\begin{aligned}\left| x^\top \left(\widehat{\theta}^{\Omega}_{a,t} - \theta_a \right) \right| &\le \sqrt{x^\top \left(V^{\Omega}_{a,t}\right)^{-1} x} \left\| \widehat{\theta}^{\Omega}_{a,t} - \theta_a \right\|_{V^{\Omega}_{a,t}} \\ &\le \beta^{\Omega}_{a,t}(\delta) \sqrt{x^\top \left(V^{\Omega}_{a,t}\right)^{-1} x}.\end{aligned} \tag{410}$$

The second display applies Cauchy–Schwarz in the $V^{\Omega}_{a,t}$-inner product. □

A concrete confidence-bound computation is useful. Suppose

$$\frac{\det(V_{a,t}^{\Omega})^{1/2}}{\det(\lambda\Omega^{-1})^{1/2}} \;=\; 10, \tag{411}$$

$$\sigma \;=\; 0.05, \tag{412}$$

$$\delta \;=\; 0.05, \tag{413}$$

$$\sqrt{\lambda}S_{\Omega} \;=\; 0.20. \tag{414}$$

Then the confidence radius is

$$\begin{aligned}\beta_{a,t}^{\Omega}(\delta) &= 0.05\sqrt{2\log(10/0.05)} + 0.20 \\ &\approx 0.05\sqrt{10.596} + 0.20 \\ &\approx 0.363.\end{aligned} \tag{415}$$

If a candidate has width 0.10, Lemma 45 gives

$$\sqrt{x^{\top}(V_{a,t}^{\Omega})^{-1}x} \;=\; 0.10, \tag{416}$$

$$\left|x^{\top}\left(\widehat{\theta}_{a,t}^{\Omega} - \theta_a\right)\right| \;\le\; 0.363 \cdot 0.10 \;=\; 0.0363. \tag{417}$$

J.4 Elliptical potential in the Ω geometry

The next lemma is the step that converts per-round confidence widths into a dimension-dependent exploration cost. Its role is the same as the elliptical-potential lemma in LinUCB, except that the determinant is measured relative to $\lambda\Omega^{-1}$, and the resulting log-determinant is controlled by $d_{\text{eff}}(\Omega; X)$ up to a logarithmic factor.

Lemma 46 (Ω-elliptical potential and effective dimension)**.** *Fix one candidate a. Write its selected features in chronological order and define*

$$x_1, \ldots, x_n, \tag{418}$$

$$V_0 \;=\; \lambda\Omega^{-1}, \tag{419}$$

$$V_s \;=\; V_{s-1} + x_s x_s^{\top}, \tag{420}$$

$$q_s \;=\; x_s^{\top} V_{s-1}^{-1} x_s, \tag{421}$$

$$X \;=\; \begin{bmatrix} x_1^{\top} \\ \vdots \\ x_n^{\top} \end{bmatrix}. \tag{422}$$

For the covariance-scaled Gram matrix, let

$$M \;=\; \lambda^{-1}\Omega^{1/2}X^{\top}X\Omega^{1/2}. \tag{423}$$

$$R \;=\; \lambda_{\max}(M), \tag{424}$$

$$c(R) = \begin{cases} 1, & R = 0, \\ \dfrac{1+R}{R}\log(1+R), & R > 0. \end{cases} \tag{425}$$

Then

$$\sum_{s=1}^{n} \min\{1, q_s\} \;\le\; 2\,c(R)\,d_{\text{eff}}(\Omega; X). \tag{426}$$

If additionally $x_s^{\top}\Omega x_s \le L_{\Omega}^2$, then

$$R \;\le\; \frac{nL_{\Omega}^2}{\lambda}, \tag{427}$$

$$\sum_{s=1}^{n} \min\{1, q_s\} \;\le\; 2\left[1 + \log\left(1 + \frac{nL_{\Omega}^2}{\lambda}\right)\right] d_{\text{eff}}(\Omega; X). \tag{428}$$

Proof. The matrix determinant lemma, its product over s, and the logarithm of that product give

$$\begin{aligned} \det(V_s) \;&=\; \det(V_{s-1})\left(1 + x_s^\top V_{s-1}^{-1} x_s\right) \;=\; \det(V_{s-1})(1+q_s), \\ \frac{\det(V_n)}{\det(V_0)} \;&=\; \prod_{s=1}^{n}(1+q_s), \\ \log\frac{\det(V_n)}{\det(V_0)} \;&=\; \sum_{s=1}^{n}\log(1+q_s). \end{aligned} \tag{429}$$

For every $q \geq 0$, $\min\{1, q\} \leq 2\log(1+q)$. Hence

$$\begin{aligned} \min\{1,q\} \;&\leq\; 2\log(1+q), \\ \sum_{s=1}^{n}\min\{1,q_s\} \;&\leq\; 2\sum_{s=1}^{n}\log(1+q_s) \;=\; 2\log\frac{\det(V_n)}{\det(V_0)}. \end{aligned} \tag{430}$$

The determinant ratio can be rewritten in the covariance-scaled coordinates:

$$\begin{aligned} \log\frac{\det(V_n)}{\det(V_0)} &= \log\det\left(I + \lambda^{-1}\Omega^{1/2}X^\top X\Omega^{1/2}\right) \\ &= \log\det(I+M). \end{aligned} \tag{431}$$

Let $\nu_1, \dots, \nu_d$ be the eigenvalues of M. Then

$$\begin{gathered} \nu_1, \dots, \nu_d, \\ 0 \;\leq\; \nu_i \;\leq\; R, \\ \log\det(I+M) \;=\; \sum_{i=1}^{d}\log(1+\nu_i). \end{gathered} \tag{432}$$

The effective dimension is

$$\begin{aligned} d_{\text{eff}}(\Omega; X) &= \operatorname{tr}\left[X^\top X\left(X^\top X + \lambda\Omega^{-1}\right)^{-1}\right] \\ &= \operatorname{tr}\left[\Omega^{1/2}X^\top X\Omega^{1/2}\left(\Omega^{1/2}X^\top X\Omega^{1/2} + \lambda I\right)^{-1}\right] \\ &= \sum_{i=1}^{d}\frac{\nu_i}{1+\nu_i}. \end{aligned} \tag{433}$$

For $z > 0$, define

$$g(z) = \frac{1+z}{z}\log(1+z).$$

$$\begin{aligned} g'(z) \;&=\; -\frac{\log(1+z)}{z^2} + \frac{1+1/z}{1+z} \;=\; \frac{z - \log(1+z)}{z^2} \;\geq\; 0, \\ \log(1+z) \;&\leq\; c(R)\frac{z}{1+z} \qquad \text{for } 0 \leq z \leq R. \end{aligned} \tag{434}$$

The last row is equivalent to $g(z) \leq g(R)$.

Applying this scalar bound eigenvalue by eigenvalue gives

$$\begin{aligned} \log\det(I+M) &= \sum_{i=1}^{d}\log(1+\nu_i) \\ &\leq c(R)\sum_{i=1}^{d}\frac{\nu_i}{1+\nu_i} \\ &= c(R)d_{\text{eff}}(\Omega; X). \end{aligned} \tag{435}$$

Combining equation 430 and equation 435 proves

$$\sum_{s=1}^{n} \min\{1, q_s\} \le 2c(R) d_{\text{eff}}(\Omega; X). \tag{436}$$

It remains to bound R. Since $M \succeq 0$,

$$\begin{aligned} R \;\le\; \operatorname{tr}(M) \;=\; \lambda^{-1} \operatorname{tr}\Big(\Omega^{1/2} X^\top X \Omega^{1/2}\Big) \\ =\; \lambda^{-1} \sum_{s=1}^{n} x_s^\top \Omega x_s \;\le\; \frac{n L_\Omega^2}{\lambda}. \end{aligned} \tag{437}$$

$$\begin{aligned} c(R) \;&\le\; 1 + \log(1 + R) \\ &\le\; 1 + \log\left(1 + \frac{n L_\Omega^2}{\lambda}\right). \end{aligned} \tag{438}$$

The claimed logarithmic bound follows. □

K ORACLE HYPER-COVARIANCE: PROOF OF THE EFFECTIVE-DIMENSION COST

We now combine Lemma 45 and Lemma 46. The result is an oracle statement: the router is allowed to use the true hyper-covariance Ω. The estimated-Ω case is treated afterward as a perturbation of this oracle bound.

Assumption 18 (Oracle covariance and explicit confidence radius)**.** *The router uses the true positive-definite hyper-covariance. For the fixed-user route, define the realized candidate union and assume*

$$\widehat{\Omega}_t \;=\; \Omega, \tag{439}$$

$$\Omega \;\succ\; 0, \tag{440}$$

$$\mathcal{A}_T(U) \;:=\; \bigcup_{t=1}^{T} A_t(U). \tag{441}$$

$$|\mathcal{A}_T(U)| \;\le\; K_{\max}, \tag{442}$$

$$x_t(a)^\top \Omega x_t(a) \;\le\; L_\Omega^2. \tag{443}$$

The archive uses the predictable birth-order slots of Lemma 4. Its realized candidate designs obey the deterministic cap

$$\max_{1 \le s \le T+1} \max_{a \in \mathcal{A}_T(U)} d_{\text{eff}}(\Omega; X_{a,s}) \;\le\; \bar{d}_{\text{eff},T}(\Omega) \quad \textit{almost surely.} \tag{444}$$

For $\delta \in (0, 1)$, define

$$c_T^\Omega \;:=\; 1 + \log\left(1 + \frac{T L_\Omega^2}{\lambda}\right) \tag{445}$$

$$\alpha_{\Omega,T}(\delta) \;:=\; \max\left\{1,\, \sigma\sqrt{c_T^\Omega \bar{d}_{\text{eff},T}(\Omega) + 2\log(K_{\max}/\delta)} + \sqrt{\lambda} S_\Omega\right\}. \tag{446}$$

The bound uses the largest realized candidate-wise effective dimension. For each a, collect its selected feature rows and define

$$X_{a,T+1} \;:=\; \left[x_s(a_s)^\top\right]_{s \le T:\, a_s = a}, \tag{447}$$

$$d_{\text{eff}}(\Omega; X_{a,T+1}) \;:=\; \operatorname{tr}\left[X_{a,T+1}^\top X_{a,T+1} \left(X_{a,T+1}^\top X_{a,T+1} + \lambda \Omega^{-1}\right)^{-1}\right], \tag{448}$$

$$d_{\text{eff},T}^{\max}(\Omega) \;:=\; \max_{a \in \mathcal{A}_T(U)} d_{\text{eff}}(\Omega; X_{a,T+1}). \tag{449}$$

Assumption 18 requires $d_{\text{eff},T}^{\max}(\Omega) \le \bar{d}_{\text{eff},T}(\Omega)$ almost surely, and imposes the same cap on every earlier design prefix.

Proposition 47 (Known-Ω effective-dimension exploration cost)**.** *Under Assumptions 17 and 18, let the oracle Hyper-Ridge router use multiplier $\alpha_{\Omega,T}(\delta)$. For the result used in the four-term decomposition, $U_t = U$ for all t. There is an event $\mathcal{E}_\Omega$ with probability at least $1-\delta$ on which*

$$\sum_{t=1}^{T} L_t(U) \le 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\mathrm{eff},T}(\Omega)}. \tag{450}$$

Consequently, its expected exploration cost satisfies

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\mathrm{eff},T}(\Omega)} + \delta T. \tag{451}$$

For $T \ge 2$, fixed problem parameters, and $\delta = 1/T$,

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] = O\left(\sqrt{T}\log T\right). \tag{452}$$

For fixed δ, the same order holds for the pathwise right-hand side of equation 450 on its probability-$(1-\delta)$ event, but the expectation bound equation 451 retains the linear term δT.

Proof. Index all candidates by the predictable birth-order slots in Lemma 4, using a dummy process for any slot that is never filled. For each of the at most $K_{\max}$ slots, apply Lemma 45 with failure probability $\delta/K_{\max}$. For every realized slot and every $t \le T$, the determinant calculation in Lemma 46 gives

$$\begin{aligned} \log\frac{\det(V_{a,t}^{\Omega})}{\det(\lambda\Omega^{-1})} &\le c_T^{\Omega} d_{\mathrm{eff}}(\Omega; X_{a,t}) \\ &\le c_T^{\Omega}\bar{d}_{\mathrm{eff},T}(\Omega). \end{aligned} \tag{453}$$

Thus the confidence radius in Lemma 45, with $\delta/K_{\max}$ in place of δ, is at most $\alpha_{\Omega,T}(\delta)$. A union bound over the slots defines an event $\mathcal{E}_\Omega$ with probability at least $1-\delta$ on which these confidence intervals hold for every candidate after its birth and every $t \le T$.

On $\mathcal{E}_\Omega$, define the current-archive oracle action, its exploration cost, and the confidence quadratic form by

$$a_t^\star \in \arg\max_{a\in A_t(U)} x_t(a)^\top \theta_a, \tag{454}$$

$$L_t(U) = x_t(a_t^\star)^\top\theta_{a_t^\star} - x_t(a_t)^\top\theta_{a_t}, \tag{455}$$

$$q_t(a) := x_t(a)^\top \left(V_{a,t}^{\Omega}\right)^{-1} x_t(a), \qquad q_t := q_t(a_t). \tag{456}$$

The confidence bounds and the router's UCB choice give the single chain

$$\begin{aligned} x_t(a_t^\star)^\top\theta_{a_t^\star} &\le x_t(a_t^\star)^\top\widehat{\theta}^{\Omega}_{a_t^\star,t} + \alpha_{\Omega,T}(\delta)\sqrt{q_t(a_t^\star)} \\ &\le x_t(a_t)^\top\widehat{\theta}^{\Omega}_{a_t,t} + \alpha_{\Omega,T}(\delta)\sqrt{q_t} \\ &\le x_t(a_t)^\top\theta_{a_t} + 2\alpha_{\Omega,T}(\delta)\sqrt{q_t}. \end{aligned} \tag{457}$$

Subtracting $x_t(a_t)^\top\theta_{a_t}$, then using bounded rewards and $\alpha_{\Omega,T}(\delta) \ge 1$, yields

$$\begin{aligned} L_t(U) &\le 2\alpha_{\Omega,T}(\delta)\sqrt{q_t}, \\ 0 &\le L_t(U) \le 1, \\ L_t(U) &\le \min\{1, 2\alpha_{\Omega,T}(\delta)\sqrt{q_t}\} \le 2\alpha_{\Omega,T}(\delta)\min\{1,\sqrt{q_t}\}, \\ \sum_{t=1}^{T} L_t(U) &\le 2\alpha_{\Omega,T}(\delta)\sum_{t=1}^{T}\min\{1,\sqrt{q_t}\}. \end{aligned} \tag{458}$$

Partition the selected rounds by candidate. Let n_a be the number of selections of a, and let $q_{a,s}$ be its uncertainty at its s-th selection:

$$n_a := \sum_{t=1}^{T} \mathbf{1}\{a_t = a\}. \tag{459}$$

$$q_{a,s} := x_{a,s}^\top \left(V_{a,s-1}^{\Omega}\right)^{-1} x_{a,s}. \tag{460}$$

Then Cauchy–Schwarz gives

$$\begin{aligned}\sum_{t=1}^{T}\min\{1,\sqrt{q_t}\} &= \sum_{a\in\mathcal{A}_T(U)}\sum_{s=1}^{n_a}\min\{1,\sqrt{q_{a,s}}\}\\ &\le \sqrt{\left(\sum_{a\in\mathcal{A}_T(U)} n_a\right)\left(\sum_{a\in\mathcal{A}_T(U)}\sum_{s=1}^{n_a}\min\{1,q_{a,s}\}\right)}\\ &= \sqrt{T\sum_{a\in\mathcal{A}_T(U)}\sum_{s=1}^{n_a}\min\{1,q_{a,s}\}}.\end{aligned} \tag{461}$$

Lemma 46 applied separately to each candidate yields

$$\begin{aligned}\sum_{s=1}^{n_a}\min\{1,q_{a,s}\} &\le 2\left[1+\log\left(1+\frac{n_a L_\Omega^2}{\lambda}\right)\right] d_{\mathrm{eff}}(\Omega; X_{a,T+1})\\ &\le 2c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega),\end{aligned} \tag{462}$$

$$\sum_{a\in\mathcal{A}_T(U)}\sum_{s=1}^{n_a}\min\{1,q_{a,s}\} \le 2K_{\max}c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega). \tag{463}$$

Substituting equation 463 into equation 461 and then into equation 458 proves

$$\sum_{t=1}^{T} L_t(U) \le 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega)}. \tag{464}$$

This is equation 450. Taking expectations and splitting on $\mathcal{E}_\Omega$ gives

$$\begin{aligned}\sum_{t=1}^{T}\mathbb{E}[L_t(U)] &= \mathbb{E}\left[\mathbf{1}_{\mathcal{E}_\Omega}\sum_{t=1}^{T}L_t(U)\right] + \mathbb{E}\left[\mathbf{1}_{\mathcal{E}_\Omega^c}\sum_{t=1}^{T}L_t(U)\right]\\ &\le 2\alpha_{\Omega,T}(\delta)\sqrt{2K_{\max}T\, c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega)} + \delta T.\end{aligned} \tag{465}$$

This is equation 451. For fixed problem parameters, $c_T^\Omega = O(\log T)$ and $\alpha_{\Omega,T}(\delta) = O(\sqrt{\log T})$ for either fixed δ or $\delta = 1/T$ with $T \ge 2$. Thus the pathwise right-hand side is $O(\sqrt{T}\log T)$ in both cases. For the expectation bound, choosing $\delta = 1/T$ with $T \ge 2$ also makes the complement term $\delta T = 1$, which proves equation 452. With fixed δ, that complement term is linear and no sublinear expected bound follows from this conversion. □

The geometry changes the potential factor from an ambient-dimension term to $\bar{d}_{\mathrm{eff},T}(\Omega)$. This does not by itself determine which router has the smaller certificate: the full comparison must also include $\alpha_{\Omega,T}(\delta)$, c_T^Ω, and their LinUCB counterparts.

K.1 Estimated hyper-covariance: perturbation reduction and the remaining gap

The actual Hyper-Ridge PersonalEvo router uses an estimate $\widehat{\Omega}_t$. We therefore separate two issues. First, if $\widehat{\Omega}_t$ is close to Ω in relative spectral norm, the oracle proof above is stable. Second, proving that an online archive estimator achieves this closeness is a separate statistical problem; we state it as Conjecture 1 rather than claiming it as proven.

Define the pointwise and uniform relative covariance errors, together with the estimated and oracle design matrices, by

$$\rho_{\Omega,t} := \left\|\Omega^{-1/2}\left(\widehat{\Omega}_t - \Omega\right)\Omega^{-1/2}\right\|_{\mathrm{op}}, \tag{466}$$

$$\rho_{\Omega,T} := \max_{1\le t\le T}\rho_{\Omega,t}, \tag{467}$$

$$V_{a,t}^{\widehat{\Omega}} := X_{a,t}^\top X_{a,t} + \lambda\widehat{\Omega}_t^{-1}, \tag{468}$$

$$V_{a,t}^{\Omega} := X_{a,t}^\top X_{a,t} + \lambda\Omega^{-1}. \tag{469}$$

For a multiplier $\widetilde{\alpha}_T \geq 1$, define the excess width caused by using $\widehat{\Omega}_t$ instead of Ω:

$$\begin{aligned} \widehat{w}_t &\coloneqq \min\left\{1, \sqrt{x_t(a_t)^\top \left(V^{\widehat{\Omega}}_{a_t,t}\right)^{-1} x_t(a_t)}\right\}, \\ w^{\Omega}_t &\coloneqq \min\left\{1, \sqrt{x_t(a_t)^\top \left(V^{\Omega}_{a_t,t}\right)^{-1} x_t(a_t)}\right\}, \\ [z]_+ &\coloneqq \max\{z, 0\}, \end{aligned} \tag{470}$$

$$\mathrm{WidthErr}_T(\widehat{\Omega}, \Omega) \coloneqq 2\widetilde{\alpha}_T \sum_{t=1}^{T} \left[\widehat{w}_t - w^{\Omega}_t\right]_+ . \tag{471}$$

Lemma 48 (Relative spectral error implies Loewner control)**.** *Assume*

$$\rho_{\Omega,T} \leq \varepsilon, \tag{472}$$

$$0 \leq \varepsilon < 1. \tag{473}$$

Then, for every candidate a and round t,

$$(1-\varepsilon)\Omega \preceq \widehat{\Omega}_t \preceq (1+\varepsilon)\Omega. \tag{474}$$

$$\frac{1}{1+\varepsilon}\Omega^{-1} \preceq \widehat{\Omega}_t^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}. \tag{475}$$

$$V^{\widehat{\Omega}}_{a,t} \succeq \frac{1}{1+\varepsilon} V^{\Omega}_{a,t}. \tag{476}$$

Proof. The definition of $\rho_{\Omega,T}$, followed by adding I, congruence by $\Omega^{1/2}$, and order-reversing inversion, gives

$$\begin{aligned} -\varepsilon I &\preceq \Omega^{-1/2}\left(\widehat{\Omega}_t - \Omega\right)\Omega^{-1/2} \preceq \varepsilon I. \\ (1-\varepsilon)I &\preceq \Omega^{-1/2}\widehat{\Omega}_t\Omega^{-1/2} \preceq (1+\varepsilon)I. \\ (1-\varepsilon)\Omega &\preceq \widehat{\Omega}_t \preceq (1+\varepsilon)\Omega. \\ \frac{1}{1+\varepsilon}\Omega^{-1} &\preceq \widehat{\Omega}_t^{-1} \preceq \frac{1}{1-\varepsilon}\Omega^{-1}. \end{aligned} \tag{477}$$

Substituting the lower inverse bound into the design matrix gives

$$\begin{aligned} V^{\widehat{\Omega}}_{a,t} &= X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}_t^{-1} \\ &\succeq X_{a,t}^\top X_{a,t} + \frac{\lambda}{1+\varepsilon}\Omega^{-1} \\ &\succeq \frac{1}{1+\varepsilon}\left(X_{a,t}^\top X_{a,t} + \lambda\Omega^{-1}\right) \\ &= \frac{1}{1+\varepsilon} V^{\Omega}_{a,t}. \end{aligned} \tag{478}$$

□

Lemma 49 (Width and effective-dimension stability)**.** *Under the assumptions of Lemma 48, every feature vector x and fixed design matrix X satisfy*

$$x^\top \left(V^{\widehat{\Omega}}_{a,t}\right)^{-1} x \leq (1+\varepsilon) x^\top \left(V^{\Omega}_{a,t}\right)^{-1} x. \tag{479}$$

$$d_{\mathrm{eff}}(\widehat{\Omega}_t; X) \leq (1+\varepsilon) d_{\mathrm{eff}}(\Omega; X). \tag{480}$$

Proof. From equation 476, inversion, quadratic-form monotonicity, and trace monotonicity with $A = X^\top X \succeq 0$ give

$$\begin{aligned} \left(V^{\widehat{\Omega}}_{a,t}\right)^{-1} &\preceq (1+\varepsilon)\left(V^{\Omega}_{a,t}\right)^{-1}. \\ x^\top \left(V^{\widehat{\Omega}}_{a,t}\right)^{-1} x &\leq (1+\varepsilon) x^\top \left(V^{\Omega}_{a,t}\right)^{-1} x. \end{aligned} \tag{481}$$

$$
\begin{aligned}
d_{\text{eff}}(\widehat{\Omega}_t; X) &= \operatorname{tr}\left[A\left(A+\lambda\widehat{\Omega}_t^{-1}\right)^{-1}\right] \\
&\le (1+\varepsilon)\operatorname{tr}\left[A\left(A+\lambda\Omega^{-1}\right)^{-1}\right] \\
&= (1+\varepsilon) d_{\text{eff}}(\Omega; X).
\end{aligned} \tag{482}
$$

□

The spectral comparison controls confidence widths. A regret statement also needs valid confidence intervals in the estimated geometry.

Proposition 50 (Conditional perturbation bound for estimated hyper-ridge)**.** *Assume Assumption 17, suppose at most $K_{\max}$ candidates appear through round T, and let $\widehat{\Omega}_t \succ 0$ for every t. Let an estimated-covariance router use a multiplier $\widetilde{\alpha}_T \ge 1$, and suppose the event $\mathcal{E}_{\text{conf}}(\widehat{\Omega})$ gives simultaneous confidence:*

$$
\left|x^\top\left(\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a\right)\right| \le \widetilde{\alpha}_T \sqrt{x^\top\left(V^{\widehat{\Omega}}_{a,t}\right)^{-1} x} \tag{483}
$$

for every available candidate a, every $t \le T$, and every predictable test feature x. Assume also that

$$
\rho_{\Omega,T} \le \varepsilon < 1. \tag{484}
$$

On the intersection of these two events,

$$
\operatorname{WidthErr}_T(\widehat{\Omega}, \Omega) \le 2\widetilde{\alpha}_T\left(\sqrt{1+\varepsilon} - 1\right)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega)}, \tag{485}
$$

$$
\sum_{t=1}^{T} L_t(U) \le 2\widetilde{\alpha}_T\sqrt{1+\varepsilon}\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega)}. \tag{486}
$$

If the intersection has probability at least $1-\eta$, then the right-hand side of equation 486 plus ηT bounds the expected cumulative exploration cost.

Proof. Lemma 49 and $\min\{1, cz\} \le c\min\{1, z\}$ for $c \ge 1$ give

$$
\begin{aligned}
\widehat{w}_t &\le \sqrt{1+\varepsilon}\, w_t^{\Omega}, \\
\left[\widehat{w}_t - w_t^{\Omega}\right]_+ &\le \left(\sqrt{1+\varepsilon} - 1\right) w_t^{\Omega}.
\end{aligned} \tag{487}
$$

The definition of $\operatorname{WidthErr}_T$ and the oracle width-sum bound from the proof of Proposition 47 now give

$$
\begin{aligned}
\operatorname{WidthErr}_T(\widehat{\Omega}, \Omega) &\le 2\widetilde{\alpha}_T\left(\sqrt{1+\varepsilon} - 1\right)\sum_{t=1}^{T} w_t^{\Omega} \\
&\le 2\widetilde{\alpha}_T\left(\sqrt{1+\varepsilon} - 1\right)\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega)}.
\end{aligned} \tag{488}
$$

On $\mathcal{E}_{\text{conf}}(\widehat{\Omega})$, the same UCB argument and the first row above yield

$$
\begin{aligned}
L_t(U) &\le 2\widetilde{\alpha}_T\widehat{w}_t \le 2\widetilde{\alpha}_T\sqrt{1+\varepsilon}\, w_t^{\Omega}, \\
\sum_{t=1}^{T} L_t(U) &\le 2\widetilde{\alpha}_T\sqrt{1+\varepsilon}\sum_{t=1}^{T} w_t^{\Omega} \\
&\le 2\widetilde{\alpha}_T\sqrt{1+\varepsilon}\sqrt{2K_{\max}T\, c_T^{\Omega}\bar{d}_{\text{eff},T}(\Omega)}.
\end{aligned}
$$

This is equation 486. On the complement of an event with probability $1-\eta$, bounded rewards give $\sum_{t=1}^{T} L_t(U) \le T$, which adds ηT in expectation. □

A numerical perturbation example clarifies the scale:

$$
\varepsilon = 0.21, \tag{489}
$$

$$
\sqrt{1+\varepsilon} = 1.10. \tag{490}
$$

Thus the estimated-geometry width is at most 1.10 times the oracle-geometry width on the spectral event. Turning this comparison into an exploration bound also requires the confidence event. Neither event has been proved for the periodic estimator.

Conjecture 1 (Sharper estimated-covariance exploration cost)**.** *There is an estimated-covariance routing rule for which*

$$\sum_{t=1}^{T} \mathbb{E}[L_t(U)] \le C\sqrt{K_{\max}\bar{d}_{\mathrm{eff},T}(\Omega)T\log(1+T)} + o(\sqrt{T}), \tag{491}$$

under assumptions comparable to those used here. The explicit frequentist radius in Theorem 57 does not prove this rate.

The conjecture requires a sharper confidence analysis or a different routing rule. The periodic estimator and its diagnostics do not establish it.

K.2 Map of hyper-ridge claims to proofs and open steps

Table 33 summarizes which parts of the Hyper-Ridge PersonalEvo theory are proved in this section and which are conditional. The inherited four-term decomposition remains the base accounting theorem; the new analysis only sharpens the exploration-cost term $\bar{L}_T$.

Claim	Mathematical object	Status
Exact PersonalEvo accounting	$V_{\mathrm{static}} + \Delta_{\mathrm{route}} + \bar{G}_T - \bar{L}_T$	inherited from Theorem 2
Oracle hyper-ridge confidence	$V_{a,t}^{\Omega} = X_{a,t}^{\top} X_{a,t} + \lambda\Omega^{-1}$	Lemma 45
Effective-dimension potential	$\sum_s \min\{1, q_s\}$	Lemma 46
Known-Ω exploration cost	explicit radius times effective-dimension potential	Proposition 47
Estimated-Ω width stability	$\mathrm{WidthErr}_T$ under spectral error and confidence	Proposition 50
Frozen calibration estimate	finite-sample covariance and exploration bounds	Theorem 57
Periodic covariance estimate	confidence and covariance rates	open
Empirical validation of hyper-ridge gains	oracle, estimated, diagonal, and base routers	diagnostic only; full test remains open

Table 33: Theory map for Hyper-Ridge PersonalEvo. The oracle and calibration-split variants have separate guarantees; the periodic estimator does not.

L Estimated-Covariance Control: the Calibration-Split Theorem

This section analyzes a separate calibration-split router. It estimates Ω from fresh candidates, freezes the estimate, and then routes over a fixed archive. The result does not cover the periodic estimator in Algorithm 1.

Assumption 19 (Exogenous uniform-routing calibration diversity)**.** *Let $\Omega \succ 0$. The calibration phase uses N candidates and collects m rewards per candidate under an exogenous uniform-routing policy. For $i \in \{1, \dots, N\}$ and $j \in \{1, \dots, m\}$, define*

$$z_{i,j} = \Omega^{1/2} x_{i,j}, \quad X_i = \begin{bmatrix} x_{i,1}^{\top} \\ \vdots \\ x_{i,m}^{\top} \end{bmatrix}. \tag{492}$$

The arrays $X_1, \dots, X_N$ are independent. Within block i, the vectors $z_{i,1}, \dots, z_{i,m}$ are independent and, for deterministic $\kappa_{\mathrm{cal}} > 0$ and $L_{\mathrm{cal}} < \infty$,

$$\mathbb{E}[z_{i,j} z_{i,j}^{\top}] \succeq \kappa_{\mathrm{cal}} I_d, \qquad \|z_{i,j}\|_2^2 \le L_{\mathrm{cal}}^2 \quad \textit{almost surely.} \tag{493}$$

The full feature array is independent of the candidate effects.

The effects $\theta_1, \dots, \theta_N$ are i.i.d. and satisfy

$$\mathbb{E}\theta_i = 0, \qquad \mathbb{E}[\theta_i \theta_i^\top] = \Omega, \qquad \|\theta_i\|_{\Omega^{-1}} \le S_\Omega \quad \textit{almost surely}. \tag{494}$$

The reward and centered response are

$$r_{i,j} \;=\; b(u_{i,j}) + x_{i,j}^\top \theta_i + \xi_{i,j}, \quad y_{i,j} \;=\; r_{i,j} - b(u_{i,j}). \tag{495}$$

Conditional on the full feature array, the pairs (θ_i, ξ_i), with $\xi_i = (\xi_{i,1}, \dots, \xi_{i,m})^\top$, are independent across i. For each block there is a filtration $\mathcal{H}_{i,0} \subseteq \cdots \subseteq \mathcal{H}_{i,m}$ such that the full feature array and θ_i are $\mathcal{H}_{i,0}$-measurable, $\xi_{i,j}$ is $\mathcal{H}_{i,j}$-measurable, and

$$\mathbb{E}[\exp(\eta \xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{\eta^2 \sigma^2}{2}\right) \quad \textit{almost surely for every } \eta \in \mathbb{R}. \tag{496}$$

The filtration is used only for conditioning: the estimator observes (X_i, y_i), not θ_i.

Lemma 51 (Net reduction for symmetric quadratic forms)**.** *Let $A \in \mathbb{R}^{d\times d}$ be symmetric. Let $\mathcal{N} \subset \mathbb{S}^{d-1}$ be a $1/4$-net of the Euclidean unit sphere. Then*

$$\|A\|_{\mathrm{op}} \le 2 \sup_{v \in \mathcal{N}} |v^\top A v|. \tag{497}$$

Moreover, there exists such a $1/4$-net with cardinality at most 9^d.

Proof. Set $M = \sup_{v\in\mathcal{N}} |v^\top A v|$. For any $u \in \mathbb{S}^{d-1}$, choose $v \in \mathcal{N}$ with $\|u - v\|_2 \le 1/4$. Symmetry gives

$$\begin{aligned} |u^\top A u - v^\top A v| \;&=\; |(u-v)^\top A u + v^\top A (u - v)| \\ &\le\; 2\|u - v\|_2 \|A\|_{\mathrm{op}} \;\le\; \tfrac{1}{2}\|A\|_{\mathrm{op}}. \end{aligned} \tag{498}$$

Thus $|u^\top A u| \le M + \|A\|_{\mathrm{op}}/2$. Taking the supremum over u and using $\|A\|_{\mathrm{op}} = \sup_{\|u\|_2=1} |u^\top A u|$ gives $\|A\|_{\mathrm{op}} \le 2M$.

For the covering-number claim, let $\mathcal{N}$ be a maximal subset of $\mathbb{S}^{d-1}$ such that distinct points of $\mathcal{N}$ have Euclidean distance strictly larger than $1/4$. Maximality implies that $\mathcal{N}$ is a $1/4$-net. The balls

$$B(v, 1/8) = \{w \in \mathbb{R}^d : \|w - v\|_2 \le 1/8\}, \qquad v \in \mathcal{N}, \tag{499}$$

are pairwise disjoint and contained in $B(0, 9/8)$. Volume comparison yields

$$|\mathcal{N}| \left(\frac{1}{8}\right)^d \le \left(\frac{9}{8}\right)^d. \tag{500}$$

Thus $|\mathcal{N}| \le 9^d$. □

Lemma 52 (Independent sub-Gaussian sample covariance)**.** *There is a universal numerical constant $C_{\mathrm{cov}} > 0$ with the following property. Let $Z_1, \dots, Z_N \in \mathbb{R}^d$ be independent mean-zero random vectors such that, for every i, every $u \in \mathbb{S}^{d-1}$, and every $s \in \mathbb{R}$,*

$$\mathbb{E}\exp(s u^\top Z_i) \le \exp\left(\frac{s^2 K^2}{2}\right). \tag{501}$$

Let $\Sigma_i = \mathbb{E}[Z_i Z_i^\top]$. Then, for every $\delta \in (0,1)$, with probability at least $1 - \delta$,

$$\left\| \frac{1}{N} \sum_{i=1}^N \left(Z_i Z_i^\top - \Sigma_i\right) \right\|_{\mathrm{op}} \le C_{\mathrm{cov}} K^2 \left[\sqrt{\frac{d + \log(2/\delta)}{N}} + \frac{d + \log(2/\delta)}{N} \right]. \tag{502}$$

The same conclusion holds conditionally on any sigma-field with respect to which the displayed independence and moment-generating-function hypotheses hold with the same deterministic value of K.

Proof. Fix $u \in \mathbb{S}^{d-1}$. Chernoff's method, applied also to $-u$, gives

$$\Pr(|u^\top Z_i| \geq t) \;\leq\; 2\exp\left(-\frac{t^2}{2K^2}\right) \qquad t \geq 0. \tag{503}$$

Equivalently, $u^\top Z_i$ is sub-Gaussian with sub-Gaussian norm bounded by a universal multiple of K. Lemma 2.7.7 of Vershynin, *High-Dimensional Probability*, Cambridge University Press, 2018, shows that $Y_i = (u^\top Z_i)^2 - \mathbb{E}[(u^\top Z_i)^2]$ is centered sub-exponential with norm at most a universal multiple of K^2. Theorem 2.8.1 of the same reference gives a universal $c_1 > 0$ such that

$$\Pr\left(\left|\frac{1}{N}\sum_{i=1}^{N}\left[(u^\top Z_i)^2 - \mathbb{E}(u^\top Z_i)^2\right]\right| > c_1 K^2\left[\sqrt{\frac{t}{N}} + \frac{t}{N}\right]\right) \leq 2e^{-t}. \tag{504}$$

Let $\mathcal{N}$ be a $1/4$-net of $\mathbb{S}^{d-1}$ with $|\mathcal{N}| \leq 9^d$, whose existence follows from Lemma 51. Applying the preceding scalar inequality with $t = d\log 9 + \log(2/\delta)$, followed by a union bound, gives

$$\sup_{u\in\mathcal{N}}\left|u^\top\left[\frac{1}{N}\sum_{i=1}^{N}\left(Z_iZ_i^\top - \Sigma_i\right)\right]u\right| \leq c_1K^2\left[\sqrt{\frac{d\log 9 + \log(2/\delta)}{N}} + \frac{d\log 9 + \log(2/\delta)}{N}\right]. \tag{505}$$

Applying Lemma 51 to the symmetric matrix inside the brackets yields

$$\left\|\frac{1}{N}\sum_{i=1}^{N}\left(Z_iZ_i^\top - \Sigma_i\right)\right\|_{\mathrm{op}} \leq 2c_1K^2\left[\sqrt{\frac{d\log 9 + \log(2/\delta)}{N}} + \frac{d\log 9 + \log(2/\delta)}{N}\right]. \tag{506}$$

Since $\log 9 < 3$, increasing the universal constant proves the stated bound.

For the conditional version, fix a regular conditional distribution given the conditioning sigma-field. Under that conditional distribution, the same proof applies with the same deterministic K and constants. Integrating its failure probability completes the conditional claim. $\square$

Lemma 53 (Calibration Gram lower tail under exogenous diversity)**.** *Assume Assumption 19. For each calibration candidate define*

$$G_i = X_i^\top X_i, \qquad H_i = \Omega^{1/2}G_i\Omega^{1/2} = \sum_{j=1}^{m} z_{i,j}z_{i,j}^\top. \tag{507}$$

For every $\delta_{\mathrm{gram}} \in (0,1)$, if

$$m \geq \frac{8L_{\mathrm{cal}}^2}{\kappa_{\mathrm{cal}}}\log\left(\frac{dN}{\delta_{\mathrm{gram}}}\right), \tag{508}$$

then, with probability at least $1-\delta_{\mathrm{gram}}$,

$$\begin{aligned} H_i &\succeq \frac{\kappa_{\mathrm{cal}}m}{2}I_d \qquad (i = 1,\ldots,N), \\ \left\|\Omega^{-1/2}G_i^{-1}\Omega^{-1/2}\right\|_{\mathrm{op}} &\leq \frac{2}{\kappa_{\mathrm{cal}}m} \qquad\qquad (i = 1,\ldots,N). \end{aligned} \tag{509}$$

Every G_i is invertible on this event.

Proof. Fix i and set $A_{i,j} = z_{i,j}z_{i,j}^\top$. By Assumption 19, the matrices $A_{i,1},\ldots,A_{i,m}$ are independent positive-semidefinite random matrices satisfying

$$0 \preceq A_{i,j} \preceq L_{\mathrm{cal}}^2 I_d, \qquad \lambda_{\min}\left(\sum_{j=1}^{m}\mathbb{E}A_{i,j}\right) \geq \kappa_{\mathrm{cal}}m. \tag{510}$$

Tropp's matrix Chernoff lower-tail inequality for independent positive-semidefinite summands bounded above by RI_d states that, with $\mu_{\min} = \lambda_{\min}(\sum_j \mathbb{E}A_{i,j})$, for every $\varepsilon \in [0,1]$,

$$\Pr\left(\lambda_{\min}\left(\sum_{j=1}^{m}A_{i,j}\right) \leq (1-\varepsilon)\mu_{\min}\right) \leq d\exp\left(-\frac{\varepsilon^2\mu_{\min}}{2R}\right). \tag{511}$$

This is Theorem 5.1.1 of Tropp, *An Introduction to Matrix Concentration Inequalities*, Foundations and Trends in Machine Learning, 2015, applied with $R = L_{\mathrm{cal}}^2$. Taking $\varepsilon = 1/2$ gives

$$\Pr\left(H_i \not\succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d\right) \le d \exp\left(-\frac{\kappa_{\mathrm{cal}} m}{8 L_{\mathrm{cal}}^2}\right). \tag{512}$$

The sample-size condition makes the right-hand side at most δ_{gram}/N; a union bound proves the simultaneous Gram event.

On this event,

$$\Omega^{1/2} G_i \Omega^{1/2} \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d. \tag{513}$$

Thus G_i is invertible. Inverting the Loewner inequality and identifying the inverse give

$$\begin{aligned}(\Omega^{1/2} G_i \Omega^{1/2})^{-1} &\preceq \frac{2}{\kappa_{\mathrm{cal}} m} I_d, \\ (\Omega^{1/2} G_i \Omega^{1/2})^{-1} &= \Omega^{-1/2} G_i^{-1} \Omega^{-1/2},\end{aligned} \tag{514}$$

the asserted operator-norm bound follows. □

Lemma 54 (Conditional calibration-block covariance control)**.** *Assume Assumption 19. Write $X_{1:N} = (X_1, \ldots, X_N)$. Condition on a realization of the full calibration feature array for which*

$$H_i = \Omega^{1/2} X_i^\top X_i \Omega^{1/2} \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d \qquad \text{for every } i = 1, \ldots, N. \tag{515}$$

For each i, define

$$y_i = \begin{bmatrix} r_{i,1} - b(u_{i,1}) \\ \vdots \\ r_{i,m} - b(u_{i,m}) \end{bmatrix} = X_i \theta_i + \xi_i, \tag{516}$$

and

$$\begin{aligned}\widetilde{\theta}_i &= G_i^{-1} X_i^\top y_i, \qquad & Z_i &= \Omega^{-1/2} \widetilde{\theta}_i, \\ \tau_m &= \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad & K_m^2 &= S_\Omega^2 + \tau_m.\end{aligned} \tag{517}$$

Then, conditional on the feature array, $Z_1, \ldots, Z_N$ are independent mean-zero random vectors. Moreover,

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \qquad R_i \succeq 0, \qquad \|R_i\|_{\mathrm{op}} \le \tau_m, \tag{518}$$

and, for every $u \in \mathbb{S}^{d-1}$ and every $s \in \mathbb{R}$,

$$\mathbb{E}\big[\exp(s u^\top Z_i) \mid X_{1:N}\big] \le \exp\left(\frac{s^2 K_m^2}{2}\right). \tag{519}$$

Proof. The least-squares identity gives

$$\begin{aligned}\widetilde{\theta}_i &= G_i^{-1} X_i^\top (X_i \theta_i + \xi_i) \\ &= \theta_i + G_i^{-1} X_i^\top \xi_i.\end{aligned} \tag{520}$$

Set

$$\begin{aligned}U_i &= \Omega^{-1/2} \theta_i, \qquad & Z_i &= U_i + E_i, \\ E_i &= \Omega^{-1/2} G_i^{-1} X_i^\top \xi_i.\end{aligned} \tag{521}$$

The random-effects assumptions imply, conditional on the full calibration feature array,

$$\mathbb{E}[U_i \mid X_{1:N}] = 0, \qquad \mathbb{E}[U_i U_i^\top \mid X_{1:N}] = I_d, \qquad \|U_i\|_2 \le S_\Omega \quad \text{almost surely.} \tag{522}$$

The conditional sub-Gaussian noise assumption implies $\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] = 0$. Indeed, Jensen's inequality gives, for $h > 0$,

$$\exp(h\, \mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}]) \le \mathbb{E}[\exp(h \xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{h^2 \sigma^2}{2}\right), \tag{523}$$

so $\mathbb{E}[\xi_{i,j} \mid \mathcal{H}_{i,j-1}] \le h\sigma^2/2$. Letting $h \downarrow 0$ gives the upper bound 0. Applying the same argument to $-\xi_{i,j}$ gives the lower bound 0. Therefore the conditional mean is zero. By iterated conditioning,

$$\mathbb{E}[\xi_i \mid X_{1:N}, \theta_i] = 0, \qquad \mathbb{E}[E_i \mid X_{1:N}, \theta_i] = 0. \tag{524}$$

Consequently, the cross terms vanish:

$$\mathbb{E}[U_i E_i^\top \mid X_{1:N}] = 0, \qquad \mathbb{E}[E_i U_i^\top \mid X_{1:N}] = 0. \tag{525}$$

Thus

$$\mathbb{E}[Z_i Z_i^\top \mid X_{1:N}] = I_d + R_i, \qquad R_i = \mathbb{E}[E_i E_i^\top \mid X_{1:N}] \succeq 0. \tag{526}$$

To bound R_i, fix $v \in \mathbb{S}^{d-1}$ and set $b_i = X_i G_i^{-1} \Omega^{-1/2} v$. Conditional on the feature array, b_i is deterministic and

$$v^\top R_i v = \mathbb{E}\left[(b_i^\top \xi_i)^2 \mid X_{1:N}\right]. \tag{527}$$

For $j < k$, the random variable $\xi_{i,j}$ is $\mathcal{H}_{i,k-1}$-measurable. Hence

$$\begin{aligned}\mathbb{E}[\xi_{i,j}\xi_{i,k} \mid X_{1:N}] &= \mathbb{E}[\xi_{i,j}\, \mathbb{E}[\xi_{i,k} \mid \mathcal{H}_{i,k-1}] \mid X_{1:N}] \\ &= 0.\end{aligned} \tag{528}$$

The conditional sub-Gaussian inequality also implies $\mathbb{E}[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}] \le \sigma^2$. To see this, average the MGF bounds at h and $-h$:

$$\mathbb{E}[\cosh(h\xi_{i,j}) \mid \mathcal{H}_{i,j-1}] \le \exp\left(\frac{h^2\sigma^2}{2}\right). \tag{529}$$

Conditional Fatou's lemma and $2(\cosh(hx) - 1)/h^2 \to x^2$ yield

$$\begin{aligned}\mathbb{E}\left[\xi_{i,j}^2 \mid \mathcal{H}_{i,j-1}\right] &\le \liminf_{h\downarrow 0} \frac{2}{h^2}\left[\exp\left(\frac{h^2\sigma^2}{2}\right) - 1\right] \\ &= \sigma^2.\end{aligned} \tag{530}$$

The zero cross-covariances and variance bound now give

$$\begin{aligned} v^\top R_i v &\le \sigma^2 \|b_i\|_2^2, \\ \|b_i\|_2^2 &= v^\top \Omega^{-1/2} G_i^{-1} X_i^\top X_i G_i^{-1} \Omega^{-1/2} v \\ &= v^\top \Omega^{-1/2} G_i^{-1} \Omega^{-1/2} v.\end{aligned} \tag{531}$$

On the conditioned Gram event,

$$\Omega^{-1/2} G_i^{-1} \Omega^{-1/2} = (\Omega^{1/2} G_i \Omega^{1/2})^{-1} \preceq \frac{2}{\kappa_{\mathrm{cal}} m} I_d. \tag{532}$$

Therefore

$$v^\top R_i v \le \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m} = \tau_m. \tag{533}$$

Taking the supremum over v gives $\|R_i\|_{\mathrm{op}} \le \tau_m$.

It remains to prove the one-dimensional moment-generating-function bound. Fix $u \in \mathbb{S}^{d-1}$. Since $\mathbb{E}[u^\top U_i \mid X_{1:N}] = 0$ and $|u^\top U_i| \le \|U_i\|_2 \le S_\Omega$ almost surely, Hoeffding's lemma gives

$$\mathbb{E}\left[\exp(s u^\top U_i) \mid X_{1:N}\right] \le \exp\left(\frac{s^2 S_\Omega^2}{2}\right). \tag{534}$$

For the noise part, set $b_i = X_i G_i^{-1} \Omega^{-1/2} u$, which is deterministic conditional on the full feature array. Iterating the conditional sub-Gaussian inequality over $j = m, m-1, \ldots, 1$ gives

$$\mathbb{E}\left[\exp(s b_i^\top \xi_i) \mid X_{1:N}, \theta_i\right] \le \exp\left(\frac{s^2\sigma^2 \|b_i\|_2^2}{2}\right). \tag{535}$$

The Gram-event calculation above gives $\|b_i\|_2^2 \le 2/(\kappa_{\mathrm{cal}} m)$. Therefore,

$$\mathbb{E}\left[\exp(s u^\top E_i) \mid X_{1:N}, \theta_i\right] \le \exp\left(\frac{s^2 \tau_m}{2}\right). \tag{536}$$

Since U_i is a function of θ_i,

$$\begin{aligned}\mathbb{E}\big[\exp(su^\top Z_i) \mid X_{1:N}\big] &= \mathbb{E}\big[\exp(su^\top U_i)\mathbb{E}\big[\exp(su^\top E_i) \mid X_{1:N}, \theta_i\big] \mid X_{1:N}\big] \\ &\le \exp\left(\frac{s^2\tau_m}{2}\right)\mathbb{E}\big[\exp(su^\top U_i) \mid X_{1:N}\big] \\ &\le \exp\left(\frac{s^2(S_\Omega^2+\tau_m)}{2}\right).\end{aligned} \tag{537}$$

Finally, Assumption 19 makes the pairs (θ_i, ξ_i) independent across i, conditional on the feature array. Since Z_i is a function of (X_i, θ_i, ξ_i), the Z_i are conditionally independent. $\square$

Lemma 55 (Relative covariance error for the calibration estimator)**.** *Assume Assumption 19. Define*

$$\widehat{\Omega} = \frac{1}{N}\sum_{i=1}^N \widetilde{\theta}_i\widetilde{\theta}_i^\top + \gamma I_d, \qquad \gamma > 0, \tag{538}$$

where $\widetilde{\theta}_i = G_i^{-1}X_i^\top y_i$ on the event that G_i is invertible, and $\widetilde{\theta}_i = 0$ otherwise. Let

$$\tau_m = \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad K_m^2 = S_\Omega^2 + \tau_m. \tag{539}$$

For $\delta_{\mathrm{gram}}, \delta_{\mathrm{cov}} \in (0,1)$, suppose

$$m \ge \frac{8L_{\mathrm{cal}}^2}{\kappa_{\mathrm{cal}}}\log\left(\frac{dN}{\delta_{\mathrm{gram}}}\right). \tag{540}$$

Define

$$\begin{aligned}\varepsilon_{\mathrm{cov}}(N, m, \gamma, \delta_{\mathrm{cov}}) &= C_{\mathrm{cov}}K_m^2\left[\sqrt{\frac{d+\log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d+\log(2/\delta_{\mathrm{cov}})}{N}\right] \\ &\quad + \tau_m + \gamma\|\Omega^{-1}\|_{\mathrm{op}}.\end{aligned} \tag{541}$$

Then, with probability at least $1-\delta_{\mathrm{gram}}-\delta_{\mathrm{cov}}$,

$$\left\|\Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2}\right\|_{\mathrm{op}} \le \varepsilon_{\mathrm{cov}}(N, m, \gamma, \delta_{\mathrm{cov}}). \tag{542}$$

Proof. Let

$$\mathcal{E}_{\mathrm{gram}} = \left\{H_i \succeq \frac{\kappa_{\mathrm{cal}} m}{2} I_d \text{ for every } i = 1, \ldots, N\right\}. \tag{543}$$

Lemma 53 gives $\Pr(\mathcal{E}_{\mathrm{gram}}) \ge 1-\delta_{\mathrm{gram}}$. Condition on a realization of the full calibration feature array in $\mathcal{E}_{\mathrm{gram}}$. By Lemma 54, the vectors $Z_i = \Omega^{-1/2}\widetilde{\theta}_i$ are conditionally independent and satisfy

$$\mathbb{E}[Z_iZ_i^\top \mid X_{1:N}] = I_d + R_i, \qquad R_i \succeq 0, \qquad \|R_i\|_{\mathrm{op}} \le \tau_m, \tag{544}$$

and, for every $u \in \mathbb{S}^{d-1}$ and $s \in \mathbb{R}$,

$$\mathbb{E}\big[\exp(su^\top Z_i) \mid X_{1:N}\big] \le \exp\left(\frac{s^2K_m^2}{2}\right). \tag{545}$$

The conditional form of Lemma 52 therefore gives an event of conditional probability at least $1-\delta_{\mathrm{cov}}$ on which

$$\begin{aligned}\left\|\frac{1}{N}\sum_{i=1}^N\big[Z_iZ_i^\top - \mathbb{E}(Z_iZ_i^\top \mid X_{1:N})\big]\right\|_{\mathrm{op}} &\le C_{\mathrm{cov}}K_m^2\Bigg[\sqrt{\frac{d+\log(2/\delta_{\mathrm{cov}})}{N}} \\ &\qquad + \frac{d+\log(2/\delta_{\mathrm{cov}})}{N}\Bigg].\end{aligned} \tag{546}$$

On the intersection of this event and $\mathcal{E}_{\text{gram}}$,

$$\begin{aligned}\left\|\frac{1}{N}\sum_{i=1}^{N} Z_iZ_i^\top - I_d\right\|_{\text{op}} &\le \left\|\frac{1}{N}\sum_{i=1}^{N}\left[Z_iZ_i^\top - \mathbb{E}(Z_iZ_i^\top \mid X_{1:N})\right]\right\|_{\text{op}} + \left\|\frac{1}{N}\sum_{i=1}^{N} R_i\right\|_{\text{op}} \\ &\le C_{\text{cov}}K_m^2\left[\sqrt{\frac{d+\log(2/\delta_{\text{cov}})}{N}} + \frac{d+\log(2/\delta_{\text{cov}})}{N}\right] + \tau_m.\end{aligned} \tag{547}$$

Here

$$\left\|\frac{1}{N}\sum_{i=1}^{N} R_i\right\|_{\text{op}} \le \frac{1}{N}\sum_{i=1}^{N}\|R_i\|_{\text{op}} \le \tau_m. \tag{548}$$

Finally,

$$\begin{aligned}\Omega^{-1/2}\widehat{\Omega}\Omega^{-1/2} &= \frac{1}{N}\sum_{i=1}^{N} Z_iZ_i^\top + \gamma\Omega^{-1}, \\ \Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2} &= \left(\frac{1}{N}\sum_{i=1}^{N} Z_iZ_i^\top - I_d\right) + \gamma\Omega^{-1}.\end{aligned} \tag{549}$$

The triangle inequality gives

$$\left\|\Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2}\right\|_{\text{op}} \le \varepsilon_{\text{cov}}(N, m, \gamma, \delta_{\text{cov}}). \tag{550}$$

The unconditional probability of the intersection of the Gram event and the conditional covariance event is at least $1 - \delta_{\text{gram}} - \delta_{\text{cov}}$ by the law of total probability and a union bound. $\square$

Lemma 56 (Confidence under a frozen estimated covariance)**.** *Assume the post-calibration routing model of Assumption 17. Assume the post-calibration routing archive is fixed before routing begins and contains at most $K_{\max}$ candidates. Let $\mathcal{F}_0^{\text{route}}$ contain all calibration data, the fixed routing archive, and the latent parameter of every routing candidate. Let $\widehat{\Omega} \succ 0$ be $\mathcal{F}_0^{\text{route}}$-measurable and frozen during routing. Assume the post-routing conditional sub-Gaussian bound equation 390 holds for a routing filtration that contains $\mathcal{F}_0^{\text{route}}$. Suppose*

$$\left\|\Omega^{-1/2}(\widehat{\Omega}-\Omega)\Omega^{-1/2}\right\|_{\text{op}} \le \varepsilon, \qquad 0 \le \varepsilon < 1. \tag{551}$$

For candidate a, define

$$V_{a,t}^{\widehat{\Omega}} = X_{a,t}^\top X_{a,t} + \lambda\widehat{\Omega}^{-1}, \quad \widehat{\theta}_{a,t}^{\widehat{\Omega}} = (V_{a,t}^{\widehat{\Omega}})^{-1}X_{a,t}^\top y_{a,t}. \tag{552}$$

Define

$$\alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\text{route}}) = \max\left\{1,\ \sigma\sqrt{d\log\left(1 + \frac{(1+\varepsilon)TL_\Omega^2}{\lambda}\right) + 2\log\left(\frac{K_{\max}}{\delta_{\text{route}}}\right)} + \sqrt{\frac{\lambda}{1-\varepsilon}}\,S_\Omega\right\}. \tag{553}$$

Then, conditional on $\mathcal{F}_0^{\text{route}}$, with probability at least $1-\delta_{\text{route}}$, uniformly over all routing candidates a, all $t \le T$, and all predictable test features x,

$$\left|x^\top(\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a)\right| \le \alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\text{route}})\sqrt{x^\top(V_{a,t}^{\widehat{\Omega}})^{-1}x}. \tag{554}$$

Proof. Fix a routing candidate a. Conditional on $\mathcal{F}_0^{\text{route}}$, the matrix $A = \lambda\widehat{\Omega}^{-1}$ is deterministic and positive definite. Set

$$S_{a,t} = \sum_{s\in\mathcal{T}_{a,t}} x_s(a)\xi_s, \quad V_{a,t}^{\widehat{\Omega}} = A + \sum_{s\in\mathcal{T}_{a,t}} x_s(a)x_s(a)^\top. \tag{555}$$

The reward model and the triangle inequality give

$$\begin{aligned}\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a &= (V_{a,t}^{\widehat{\Omega}})^{-1}S_{a,t} - (V_{a,t}^{\widehat{\Omega}})^{-1}A\theta_a. \\ \|\widehat{\theta}_{a,t}^{\widehat{\Omega}} - \theta_a\|_{V_{a,t}^{\widehat{\Omega}}} &\le \|S_{a,t}\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}} + \|A\theta_a\|_{(V_{a,t}^{\widehat{\Omega}})^{-1}}.\end{aligned} \tag{556}$$

Because $V^{\widehat{\Omega}}_{a,t} \succeq A$,

$$
\begin{aligned}
\|A\theta_a\|^2_{(V^{\widehat{\Omega}}_{a,t})^{-1}} \;&\le\; \theta_a^\top A\theta_a \;=\; \lambda\theta_a^\top \widehat{\Omega}^{-1}\theta_a. \\
(1-\varepsilon)\Omega \;\preceq\; \widehat{\Omega} \;\preceq\; (1+\varepsilon)\Omega, \qquad \widehat{\Omega}^{-1} \;&\preceq\; \frac{1}{1-\varepsilon}\Omega^{-1}. \\
\|A\theta_a\|_{(V^{\widehat{\Omega}}_{a,t})^{-1}} \;&\le\; \sqrt{\frac{\lambda}{1-\varepsilon}}\, S_\Omega.
\end{aligned}
\tag{557}
$$

It remains to control the martingale term. If $\sigma = 0$, conditional sub-Gaussianity implies $S_{a,t} = 0$ almost surely and the claim follows from the bias bound. Suppose $\sigma > 0$. For fixed $q \in \mathbb{R}^d$, define

$$
\begin{aligned}
M_t(q) \;&=\; \exp\!\left(\frac{q^\top S_{a,t}}{\sigma} - \frac{1}{2}q^\top \left(V^{\widehat{\Omega}}_{a,t} - A\right) q\right). \\
\mathbb{E}\!\left[\exp\!\left(\frac{q^\top x_t(a)\xi_t}{\sigma} - \frac{(q^\top x_t(a))^2}{2}\right)\middle|\mathcal{F}_{t-1}\right] \;&\le\; 1 \quad \text{when round } t \text{ selects } a.
\end{aligned}
\tag{558}
$$

If round t does not select a, then $S_{a,t}$ and $V^{\widehat{\Omega}}_{a,t}$ do not change. Therefore $(M_t(q))_{t\le T}$ is a nonnegative supermartingale with initial value one.

Let Q be Gaussian with precision A. Its density and the corresponding mixture process are

$$
\varphi_A(q) \;=\; \frac{\det(A)^{1/2}}{(2\pi)^{d/2}} \exp\!\left(-\frac{1}{2}q^\top A q\right), \quad \mathcal{M}_t \;=\; \int_{\mathbb{R}^d} M_t(q)\,\varphi_A(q)\,dq.
\tag{559}
$$

Tonelli's theorem and the supermartingale property of $M_t(q)$ imply that $(\mathcal{M}_t)_{t\le T}$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Completing the square and then applying Ville's inequality give

$$
\begin{aligned}
\mathcal{M}_t \;&=\; \frac{\det(A)^{1/2}}{\det(V^{\widehat{\Omega}}_{a,t})^{1/2}} \exp\!\left(\frac{\|S_{a,t}\|^2_{(V^{\widehat{\Omega}}_{a,t})^{-1}}}{2\sigma^2}\right). \\
\Pr\!\left(\sup_{t\le T} \mathcal{M}_t \ge \frac{1}{\delta_a}\right) \;&\le\; \delta_a, \\
\|S_{a,t}\|_{(V^{\widehat{\Omega}}_{a,t})^{-1}} \;&\le\; \sigma\sqrt{2\log\!\left(\frac{\det(V^{\widehat{\Omega}}_{a,t})^{1/2}}{\det(A)^{1/2}\delta_a}\right)} \quad \text{uniformly over } t \le T.
\end{aligned}
\tag{560}
$$

The last row holds with probability at least $1-\delta_a$. To bound its determinant ratio, use $A = \lambda\widehat{\Omega}^{-1}$ and $\widehat{\Omega} \preceq (1+\varepsilon)\Omega$:

$$
\begin{aligned}
\frac{\det(V^{\widehat{\Omega}}_{a,t})}{\det(A)} \;&=\; \det\!\left(I + \lambda^{-1}\widehat{\Omega}^{1/2} X_{a,t}^\top X_{a,t} \widehat{\Omega}^{1/2}\right). \\
\operatorname{tr}\!\left(\lambda^{-1}\widehat{\Omega}^{1/2} X_{a,t}^\top X_{a,t}\widehat{\Omega}^{1/2}\right) \;&=\; \lambda^{-1}\operatorname{tr}(X_{a,t}\widehat{\Omega}X_{a,t}^\top) \\
&\le\; \frac{1+\varepsilon}{\lambda}\sum_{s\in\mathcal{T}_{a,t}} x_s(a)^\top \Omega x_s(a) \;\le\; \frac{(1+\varepsilon)TL_\Omega^2}{\lambda}.
\end{aligned}
\tag{561}
$$

For any positive-semidefinite $d\times d$ matrix M with eigenvalues $\nu_1,\dots,\nu_d$,

$$
\begin{aligned}
\det(I+M) \;&=\; \prod_{k=1}^d (1+\nu_k) \;\le\; \left(1 + \sum_{k=1}^d \nu_k\right)^d \;=\; (1+\operatorname{tr} M)^d. \\
2\log\!\left(\frac{\det(V^{\widehat{\Omega}}_{a,t})^{1/2}}{\det(A)^{1/2}\delta_a}\right) \;&\le\; d\log\!\left(1 + \frac{(1+\varepsilon)TL_\Omega^2}{\lambda}\right) + 2\log(1/\delta_a).
\end{aligned}
\tag{562}
$$

Taking $\delta_a = \delta_{\mathrm{route}}/K_{\max}$ and applying a union bound over the fixed routing archive gives, with probability at least $1-\delta_{\mathrm{route}}$, the simultaneous bound

$$
\begin{aligned}
\|\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a\|_{V^{\widehat{\Omega}}_{a,t}} \;&\le\; \alpha_{\widehat{\Omega},T}(\varepsilon, \delta_{\mathrm{route}}) \\
\left|x^\top(\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a)\right| \;&\le\; \|\widehat{\theta}^{\widehat{\Omega}}_{a,t} - \theta_a\|_{V^{\widehat{\Omega}}_{a,t}} \sqrt{x^\top (V^{\widehat{\Omega}}_{a,t})^{-1} x}.
\end{aligned}
\tag{563}
$$

The two rows hold for every candidate a, every $t \le T$, and every predictable test feature x. The second row is Cauchy–Schwarz in the $V^{\widehat{\Omega}}_{a,t}$-inner product. Substituting the preceding simultaneous parameter bound proves the claim. $\square$

Theorem 57 (Calibration-split frozen-covariance Hyper-Ridge)**.** *Fix a post-calibration routing horizon T. Consider the following modified Hyper-Ridge PersonalEvo procedure.*

For the version combined with the four-term decomposition, draw one routing user U before the post-calibration phase and hold that user fixed for all T routing rounds.

Before routing begins, the algorithm runs a calibration phase on $N = N_T$ calibration candidates that are not used in the subsequent routing archive. For each calibration candidate i, the algorithm collects $m = m_T$ calibration rewards under the exogenous uniform-routing calibration policy:

$$y_{i,j} \coloneqq r_{i,j} - b(u_{i,j}) \;=\; x_{i,j}^\top \theta_i + \xi_{i,j}, \qquad j = 1, \dots, m. \tag{564}$$

Let

$$\begin{aligned} X_i \;&=\; \begin{bmatrix} x_{i,1}^\top \\ \vdots \\ x_{i,m}^\top \end{bmatrix}, \qquad\qquad G_i \;=\; X_i^\top X_i, \\ \widetilde{\theta}_i \;&=\; \begin{cases} G_i^{-1} X_i^\top y_i, & G_i \text{ invertible}, \\ 0, & \text{otherwise}. \end{cases} \end{aligned} \tag{565}$$

For a deterministic ridge floor $\gamma = \gamma_T > 0$, define

$$\begin{aligned} \widehat{\Omega} \;&=\; \frac{1}{N} \sum_{i=1}^{N} \widetilde{\theta}_i \widetilde{\theta}_i^\top + \gamma I_d. \\ \widehat{\Omega}_t \;&=\; \widehat{\Omega}, \qquad t = 1, \dots, T. \end{aligned} \tag{566}$$

The post-calibration routing archive is fixed before routing begins and has cardinality at most $K_{\max}$. The routing UCB rule uses

$$\alpha_T \;=\; \alpha_{\widehat{\Omega},T}(\varepsilon_*, \delta_{\mathrm{route}}), \tag{567}$$

where $\alpha_{\widehat{\Omega},T}$ is defined in Lemma 56, and where ε_* is the deterministic quantity defined below.

Assume Assumption 19 and the post-calibration routing assumptions of Assumption 17. Let

$$\begin{aligned} \tau_m \;&=\; \frac{2\sigma^2}{\kappa_{\mathrm{cal}} m}, \qquad\qquad K_m^2 \;=\; S_\Omega^2 + \tau_m, \\ m \;&\ge\; \frac{8L_{\mathrm{cal}}^2}{\kappa_{\mathrm{cal}}} \log\!\left(\frac{dN}{\delta_{\mathrm{gram}}}\right). \end{aligned} \tag{568}$$

Here $\delta_{\mathrm{gram}}, \delta_{\mathrm{cov}}, \delta_{\mathrm{route}} \in (0,1)$. Define

$$\begin{aligned} \varepsilon_* \;=\; & C_{\mathrm{cov}} K_m^2 \left[\sqrt{\frac{d + \log(2/\delta_{\mathrm{cov}})}{N}} + \frac{d + \log(2/\delta_{\mathrm{cov}})}{N} \right] \\ & + \tau_m + \gamma \|\Omega^{-1}\|_{\mathrm{op}}. \end{aligned} \tag{569}$$

Computing this radius requires supplied deterministic bounds for $S_\Omega, \sigma, \kappa_{\mathrm{cal}}, L_{\mathrm{cal}}, L_\Omega$, and $\|\Omega^{-1}\|_{\mathrm{op}}$. Without them, the certificate is oracle-tuned. Let $\mathcal{F}_0^{\mathrm{route}}$ contain all calibration data, the fixed post-calibration archive, and the latent parameter of every routing candidate. Assume the post-routing conditional sub-Gaussian bound holds after this sigma-field is added to the routing filtration, and assume $\varepsilon_* < 1$. Define

$$c_T^\Omega \;=\; 1 + \log\!\left(1 + \frac{TL_\Omega^2}{\lambda}\right), \quad B_{T,0}^{\mathrm{cal}} \;=\; 2\alpha_T \sqrt{2K_{\max} T\, c_T^\Omega \bar{d}_{\mathrm{eff},T}(\Omega)}. \tag{570}$$

Then, with probability at least $1 - \delta_{\mathrm{gram}} - \delta_{\mathrm{cov}}$,

$$\rho_{\Omega,T} \coloneqq \max_{1 \le t \le T} \left\| \Omega^{-1/2}(\widehat{\Omega}_t - \Omega)\Omega^{-1/2} \right\|_{\mathrm{op}} \;\le\; \varepsilon_*. \tag{571}$$

In particular, taking $\delta_{\mathrm{gram}} = \delta_{\mathrm{cov}} = \delta/2$ gives the covariance-error control with probability at least $1-\delta$.

With probability at least $1-\delta_{\mathrm{gram}}-\delta_{\mathrm{cov}}-\delta_{\mathrm{route}}$,

$$\sum_{t=1}^{T} L_t(U) \;\le\; \sqrt{1+\varepsilon_*}\, B^{\mathrm{cal}}_{T,0}. \tag{572}$$

Define the uncharged and charged remainder terms and record the resulting expectation bounds:

$$\begin{aligned} \mathrm{Rem}^{\mathrm{cal}}_T \;&=\; \big(\sqrt{1+\varepsilon_*}-1\big)\, B^{\mathrm{cal}}_{T,0} + (\delta_{\mathrm{gram}}+\delta_{\mathrm{cov}}+\delta_{\mathrm{route}})T. \\ \sum_{t=1}^{T}\mathbb{E}[L_t(U)] \;&\le\; B^{\mathrm{cal}}_{T,0} + \mathrm{Rem}^{\mathrm{cal}}_T\,. \\ \mathrm{Rem}^{\mathrm{charged}}_T \;&=\; Nm + \mathrm{Rem}^{\mathrm{cal}}_T\,. \\ \sum_{t=1}^{T}\mathbb{E}[L_t(U)] + Nm \;&\le\; B^{\mathrm{cal}}_{T,0} + \mathrm{Rem}^{\mathrm{charged}}_T\,. \end{aligned} \tag{573}$$

The last two rows assign each of the Nm calibration pulls a worst-case unit opportunity-cost charge. For fixed $d, K_{\max}, S_\Omega, \sigma, \kappa_{\mathrm{cal}}, L_{\mathrm{cal}}, L_\Omega, \lambda$, and $\|\Omega^{-1}\|_{\mathrm{op}}$, and uniformly bounded $\bar d_{\mathrm{eff},T}(\Omega)$, take

$$\begin{aligned} N \;&=\; \lceil T^{1/5}\rceil, \qquad\qquad m \;=\; \lceil T^{1/5}\rceil, \quad \gamma \;=\; T^{-1/5}, \\ \delta_{\mathrm{gram}} \;&=\; \delta_{\mathrm{cov}} \;=\; \delta_{\mathrm{route}} \;=\; T^{-2}. \end{aligned} \tag{574}$$

For all sufficiently large T, the resulting rates are

$$\begin{aligned} \varepsilon_* \;&<\; 1, & \varepsilon_* \;&=\; O\Big(T^{-1/10}\sqrt{\log T}\Big)\,. \\ \alpha_T \;&=\; O(\sqrt{d\log T}), & B^{\mathrm{cal}}_{T,0} \;&=\; O\bigg(\sqrt{dK_{\max}\bar d_{\mathrm{eff},T}(\Omega)T}\,\log T\bigg), \\ \mathrm{Rem}^{\mathrm{cal}}_T \;&=\; O\Big(T^{2/5}(\log T)^{3/2}\Big) \;=\; o(\sqrt{T}). \end{aligned} \tag{575}$$

The same rate holds for $\mathrm{Rem}^{\mathrm{charged}}_T$.

Proof. Because $\widehat\Omega_t = \widehat\Omega$ for every post-calibration routing round, Lemma 55 gives the event

$$\mathcal{E}_{\mathrm{cov}} \;:=\; \left\{\left\|\Omega^{-1/2}(\widehat\Omega-\Omega)\Omega^{-1/2}\right\|_{\mathrm{op}} \le \varepsilon_*\right\}, \quad \Pr(\mathcal{E}_{\mathrm{cov}}) \;\ge\; 1-\delta_{\mathrm{gram}}-\delta_{\mathrm{cov}}. \tag{576}$$

On $\mathcal{E}_{\mathrm{cov}}$, $\rho_{\Omega,T} \le \varepsilon_*$, and

$$(1-\varepsilon_*)\Omega \;\preceq\; \widehat\Omega \;\preceq\; (1+\varepsilon_*)\Omega. \tag{577}$$

By Lemma 56, conditional on $\mathcal{F}^{\mathrm{route}}_0$, with probability at least $1-\delta_{\mathrm{route}}$, the simultaneous routing confidence event

$$\mathcal{E}_{\mathrm{route}} \;:=\; \left\{\left|x^\top(\widehat\theta^{\widehat\Omega}_{a,t}-\theta_a)\right| \;\le\; \alpha_T\sqrt{x^\top (V^{\widehat\Omega}_{a,t})^{-1}x} \text{ for all } a,t,x\right\} \tag{578}$$

holds, where x ranges over predictable test features. Its conditional probability bound integrates to an unconditional one. Hence $\mathcal{E} = \mathcal{E}_{\mathrm{cov}} \cap \mathcal{E}_{\mathrm{route}}$ satisfies $\Pr(\mathcal{E}) \ge 1-\delta_{\mathrm{gram}}-\delta_{\mathrm{cov}}-\delta_{\mathrm{route}}$.

On $\mathcal{E}$, fix t, let $a^\star_t \in \arg\max_{a\in A_t(U)} x_t(a)^\top\theta_a$, and define the local confidence width

$$w^{\widehat\Omega}_{a,t} \;:=\; \sqrt{x_t(a)^\top (V^{\widehat\Omega}_{a,t})^{-1} x_t(a)}. \tag{579}$$

The confidence event and the UCB choice of a_t yield

$$\begin{aligned} x_t(a^\star_t)^\top\theta_{a^\star_t} \;&\le\; x_t(a^\star_t)^\top\widehat\theta^{\widehat\Omega}_{a^\star_t,t} + \alpha_T w^{\widehat\Omega}_{a^\star_t,t} \\ &\le\; x_t(a_t)^\top\widehat\theta^{\widehat\Omega}_{a_t,t} + \alpha_T w^{\widehat\Omega}_{a_t,t} \\ &\le\; x_t(a_t)^\top\theta_{a_t} + 2\alpha_T w^{\widehat\Omega}_{a_t,t}. \end{aligned} \tag{580}$$

Thus, because $0 \le L_t(U) \le 1$ and $\alpha_T \ge 1$,

$$L_t(U) \;\le\; 2\alpha_T \min\{1, w^{\widehat{\Omega}}_{a_t,t}\}. \tag{581}$$

The covariance event gives $\widehat{\Omega}^{-1} \succeq (1+\varepsilon_*)^{-1}\Omega^{-1}$. Therefore,

$$\begin{aligned} V^{\widehat{\Omega}}_{a,t} \;&=\; X_{a,t}^\top X_{a,t} + \lambda \widehat{\Omega}^{-1} \\ &\succeq\; X_{a,t}^\top X_{a,t} + \frac{\lambda}{1+\varepsilon_*}\Omega^{-1} \\ &\succeq\; \frac{1}{1+\varepsilon_*}\left(X_{a,t}^\top X_{a,t} + \lambda\Omega^{-1}\right) \;=\; \frac{1}{1+\varepsilon_*} V^{\Omega}_{a,t}. \\ (V^{\widehat{\Omega}}_{a,t})^{-1} \;&\preceq\; (1+\varepsilon_*)(V^{\Omega}_{a,t})^{-1}. \\ w^{\widehat{\Omega}}_{a,t} \;&\le\; \sqrt{1+\varepsilon_*}\, w^{\Omega}_{a,t}, \qquad w^{\Omega}_{a,t} \;\coloneqq\; \sqrt{x_t(a)^\top (V^{\Omega}_{a,t})^{-1} x_t(a)}. \end{aligned} \tag{582}$$

Since $\min\{1, cz\} \le c \min\{1, z\}$ for $c \ge 1$ and $z \ge 0$,

$$L_t(U) \;\le\; 2\alpha_T \sqrt{1+\varepsilon_*}\, \min\{1, w^{\Omega}_{a_t,t}\}. \tag{583}$$

Summing over t, partitioning selected rounds by candidate, applying Cauchy–Schwarz, and then applying the already-proved Ω-elliptical-potential lemma candidate by candidate yields

$$\sum_{t=1}^{T} \min\{1, w^{\Omega}_{a_t,t}\} \;\le\; \sqrt{2K_{\max} T\, c^{\Omega}_T \bar{d}_{\mathrm{eff},T}(\Omega)}. \tag{584}$$

Thus, on $\mathcal{E}$,

$$\begin{aligned} \sum_{t=1}^{T} L_t(U) \;&\le\; \sqrt{1+\varepsilon_*}\, B^{\mathrm{cal}}_{T,0} \\ &=\; B^{\mathrm{cal}}_{T,0} + (\sqrt{1+\varepsilon_*} - 1) B^{\mathrm{cal}}_{T,0}. \end{aligned} \tag{585}$$

On $\mathcal{E}^c$, $0 \le L_t(U) \le 1$ gives $\sum_{t=1}^T L_t(U) \le T$. Taking expectations therefore yields

$$\begin{aligned} \sum_{t=1}^{T} \mathbb{E}[L_t(U)] \;\le\; & B^{\mathrm{cal}}_{T,0} + (\sqrt{1+\varepsilon_*} - 1) B^{\mathrm{cal}}_{T,0} \\ & + (\delta_{\mathrm{gram}} + \delta_{\mathrm{cov}} + \delta_{\mathrm{route}})T. \end{aligned} \tag{586}$$

This is the stated bound with $\mathrm{Rem}^{\mathrm{cal}}_T$. Under the declared unit opportunity-cost convention, the Nm calibration pulls add at most Nm, giving $\mathrm{Rem}^{\mathrm{charged}}_T$.

It remains to verify the stated asymptotic schedule. With fixed problem constants,

$$\begin{aligned} \sqrt{\frac{d + \log(2/\delta_{\mathrm{cov}})}{N}} \;&=\; O\Big(T^{-1/10}\sqrt{\log T}\Big), \quad & \frac{d + \log(2/\delta_{\mathrm{cov}})}{N} \;&=\; O(T^{-1/5}\log T), \\ \tau_m \;&=\; O(T^{-1/5}), & \gamma\|\Omega^{-1}\|_{\mathrm{op}} \;&=\; O(T^{-1/5}), \\ \varepsilon_* \;&=\; O\Big(T^{-1/10}\sqrt{\log T}\Big). \end{aligned} \tag{587}$$

Since $T^{-1/10}\sqrt{\log T} \to 0$, there exists $T_0 < \infty$ such that $\varepsilon_* < 1$ for all $T \ge T_0$. The Gram lower-tail condition also holds for all sufficiently large T, because its right-hand side is $O(\log T)$ whereas $m = \lceil T^{1/5} \rceil$.

The explicit radius in Lemma 56, the bound $c_T^{\Omega} = O(\log T)$, and the uniformly bounded effective-dimension cap give

$$\begin{aligned}
\alpha_T &= O(\sqrt{d \log T}), \\
B_{T,0}^{\mathrm{cal}} &= O\left(\sqrt{dK_{\max}\bar{d}_{\mathrm{eff},T}(\Omega)T}\,\log T\right). \\
\sqrt{1+\varepsilon_*} - 1 &= \frac{\varepsilon_*}{\sqrt{1+\varepsilon_*}+1} \le \varepsilon_*. \\
(\sqrt{1+\varepsilon_*} - 1)B_{T,0}^{\mathrm{cal}} &= O\left(T^{-1/10}\sqrt{\log T}\right) O(\sqrt{T}\log T) = O\left(T^{2/5}(\log T)^{3/2}\right). \\
Nm &= O(T^{2/5}), \\
(\delta_{\mathrm{gram}} + \delta_{\mathrm{cov}} + \delta_{\mathrm{route}})T &= 3T^{-1}. \\
\mathrm{Rem}_T^{\mathrm{charged}} &= O(T^{2/5}) + O\left(T^{2/5}(\log T)^{3/2}\right) + O(T^{-1}) = o(\sqrt{T}),
\end{aligned} \tag{588}$$

because $T^{-1/10}(\log T)^{3/2} \to 0$. The uncharged remainder is no larger. □

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